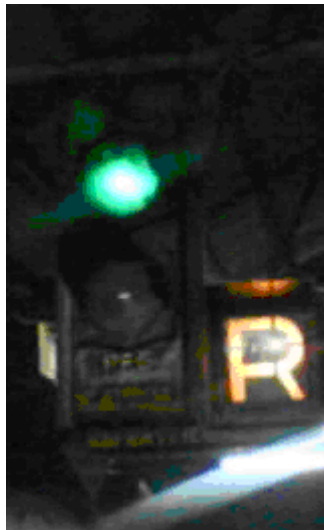


Indian Railways General Information & Safe Working Rules



Electric Loco Training Centre
South-East-Central Railway
Uslapur.

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INDEX

S/NO.	CONTENT	PAGE NO.
<i>GENERAL INFORMATION</i>		
1.	Overview of Railway Organisation	7
2.	Indian Railways: Aims and Objectives	7
3.	History of Indian Railways	7
4.	Railway Board/Zone Structure	7
5.	Organisation and Function of Departments	9
6.	Railway Transport Network	10
7.	Railway Electrification Programme	11
8.	Electrical Operations Organisation	11
9.	Strength and weakness of Railway Transport	12
10.	Competition from the mode of Transport	12
11.	Future business prospect	12
12.	Fire Fighting	13
<i>ROLE OF AN ASSISTANT LOCO PILOT</i>		
1.	Conduct Rules	19
2.	Responsibility of an Assistant Loco Pilot	19
3.	Rule books and Aids	19
<i>LOCO CHECKING AND ASSISTANT LOCO PILOT'S DUTY</i>		
1.	Assistant Loco pilots duty at Booking Point	20
2.	Signing on duties	20
3.	Crew Management System	20
4.	Checks of Various registers	22
5.	Duties while taking over charge	22
6.	Reading loco log book	22
7.	Checking electric loco while standing	23
8.	Check to be done before turning out from shed	23
9.	Procedure for taking a loco from shed	23
10.	Checks on locomotive at Yard/Pooling Points	23
11.	Checking electric locos while handling	23
12.	Duties while on run	23
13.	Coupling of locos, screw coupling etc.	24

TRANSPORTATION AND OPERATION

1.	Introduction to SWR	25
2.	Basic definition and concepts	25
3.	Difference between “Transportation” and “Operation.”	28
4.	Overview of Indian Railway Act’1890	28

SIGNALS

1.	Signal and signal types	29
2.	Introduction	29
3.	Kinds of Signals	29
4.	Aspects and indications of signals	30
5.	IB and Gate Signals	47
6.	IB Signals	47
7.	Gate Signals	48
8.	IB Signal defective	48
9.	Gate signal defective	49
10.	Working of signals and points	50
11.	Facing and Trailing points	51
12.	Locking of facing points	51
13.	Interlocking	51
14.	Isolation of points	52
15.	Track circuit and axle counter	52
16.	Defective signal working	52
17.	Introduction	52
18.	Defective approach signals	53
19.	Defective departure signals	55
20.	Conflicting signals	56
21.	Bobbing and flickering	56
22.	Defective stop signals in automatic block sections	57

STATION AND TRACK WORKING

1.	Station working	59
2.	Classification of stations & their minimum equipments	59
3.	Station Limit and Station Section	60
4.	Adequate distance, block back & block forward	62
5.	Sidings	63
6.	Permanent way	64
7.	introduction	64
8.	Boards in block section	65
9.	Boards in station section	68

10.	Speed restriction	
	69	
11.	Boards provided in speed restriction zone	71
12.	Green notice	72
13.	Caution order	72
14.	Types of caution order	72

SYSTEM OF WORKING

1.	System of working	74
2.	Absolute block system	74
3.	Automatic block system	75
4.	One train only system	76
5.	Train staff and ticket system	77
6.	Authorities for working	77

TRAIN WORKING

1.	Locomotive and Train working	79
2.	ODC and its classification	82
3.	Speed of loco during shunting	82
4.	Speed of different types of trains	83
5.	Material train working	83
6.	Precautions to be observed	84
7.	Loco and train operation	84
8.	Time table	85
9.	Types of timings	86
10.	Types of trains	86
11.	Clearing load from road side station	87
12.	Working on electrified section	90
13.	Applicability of General rules	90
14.	Special definitions	91
15.	Rule for electrified sections	91
16.	Electrical safety	92
17.	Whistle codes	92
18.	Introduction	92
19.	Types of whistle codes	92
20.	Significance of different whistle codes	92

ABNORMAL WORKING

1.	Abnormal loco working	96
2.	Defective head light	96
3.	Defective driving cab	96

4.	Defective speedometer	96
5.	Leading cab whistle/horn not sounding	97
6.	Loco Pilot incapacitated	97
7.	Flasher light working	97
8.	Reporting unusual occurrence	97
9.	Bad riding	98
10.	Wheel slipping	98
11.	Storms and cyclone	99
12.	No tension on line	99
13.	Run over cases	99
14.	Foggy weather working	100
15.	Accident and unusual occurrence	100
16.	Working in the event of accidents	101
17.	Working of LE/Train to accident site	101
18.	Singe line working introduce on D/L section	102
19.	TSL working in D/L section	103
20.	Total interruption of communication	105
21.	Train parting	110

MISCELLANEOUS

1.	Air brake system used in rolling stock	112
2.	Percentage of brake power	112
3.	Specification of vacuum on engine gauge and brake van	112
4.	BP continuity	113
5.	Releasing the brake binding	113
6.	Types of BPC	114
7.	Accidents	115
8.	Train held up at 1 st stop signal	117
9.	Authority of shunting	118
10.	Different forms used for train movement	119
11.	Precaution in gradient section	121
12.	Speed restriction on different circumstances	122
13.	New authorities	125

ANNEXURE

Annexure on correction slips	127
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OVER VIEW OF RAILWAY ORGANISATION

DEFINITION OF RAILWAY:

“ Railway means a Railway or any portion of a Railway, for the public carriage of passenger, animals or goods and include-

1. All land within the fences or other boundary marks including the limits of the land appurtenant to a Railway.
2. All lines of Railway, sidings or branches worked over for the purpose of, or in connection with, a Railway.
3. All stations, offices, warehouses, wharves, workshops, manufactories, fixed plant and machinery and other works constructed for the purpose of, or in connection with a Railway.
4. All ferries, ships, boats and crafts which are used on inland, waters for the purpose of the traffic of a Railway and belong to or hired or worked authority administering the Railway.

INDIAN RAILWAYS- AIM & OBJECTIVES:

The corporate management objectives of Railway undertaking are as under:

1. To provide all transport for both Passenger and goods adequate to meet demand in areas where Railway operation confers optimum benefit to the economy having due regard to the Government's policy of development of backward areas.
2. To provide such rail transport at the lowest cost consistent with-
 - a. Requirement of the Railway users and safety of operation.
 - b. Adequate provision of replacements of assets and some provision for development of business; and
 - c. The least amount of pollution of environment.
3. To work in association with or utilize other modes of transportation, such as pipelines and road transport and so engage in ancillary activities necessary to sub serve the above two objects.
4. To establish corporate image of the Railway as being an up to date business organization with the interests of public and of the nation as its prime objectives; and
5. To develop organizationally effective personnel with pride in their work and faith in the management.

HISTORY OF INDIAN RAILWAYS:

In India, the Railway system came into being by the then Governor General of British India, Lord Dalhousie, in July 1850. The first Railway line was started being laid down in India from Bombay to Thane for a length of about 21 miles on 16th April 1853. The second line connecting Howrah and Hoogly was opened in 1854. In south India, the first Railway line was opened between Vysarpadi and Arcot in 1856. The Indian Railway Act was enacted on 21st March 1890. After independence in 1947, the process of Nationalization of Railway system was completed in April 1950 and the administrative control of all the state owned Railways was assumed by the Government of India.

RAILWAY BOARD/ZONE STRUCTURE

President has formed rules for the convenient transaction to the business of the Government of India, the subjects allocated to the Ministry of Railway (Railway Board) are in charge of the minister of Railways, who is minister of cabinet rank. He is associated in his work by one or more ministers who are of the status of minister of state or Deputy Minister. These ministers perform such functions as may be assigned to them in relation to the business allocated to ministry.

The Railway Board is the Chief Administrative and Executive Body assisting the Minister of

Railways in the discharge of his function. It was constituted by a resolution of Government of India dated 18th February 1905. The Railway Board constituted at present consist of:

1. Chairman, Railway Board,
2. Financial Commissioner Railways; and
3. Five members each of Traffic, Staff, Mechanical Engg, Civil Engg. And Electrical Engg.

The Chairman Railway Board is ex-officio principal secretary to the Government of India in the Ministry of Railways. He is solely responsible under the Minister of Railways for arriving at decision on technical question and advising the Government of India on the matters of Railway policy. The Chairman also functions as member in respect of one or the other subjects as decided and is also responsible for co-ordination.

Financial commissioner for Railways represents the ministry of finance on the Board and also functions as ex-officio secretary to Government of India in the ministry of Railways in financial matters. He has direct contact with the Finance Minister whom he keeps informed of developments in the Ministry of Railways. In case of any disagreement with chairman or Railway Minister or any financial aspect, he has the right to refer it to the Finance Minister.

Each of the members is responsible for dealing with all aspects of the technical subject of which he is in charge. In order to be able to effectively discharge the duties and responsibilities from the in creed tempo of development works, there are additional members in the Railway Board.

Railway Board is assisted by several directorates each under an adviser, such as Adviser Works, Mechanical, Traffic, Commercial and civil engineering. Such advisors are assisted by directors in S.A. grade, Joint Directors, Deputy Directors and Assistant Directors.

The Advisers are responsible for issuing instruction direct to Railway Board and receive and deal with reference from General Public and other ministries of the Government of India with their respective jurisdiction.

ATTACHED OFFICES:

The Railway Board has the following attached and subordinate offices:

1) Attached offices:

- a. The Research, Design and Standard Organization.
- b. Office of the Railway Liaison Officer with the Directorate General of supplies and disposals.

2) Subordinate offices:

- a. Railway Recruitment Boards
- b. Railway Staff College, Vadodara
- c. Indian Railway Institute of Civil Engineering, Pune
- d. Indian Railway Institute of Signal, Engineering and Telecommunication, Hyderabad.
- e. Indian Railway Institute of Electrical Engineering, Nasik.
- f. Central Organisation for Modernisation of Workshop, Delhi
- g. Centre for Advance Maintenance Technology, Gwalior
- h. Central Organisation for Railway Electrification, Allahabad

The Railway Recruitment Boards are at Mumbai, Kolkata, Chennai, Allahabad, Muzzffarpur, Guahati, Danapur, Bhopal, Chandigarh and Ranchi. RDSO organization is at Lucknow.

RAILWAY ZONES AND MANAGEMENT:

On operational and Management consideration, the Indian Railway has been divided into 16 Zones in order to have smooth and efficient working. These are placed in the charge of General Manager who will be the chief administrative officer answerable to Railway Board for all matters affecting the working of Railways. These zones are as under :

S/No.	Zone	Head quarter
1.	Southern Railway	Chennai
2.	Central Railway	Mumbai
3.	Western Railway	Mumbai
4.	Eastern Railway	Kolkata
5.	Northern Railway	New Delhi
6.	North Eastern Railway	Gorakhpur
7.	North East Frontier Railway	Maligaon (Guwahati)
8.	South Eastern Railway	Kolkata
9.	South Central Railway	Secundarabad
10.	North Central Railway	Allahabad
11.	North Western Railway	Jaipur
12.	South Western Railway	Bangalore
13.	West Central Railway	Jabalpur
14.	East Central Railway	Hajipur
15.	East Coast Railway	Bhubaneswar
16.	South East Central Railway	Bilaspur

Apart from the above 16 Zonal Railways, there are following production units, working directly under Railway Board:

1. Chittaranjan Locomotive Works for manufacturing electric locomotives.
2. Diesel Locomotive Works at Varanasi for manufacturing diesel locomotives.
3. Integral Coach factory at Perambur near Chennai for manufacturing coaches
4. Wheel and axle plant at Bangalore.
5. Diesel component works at Patiala for spares.
6. Rail coach Factory at Kapurthala.
7. Railway Electrification at Allahabad.
8. Metro Railway at Kolkata and Delhi.

ORGANISATION AND FUNCTION OF DEPARTMENTS:

In order to discharge the administrative functions, each General Manager is assisted by two Additional General Managers, Sr. Deputy General Manager, Deputy General Managers, Chief Public Relation Officer, Chief Planning officer and heads of departments for the following functions:

- | | | |
|-----|-------------|--|
| 1. | Operating | Chief Operating Manager |
| 2. | Engineering | Chief Engineer (Civil) |
| 3. | Mechanical | Chief Mechanical Engineer |
| 4. | Electrical | Chief Electrical Engineer |
| 5. | S & T | Chief Signal and Telecommunication Engineer |
| 6. | Accounts | Financial Advisor and Chief Accounts Officer |
| 7. | Commercial | Chief Commercial Manager |
| 8. | Personnel | Chief Personnel Officer |
| 9. | Stores | Controller of Stores |
| 10. | Medical | Chief Medical Director |
| 11. | Security | Chief Security Commissioner |

The above-mentioned head of the departments are responsible for functioning of their departments efficiently and are directly responsible to the General Managers for all matters concerning to their departments. They have been provided with Deputies, Senior and Junior Scale officers to assist them in discharging their duties efficiently.

RAILWAYS TRANSPORT NETWORK:

Transportation is regarded as the lifeline of a country. The economic, social and commercial progress of a country depends upon transportation system. Railway is a surface transportation network.

The Railway network in India have been classified in to different categories by the Railway Board on the basis of importance of the route and they have been divided mainly into three categories: (1) Trunk Route (2) Main Line (3) Branch Line

- (i) The Trunk Routes: The following 6 routes on the broad gauge and three routes on the meter gauge have been classified as trunk routes:

ON THE BROAD GAUGE:

- a. Delhi-MGS-HWH
- b. Delhi-Kota-Mumbai
- c. Delhi-Jhansi-Nagpur
- d. Howrah-Nagpur-Mumbai
- e. Howrah-Vijaywara-Chennai
- f. Mumbai-Guntakal-Chennai

The following track standards have been specified for trunk routes:

Designed speed for new track	-	160 KMPH
Maximum permissible speed	-	120 KMPH
Rail section	-	50 Kg/mtr./ or heavier (60 Kg/mtr)
Sleeper density	-	N + 7 (N= Length of track)
Ballast cushion	-	25 Cms below the sleeper
Degree of curvature	-	7.5 degree

(ii) Main Line:

All the lines, other than the trunk routes where the maximum permissible speed is 100 KMPH on the broad gauge and 75 KMPH on the meter gauge

(iii) Branch lines:

These lines are classified under the following criterion:

All those broad gauge line which carry less than 10 gross million tones of goods per annum and have a maximum permissible speed of less than 100 KMPH, are termed as branch line.

CLASSIFICATION OF INDIAN RAILWAYS BASED ON SPEED CRITERIA:

- a) Group 'A' lines -Speed 160 KMPH or more
 - i) New Delhi-Mumbai, via Kota
 - ii) New Delhi-Chennai, via G.T. route
 - iii) Howrah-Mumbai, via NGP
- b) Group 'B' lines-Maximum sanctioned speed is 130 KMPH. At present 13 routes are under this category.
- c) Group 'C' lines- All sub-urban routes of Mumbai and Kolkata.

- d) Group 'D' lines-All other routes on the Broad gauge and meter gauge of the country where maximum permissible speed is 100 KMPH.
- e) Group 'E' lines-Other routes and branch lines where the maximum permissible speed is less than 100 KMPH.

RAILWAY ELECTRIFICATION PROGRAMME

Railway electrification organisation under a separate G.M. has been entrusted with electrification of railway network throughout the country. A centralised organisation known as "CENTRAL ORGANISATION FOR RAILWAY ELECTRIFICATION" was set up at Allahabad in 1979.

The I. R. Core at New Delhi has also been placed under the administration of G.M. core at New Delhi w.e.f. July, 1998. In S. E. C Railway first electrified route on main line was commissioned between JHARSUGUDA and DURGA a distance of about 351 kms. in the year 1967.

SECR comprises of 3 divisions :-

- a) BSP division - 3494.87 TKM
- b) Raipur division - 754.49 TKM
- c) Nagpur division - 741.82 TKM

DUG-NGP Section was electrified in the year 1990-91. BSP-KTE(CIC) section was electrified in 1994 with 1091.282 track km.

The electrification programme is being adopted all over the country due to

- (i) Pollution free environment
- (ii) Easy and low maintenance cost
- (iii) Saving of precious resources like coal & oil
- (iv) Less starting time
- (v) High starting torque
- (vi) Re-generation of current
- (vii) Comfortable journey

Electrical Operation Organisation:

The electrical operation organisation of our Railway is headed by Chief Electrical Loco Engineer and he is assisted by Deputy Chief Electrical Engineer (OP) and Assistant Engineer (OP) at the Zonal Head Quarter, Bilaspur. The Senior Divisional Electrical Engineer (OP) of all divisions are under CELE and are liable to report to him through Dy. CEE(OP) in all the operational matters.

STRENGTH AND WEAKNESS OF RAILWAY TRANSPORT

Indian Railway transport network is the second largest in the world. Being large in size and volume has an advantage in itself, but only when the footage of the mass is strong. The Railway network in India is spread over all parts of the country and are well con-

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“Indian Railway transport network is the second largest in the world. Being large in size and volume has an advantage in itself, but only when the footage of the mass is strong. The Railway network in India is spread over all parts of the country and are well connected and well managed. “

nected and well managed. So, if the network can be worked with co-ordination and cohesion it can build into a formidable unit and play an important role in the growth and development of the country.

But being very big in size has its disadvantage also, unless there is cohesion and co-ordination the unity may fall apart and the network may turn into weak one. So for the progress and advancement of the country and also to keep the network connected, we must work cohesively.

COMPETITION FROM OTHER MODES OF TRANSPORT

Prior to the construction of National highways, Railways was the most reliable and dependable transport system in India and in fact the Railway was reigning in this field. But with the development of roadways and heavy-duty road vehicles, now a day the Railway is facing a hard and tenacious competition from the roadways. So unless the reliability, responsibility and efficiency in Railway transportation system is enhanced, the Railways will keep on losing grounds. For this accident free and comfortable transportation network in the Railways has to be developed to hold and gain grounds in this field.

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FUTURE BUSINESS PROSPECT

With the liberalization of Government business policy and participation of corporate house and multinationals in Indian business arena, a vast avenue has opened in front of the Railways. But no longer Railways can claim to be the whole and sole authority to transportation management of the country. Road transport has turned into a stern competitor of the Railways, so to satisfy the customer Railways has to prove that it can provide better and reliable service to the customers over other means of transportation.

FIRE FIGHTING

INTRODUCTION

Fire causes considerable amount of damage to the railway and its property every year, so every railway servant should be thoroughly conversant with the process of preventing and extinguishing fire on railway property.

At the end of this lesson the trainee should be able to adopt preventing measure and be able to extinguish the fire on railway property, to save passenger's life and take necessary action there after.

Let us deal with some common questions about fire hazard and its prevention method :

1. What is fire?

In simple words we can say that the fire is the combination of combustible material, oxygen and heat at proportionate rate.

2. How can you classify the fire?

According to the types of fuel and combustible, fire may be classified in following 4 categories -

- i) **Class 'A' fire (Solid)** –These are fires involving combustible materials of organic nature, such as wood, paper, rubber and plastics etc where the cooling effect of water is essential. Class 'A' fire are the most common and the most effective agent is generally water.
- ii) **Class 'B' fire (Liquid)** –These are fires involving liquids or liquefiable solids . Some flammable liquids are miscible with water and others are immiscible with water. Taking into consideration to the fact extinguishing agent like water spray, foam, water, vaporising liquids, carbon dioxide and dry chemical powder may be used.
- iii) **Class 'C' fire (Gas)** – These are fires involving gases or liquefied gases. Foam or dry chemical powder can be used to control these type of fire.
- iv) **Class 'D' fire (Metal)** -These are fires involving metals. Extinguishing agent containing water are ineffective, and even dangerous. Carbon dioxide and carbonate classes of dry chemical powders may also be hazardous if applied to most metal fires. Normally powdered graphite, powdered talc, soda ash, limestone and dry sand are suitable for metal fires.

3. What are the factors are essential to continue combustion?

The following factors are essential to continue the combustion -

- i) The presence of fuel or combustible material.
- ii) The presence of oxygen, usually as air, and-
- iii) The attainment and maintenance of a certain minimum temperature.

4. How can you extinguish the fire?

Fire can be extinguished by the limitation of one or more factors mentioned above in item no.-3, hence methods of extinguishing fire may be described as under-

- i) Physically separating the combustible material from the flame, called Starvation.
- ii) Removing or diluting the oxygen supply, known as Smothering.
- iii) Reducing the temperature of the combustible, called Cooling.

5. What are the precautions to be observed by the staff to prevent from the fire hazardous?

i) Glowing cigarette butts and matches shall not be thrown into waste baskets, oil rag bins, or other places of fire hazard. Burning Cigarettes, Biris etc or Match sticks should not be thrown out without ensuring that they have been completely extinguished.

ii) Smoking and use of open flames shall be prohibited near oil storage , battery and places where combustible material is kept.

- iii) High standard of cleanliness shall be maintained. Waste material, oily waste or rags etc. shall be removed from the premises daily and suitably disposed of.
- iv) After wedging of any contactor ensure proper tightness of the contacts, so as to avoid chattering between them which may cause sparks and finally ignite other combustible material in the area.
- v) Keep a close watch for loose cables, oil splashing, hot axle on loco and wagons and any other abnormally on or near the track or OHE while on run.
- vi) Rules for marshalling and shunting of inflammables, explosive and other dangerous goods should be observed strictly. Take extra precaution while shunting with such types of wagons.
- vii) In case of dropping of any safety relay check carefully the equipments provided in concerned circuit for any smoke emission, burning smell of fire.
- viii) Electrical equipment shall be installed, operated and maintained properly and in such a manner as to eliminate arcs due to poor contacts in switches and fittings, damaged insulation, crossed wires, opening of switches carrying large currents, etc.
- ix) Overloading of electric circuit should be avoided and electrical installations should be checked periodically.
- x) Electric lamps shall not be surrounded by or laid on combustible material.
- xi) Flammable gases and materials shall not be stored near electrical equipment.
 - **Note – Loco Pilot, Motormen, Asst. Loco Pilot, should be familiar with the physical location of all fire fighting equipment including fire hydrants in their place of work, the correct method of operating the equipment and precautions to be observed, as well as location of and means of summoning of Railway and Municipal Fire Brigades.**

6. What is electrical fire ?

Although it may be a fire of any class as mentioned above, any fire involving or started by electrical equipment may be called electrical fire. The normal procedure in such circumstances is to cut off the electricity and use extinguishing media appropriate to what is burning. Only when this cannot be done, special extinguishing agents be required which are non conductor of electricity and non damaging to equipment. These include vaporising liquids, dry powder and carbon dioxide, although the latter's cooling and condensation may affect sensitive electronic equipment.

7. What are the most common causes of fires in electrical installations?

The most common causes of fire in electrical installations are –

- i) Short circuits, due to failure of installations.
- ii) Overheating of cables or equipment due to overloading, lack of ventilation or presence of local resistance
- iii) Ignition of flammable gases and vapours by heat or sparks occurring in the operation of electrical equipment.
- iv) Ignition of flammable substances by electro-static discharge.

8. How many types of Fire Extinguishers are there and which type of fire extinguisher commonly used in locomotive?

The types of fire extinguishers are mentioned below against each class of fire are generally most suited. In electric locomotive Dry Chemical Powder Type fire extinguisher is used commonly.

- i) Class A Fires - Water expelling type extinguishers.

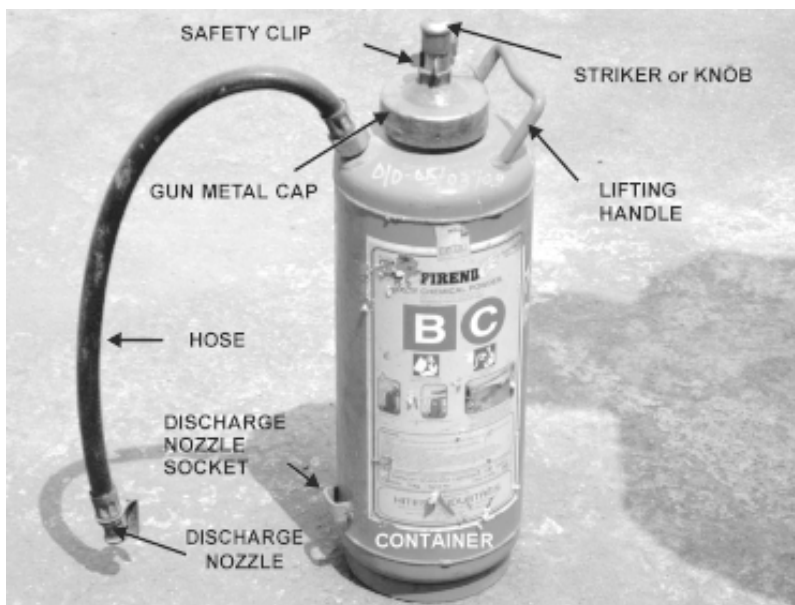
- ii) Class B Fires - Foam, dry powder, vaporizing liquid, carbon dioxide extinguishers.
- iii) Class C Fires - Dry powder and carbon dioxide extinguishers.
- v) Class D Fire - Extinguishers designed for expelling special dry chemical powder.

9. How DCPT fire extinguisher works?

The powder generally used in DCPT fire extinguisher is sodium bicarbonate with other additive to provide free flowing characteristics. Sodium bicarbonate dissociate at a temperature little higher than the boiling point of water to yield CO₂ and water vapour. It extinguishes the fire due to- Heat absorption by solid particles, heat absorbed by decomposition and emission of CO₂ and water vapour tends to smother the fire.

10. How will you use Dry Chemical Powder Type fire extinguisher in case of fire?

- i) Front view of DCPT fire extinguisher.



- ii) Break the seal and open the safety clip in the top portion of fire extinguisher.



- iii) Hold the discharge nozzle of hose pipe tightly towards the fire to be extinguished at a safe distance by one hand.



- iv) Press the striker with sufficient power to bring the extinguisher in service.



- v) Direct the jet towards the near edge of the fire. Drive the fire towards the far edge until the flames are extinguished.



CAUTION

- a) **Water not to be used in extinguisher under any circumstances.**
- b) **The person using fire extinguisher to stand in the direction of flame keeping a safe distance , so that the flame can not harm the person.**

11. What are the safety checks to be done in fire extinguisher before using it?

- i). Validity of the fire extinguisher to be checked, which is 3 months from the date of filling.
- ii). Make sure that the extinguisher is in proper condition and is not accidentally discharged. Safety clip in gun metal cap is to be in sealed and locked condition.
- iii). Condition of hose pipe to be good and nozzle outlet vent holes should not be choked.
- iv). The container to be checked for crack-ness or any other type of damage.

12. What are the duties of Running Staff in case of fire?

FIRE ON GOODS TRAIN –

- (i) Stop the train immediately and put on flasher light.
- (ii) Isolate burning vehicle at a safety distance of 45 meters on either side.
- (i) Save the wagon label card and seal card and also the contents of the vehicle.
- (iv) Extinguish the fire by any means, which is available near by.
- (v) Protect the train as per G.R.6.03.
- (iii) Draw the burning vehicle nearest to the water tank if available.
- (iv) If the fire is not controlled advice TPC to switch 'OFF' the OHE power supply and ask for fire brigade if required.

FIRE ON PASSENGER TRAIN –

- (i) Stop the train immediately and put 'ON' the flasher light.
- (ii) De-train the passengers first and attend to their safety first.
- (iii) If there is postal van on the train, every effort shall be made to save the mails.
- (iv) Disconnect the electrical couplers and remove fuse links of the batteries if possible.
- (v) Isolate the burning vehicle at a distance of 45 meters on either side
- (vi) Protect the train as per G.R.6.03
- (vii) Extinguish the fire by the fire extinguishers available in train or by using other means like sand etc.
- (viii) If the fire is out of control ask for fire brigade and inform to TPC/SCR over portable telephone post. If there is a chance of OHE being affects with fire contact TPC and advice him to suppress the tension of OHE.

FIRE ON OHE

Inform the TPC through the nearest emergency telephone tap and also nearest stationmaster about the nature of fire and the amount of damage caused by giving km. Number and between the stations.

FIRE 'ON' ELECTRIC LOCO –

- (i) Open DJ, lower pantograph, open HBA (Battery switch) and stop the train/loco.
- (ii) Try to extinguish the fire with fire extinguisher provided on the loco.
- (iii) If the fire is not controlled inform TPC to switch 'OFF' the power supply of OHE and ask for fire brigade through emergency telephone.

- (iv) Protect the train/Loco as per G.R.6.03 if it is in block section.
- (v) Inform to TLC about the fact.

Availability of fire extinguisher on train in different locations:

- DCPT fire extinguisher to be provided in the train in scale given below-
- i) Locomotive - 4 Nos. (2 in each cab)
 - ii) Each brakevan (Front and rear) – 2 Nos. each.
 - iii) In AC coaches – 2 Nos.
 - iv) Each Pantry Car – 4 Nos.
 - v) Each generator Car- 4 Nos.
 - vi) Each Motorman Cabin of EMU coaches & MEMU coaches- 4 Nos.

Note- Fire extinguisher of 5 kg capacity to be kept 4 numbers in a loco, two in each cab, with appropriate place and manners so that its hose pipe remain in unfolded condition. Its expiry date to be mentioned in log book and check the seal of safety clip.

Performance requirement of DCPT fire extinguisher of 5 KG:

Minimum period for throw of jet	- 15 seconds
Maximum period to discharge 90% of contents	- 20 seconds
Range of throw of jet	- 5 to 7.5 m
Requirement	- The whole of the powder to fall beyond 3.0 meter from the nozzle.

ROLE OF AN ASSISTANT LOCO PILOT

The role or function of an Assistant Loco Pilot is to assist the Loco Pilot to take the train safely from originating station to destination.

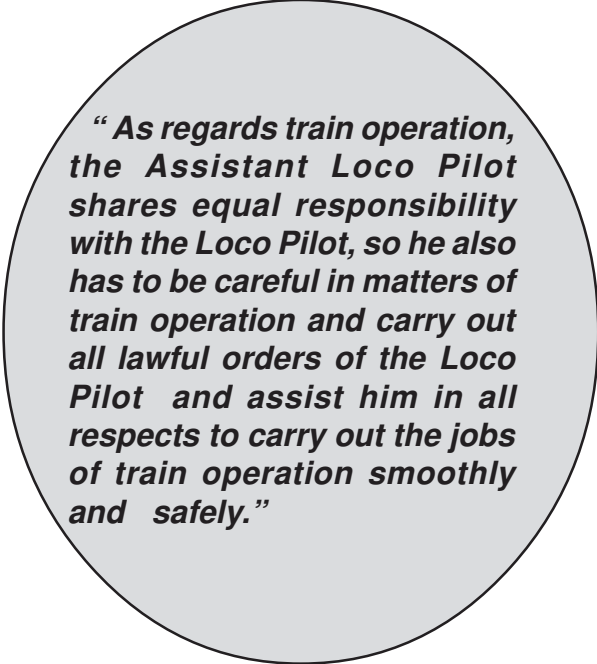
He may ever have some greater role to play in the form of driving the engine in case Loco Pilot became incapable by stopping the engine or in case of failure.

He forms the bridges of communication between Loco Pilot and guard, thus he plays a very important role in the operation of train smoothly and safely.

CONDUCT RULES(GR-2.10)

A railway servant shall –

- (i) Wear a badge and uniform if prescribed and be neat and tidy in appearance while on duty.
- (ii) Be prompt, civil and courteous.
- (iii) Not solicit or accept illegal gratification.
- (iv) Keep all reasonable assistance and be careful to give correct information to the public.
- (v) When asked, give his name and designation without any hesitation



“As regards train operation, the Assistant Loco Pilot shares equal responsibility with the Loco Pilot, so he also has to be careful in matters of train operation and carry out all lawful orders of the Loco Pilot and assist him in all respects to carry out the jobs of train operation smoothly and safely.”

RESPONSIBILITY OF AN ASSISTANT LOCO PILOT

As regards train operation, the Assistant Loco Pilot shares equal responsibility with the Loco Pilot, so he also has to be careful in matters of train operation and carry out all lawful orders of the Loco Pilot and assist him in all respects to carry out the jobs of train operation smoothly and safely; because anything going wrong the Assistant Loco Pilot will not be spared, he will be held responsible equally with the Loco Pilot.

At times of necessities and abnormal situation the Assistant Loco Pilot may have to play a role of greater importance and form the bridge of communication between the Loco Pilot and others.

RULE BOOKS AND AIDS

For the knowledge of rules and regulations, there are general rules (G.R.) and subsidiary Rules (S.R.). Except this, there are Operating Manual Accident Manual, Block-working Manual etc. for reference of rules and working system, circulars and procedure order time to time.

Special instructions and guidelines are also given through “Order Book” and by Loco Inspectors (L.I.) or Safety Instructors.

SAFETY AND TRANSPORTATION RULES

ASSISTANT LOCO PILOT'S DUTIES AT BOOKING POINT

SIGNING-ON DUTIES (GR-2.07)

Every Railway servant shall be in attendance for duty at such time and places and for such periods as may be fixed in this behalf by the Railway administration and shall also attend at any other time and place at which his service may be required.

Crew Management system

The conventional method of 'Signing-on' and 'Signing-off' in Departure and Arrival register is now being replaced by signing on and signing off in computerised system named "Crew Management System." For signing On/Off in the system following steps to be taken:

Step -1

Language selection – Crew will select the language Hindi or English.



Step-2

Crew Identification – Crew is identified by entering crew ID and password on the screen.



Step -3

Crew/Inspector Selection – Based on the ID the system recognises the crew is for a LI or Crew (Loco Pilot/Guard).

Step -4

Crew activity options – After the crew identification activity a screen with the following option will appear-

- Sign On/Sign Off
- Quick
- Crew Self Service
- Circulars
- Change password.



Step -5

Information related to the Crew and Train to be confirmed – The brief information of the crew(bio particulars) along with train information will be displayed in brief by the CMS system. Crew will see it and if the information regarding the crew particulars is not correct then the crew may try to sign on again using crew ID and password. If still it is not allowed then the crew has to approach the supervisor for further corrective action.

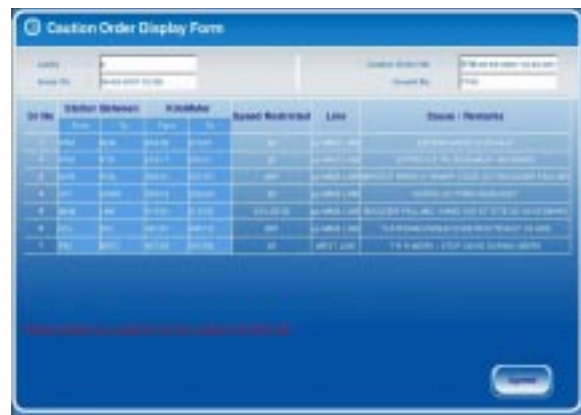


Step -6

Sign-ON – While signing ON the crew shall be offered

the following interactive options which have to be entered through check boxes.

- In case the crew is using spectacles then confirmation whether the crew is carrying two sets of spectacles.
- If crew has not read the latest circulars then he is flashed message that “you have not read the latest available circulars please read them.”
- Confirmation that the bio particulars displayed by the CMS system are correct.



Step -7

Caution order – The crew is offered the caution order of the section on which the crew has been booked. The crew has to essentially see the caution order before proceeding further.



Step -8

Sign-ON Completion – Crew will click 'Sign On' to complete the sign-on activity. He will be presented with the success message and will be greeted for the journey.

Approval by supervisor

Supervisor will approve/disapprove that the crew has met all the mandatory requirements including the BA rest. In no case the crew shall be able to sign OFF at the destination lobby unless the supervisor at the base station has given the Sign On approval.

Crew Sign Off

Crew has to Sign Off after completing his running duty. Sign Off activity is carried out at the lobbies in the presence of the Supervisors TCC/ATFR/ALS. They shall accord Sign Off approvals for completing the journey.

Sign Off screen

The crew once Signed On shall always be offered the screen to Sign Off. Unless he Signs Off and is approved by the supervisor he shall not become available for next crew booking. Based on whether the Sign Off is at HQ or outstation the different rest particulars are asked from the crew. Crew shall select the rest option and Sign Off.



CHECKS OF VARIOUS REGISTERS

After turning up on duty the Assistant Loco Pilot shall before taking over charge of the loco/train shall go through the different registers like Appearance Register, Spectacle Register (if pass with glass), General Order Book, Safety & Technical Circular Book, Learning Road Register, etc. and sign them.

DUTIES WHILE TAKING OVER CHARGE

Before taking over charge the Assistant Loco Pilot should check his Loco Pilot's name, train number, loco number, whether the loco is to be taken out from shed or it is yard pulling.

After this he must ensure that the loco, which is to be taken over charge, is standing under catenary and all wheels are on the rails. Check the intactness of all visible mechanical parts, sand boxes to be filled up with sand, all the drain cocks are closed and cut out cocks open, check oil levels of gear cases, suspension bearings, exhaustor, compressors, transformer and GR. Check the intactness of spare fuses, position of earthing switches, rotating switches, relay cover etc. and assist his driver in preparation of the locomotive. He should also go through the locomotive logbook for attention of previous bookings, check the battery voltage.

He should drain out moisture from the main reservoir, clean the loco driving cab and look out glasses. He should check the availability of the Safety items like portable telephone, fire extinguisher; flare fusee, detonators, spare head light bulb etc.

“The loco logbook provides the Loco Pilot with guidelines, which helps him in a great deal in taking corrective measures. Only after going through the loco log book the Loco Pilot can get idea about the status of the locomotive and may take action likewise.”

READING THE LOCO LOG BOOK

The loco logbook provides the Loco Pilot with guidelines, which helps him in a great deal in taking corrective measures. Only after going through the loco log book the Loco Pilot can get idea about the status of the locomotive and may take action likewise.

So on taking over charge, before energising the loco, the Loco Pilot must go through the logbook booking first.

CHECKING ON ELECTRIC LOCO WHILE STANDING

CHECKS TO BE DONE BEFORE TURNING OUT FROM SHED

While to take out a locomotive from shed, the Assistant Loco Pilot shall turn up on duty at specified time, sign the signing on and other registers, order books and circulars, read the loco logbook, and go through the breathalyzer test .

Then he shall go through the log book, checking of braking system, mechanical equipments, panel board, switch position etc. and help the driver in getting the locomotive prepared for handing over to the traffic for train service.

PROCEDURE FOR TAKING OUT A LOCO FROM SHED

After the complete preparation of the locomotive the loco is required to be handed over to the traffic for train services by giving ready report of the locomotive. Ready report is required to be given by exchanging private number. After the ready report is given, the Loco Pilot has to wait for the signal (where provided) to be given or a 'Traffic Pilot Man' to come. When the signal is taken 'OFF' or the Pilot Man turns up on the engine before starting, the Loco Pilot must ensure that everything is ready for the engine to start, all points have been correctly set and locked and correct departure signal/ shunt signal is taken 'OFF' or the traffic Pilot Man show proceed hand signals. On getting the signal to start the Loco Pilot shall start his engine and proceed cautiously for making EOT (Engine on train).

CHECKS ON LOCOMOTIVE AT YARD/POOLING POINT

The Assistant Loco Pilot shall check the following at yard/pooling point: .-
The loco is under the OHE and all wheels are on rail. Temperature of axle boxes to be normal. Intactness of bogie frame, equalizing beam, suspension spring suspension bearing, helical spring etc.

Intactness of speedometer transmitter, sand brackets with nozzles, working of sanders, intactness of both sides CBCs, TSCs, locking pins, position of B.C. isolating cocks, working of blowers, oil levels of equipments, working of sanders, head light, marker light, flasher light etc.

In addition to the above he shall check the working of cab horns and wipers and intactness of spare fuse, detonators, fire extinguishers, wooden wedges, flare fusee and portable telephone etc.

CHECKING ON ELECTRIC LOCOS WHILE HANDLING

DUTIES WHILE ON RUN

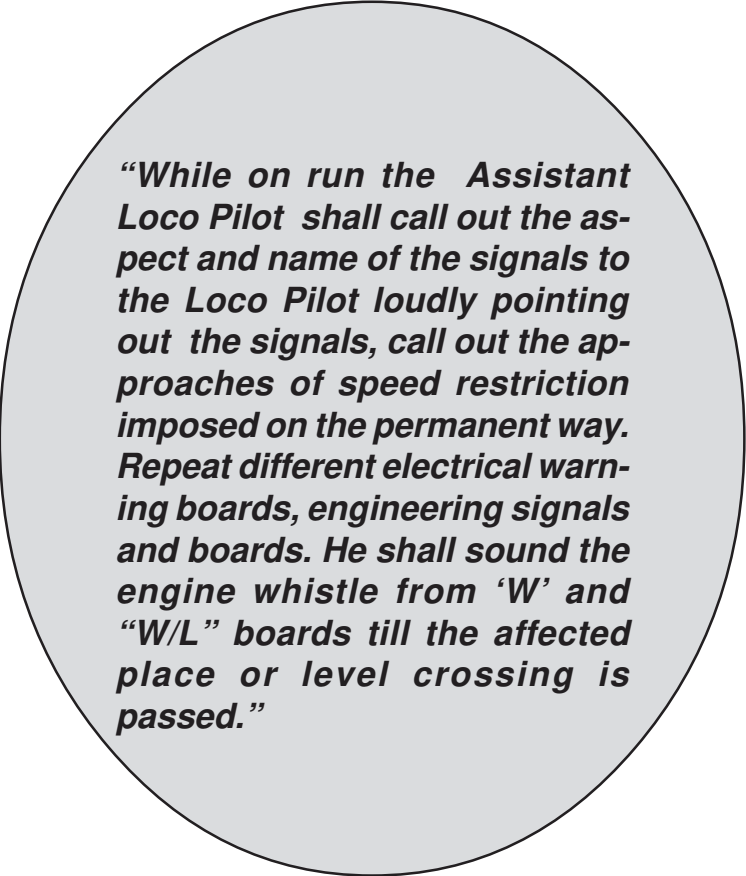
While on run the Assistant Loco Pilot shall call out the aspect and name of the signals to the Loco Pilot loudly pointing out the signals, call out the approaches of speed restriction imposed on the permanent way. Repeat different electrical warning boards, engineering signals and boards. He shall sound the engine whistle from 'W' and "W/L" boards till the affected place or level crossing is passed.

The Assistant Loco Pilot shall note down the intersection timings in the Loco Pilot's note

book, look forward on the line he is running and also on the adjacent line and OHE for any obstruction or abnormality. He shall look back frequently in curvature to check for any abnormality on train. While passing a manned level crossing he must be vigilant and notice whether any danger signal being shown until the full train crosses the gate. He shall exchange signal with station master, cabin man when they are in front of each other and he shall exchange signal with the guard at the time of starting from a station, while starting after stopping outside the station limit and also when the last vehicle is passing over the outer most trailing point or quitting the yard.

COUPLING OF LOCO, SCREW COUPLING AND CBC

Electric locomotives are provided with two kinds of coupling one in the same assemblies. CBC i.e. centre buffer coupling is provided for coupling with CBC trailing stock and TSC i.e. Transition Screw coupling for TSC trailing stock. While taking over charge of the loco, it is to be ensured that CBC and TSC with all its accessories are intact, specially the locking device and locking key of the CBC.



“While on run the Assistant Loco Pilot shall call out the aspect and name of the signals to the Loco Pilot loudly pointing out the signals, call out the approaches of speed restriction imposed on the permanent way. Repeat different electrical warning boards, engineering signals and boards. He shall sound the engine whistle from ‘W’ and “W/L” boards till the affected place or level crossing is passed.”

DUTIES ON HANDING OVER CHARGE

HANDING OVER CHARGE ON LINE

When handing over charge to the out going Loco Pilot, the in coming Loco Pilot shall apprise the out going Loco Pilot about the condition of brake power and nature and behavior of the train, amount of vacuum/BP pressure on the engine gauge, nature of any defect with the engine or the train, book loco repairs if any in the loco logbook. He shall also hand over the train BPC and caution order (if required) to the out going Loco Pilot.

STABLING A LOCO IN YARD

When the loco is required to be stabled in yard, the Loco Pilot shall take the following actions:

He shall stop the loco by clearing the fouling mark applying the loco brake, then open DJ. Lower panto and put HBA on “0” (OFF). He shall then book the loco repair in the loco logbook, close all the doors, shut the windows, keep the loco keys and loco logbook on the Loco Pilot’s desk, apply loco hand brakes and then lock the door. He shall then place skid/wooden wedge or sprag on the wheel to prevent rolling down.

TRANSPORTATION TRAINING

INTRODUCTION TO SAFE WORKING RULES

Safe working rules means the rules, which are required to be observed for working safely. Railway system is a transportation network and a very large transportation network, so unless some rules are formulated, the system can not function safely. The rules that have been formulated for the safe operation and management of Railway transportation network are called Railways safe working rules.

These rules have been contained in a book called the "INDIAN RAILWAYS (OPEN LINE) GENERAL (AMMENDMENT) RULES-2002". These rules are of two kinds- General rules or GR and Subsidiary Rules or SR.

BASIC DEFINITION AND CONCEPT

Adequate Distance - GR 1.02

Means the distance sufficient to ensure safety

Authority to Proceed - GR 1.02(6)

Means the authority given to the Loco Pilot of a train, under the system of working, to enter the block section with his train.

Block Station - GR 1.03(2)

Block station are those at which the Loco Pilot must obtain an authority to proceed under the system of working to enter the block section with his train.

Block Section -GR 1.02(10)

Means that portion of running line between two block stations on to which no running train enter until line clear has been received from the block station at the other end of the block section.

Competent Railway Servant - GR 1.02(14)

Means a railway servant duly qualified to undertake and perform the duties entrusted to him.

Centralised Traffic Controls -GR 1.02(11)

Means a system by which the working of trains over a route, to which the system applies, is governed by fixed signals remotely controlled from a designated place.

Controller- GR 1.02(16)

Means a railway servant on duty who may for the time being be responsible for regulating the working of traffic on a section of a railway provided with the system of speech communication.

Day -GR 1.02(17)

Means from Sunrise to sunset

Driver -GR 1.02(19)

Means the engine driver or any competent railway servant who may for the time being in charge of driving a train

Direction of traffic: -GR 1.02(18)

Means on a double line, the direction for which the line is signalled. On a single line, the direction for the time being established under the System of working to allow trains to move in that direction.

Facing and Trailing Point -GR 1.02(21)

Points are facing or trailing in accordance with the direction a train or vehicles moves over them. Points are said to be facing points when by their operation a train approaching them can be directly diverted from the line upon which it is running.

Fixed Signal -GR 1.02(22)

Means a signal of fixed location indicating a condition affecting the movement of a train and includes a semaphore arm or disc or fixed light for use by day and for use by night.

Fouling Mark-GR 1.02(23)

Means the mark at which the infringement of fixed standard dimensions occurs, where two or more lines cross or join one another.

Gateman -GR 1.02(26)

Means a competent railway servant posted at a level crossing for working the gate.

Gangman -GR 1.02(24)

Means a railway servant employed on permanent way or work connected there with.

Guard -GR 1.02(28)

Means the railway servant in charge of a train and includes a Brakesman or any other railway servant who may for the time being be performing the duties of a Guard.

Goods Train - GR 1.02(27)

Means a train (other than a material train) intended solely or mainly for the carriage of animals or goods.

Intermediate Block Post - GR 1.02(31)

Means a class "C" station on a double line remotely controlled from the block station in rear.

Intermediate Block Signaling:- GR 1.02(32)

Means an arrangement of signaling on a double line in which a long block section is split into two portions each constituting a separate block section by providing an Intermediate Block post.

Last Stop Signal-GR 1.02(34)

Means the fixed stop signal of a station controlling the entry of trains into the next block section.

Line Clear- GR 1.02(37)

Means the permission given from a block station to a block station in rear for a train to leave the later and approach the former or the permission obtained by a block station from a block station in advance for a train to leave the former and proceed towards the later.

Material Train -GR 1.02(39)

Means a departmental train intended solely or mainly for carriage of railway materials when picked up or put down or for execution of works either between stations or within station limits.

Mixed Train -GR 1.02(40)

Means a train intended for carriage of passenger and goods or of passengers, animals and goods.

Multi Aspect Signaling -GR 1.02(41)

Means a signaling arrangement in which signal display at any one time, any one of the three or more aspects and in which the aspect of every signal is pre-warned by the aspect of previous signal or signals.

Night -GR 1.02(42)

Means from Sunset to sunrise

Obstruction -GR 1.02(43)

It cognate expressions included a train, vehicle or obstacle on or fouling of line, or any condition, which is dangerous to trains.

Passenger Train - GR 1.02(45)

Means a train intended solely or mainly for the carriage of passengers and other coaching traffic and includes a troop train.

Point And Trap Indicators -GR 1.02

These are not signals but are appliances fitted to and working with points to indicate by day or by night the position in which the points are set.

Running Train -GR 1.02(48)

Means a train that has started under an authority to proceed and has not completed its journey.

Running Line -GR 1.02(47)

Means the line governed by one or more signals and include connections, if any used by a train when entering or leaving a station or when passing through a station or between stations.

Station -GR 1.02(51)

Means any place on a line of railway at which traffic is dealt with or at which an authority to proceed is given under the system of working.

Shunting -GR 1.02(49)

Means the movement of a vehicle or vehicles with or without an engine or of any engine or any other self-propelled vehicle for the purpose of attaching, detaching or transfer or for any other purpose.

System of Working -GR 1.02(56)

Means the system adopted for the time being for working of trains of any portion of a railway.

Special Instruction -GR 1.02(50)

Means instructions issued from time to time by the authorised officer in respect to particular case or special circumstances.

Train Examiner -GR 1.02(59)

Means a railway servant duly qualified to examine trains and certify their fitness for safe running and includes any other railway servant who may for the time being performing the duties of a train examiner.

Two Aspect signaling -GR 1.02(60)

Means a signaling arrangement in which each signal display at any one time either of the two aspects.

Difference between GR & SR

	GR	SR
1	Formed by railway board on behalf of Govt. of India.	Formed by authorised officer of individual zone on behalf of G.M.
2	Applied to all I.R.	Applied to that railway who forms them.
3	Revised by railway board only.	Revised by authorised officer.
4	The object of this rule is to regulate working and management of I.R.	The object of this rules is to supplement , clarify and simplify the G.R.
5	Being applicable to all railways they lay down basic principle of safety and do not deals with details.	Being applicable to individual railway they are framed to suit the local condition of particular railway as such to ensure greater safety.
6	Serially numbered and printed in bigger latter.	Printed below the G.R. and are in smaller letters.

DIFFERENCE BETWEEN “TRANSPORTATION AND OPERATION”

The word transport has its roots in the Latin word “Transportare” which actually is a combination of two words and means to carry across or from one place to another. So, in other words transportation means to carry men, live stock and material from one place to another to help achieve optimal utilisation of country’s resources.

‘Transport’ can be regarded as consisting of 3 elements-1) Track over which movement can take place 2) Means of loco, motion and carriage and; 3) Arrangements to be made for beginning and ending of journey, collection and distribution, loading and unloading of Tariffs be it freight or passenger in a safe or economic way.

But the operation means the process or method of working to achieve the above.

OVERVIEW OF INDIAN RAILWAY ACT’ 1890

Under section 101(b) of the Indian Railways Act (ix of 1890) all Railway servants on the Railway are bound by the General (Amendment) Rules as well as the Subsidiary Rules, Manuals and special instruction. They must make themselves fully acquainted with all these rules and instruction and any revisions made from time to time.

SIGNALS

INTRODUCTION

The first train of the world did not have the benefit of signal. Two policeman on horses were sent ahead to clear the way, later on police man's arms was used as signal and the semaphore arm signal is obtained its shape from the function of arm of the traffic signal man

OBJECT OF SIGNALING

- i) To prevent a running train to come into contact with another train or obstruction and thus to provide safety to passenger, the staff and the rolling stock.
- ii) To facilitate the flow of traffic and to increase the carrying capacity of the track.

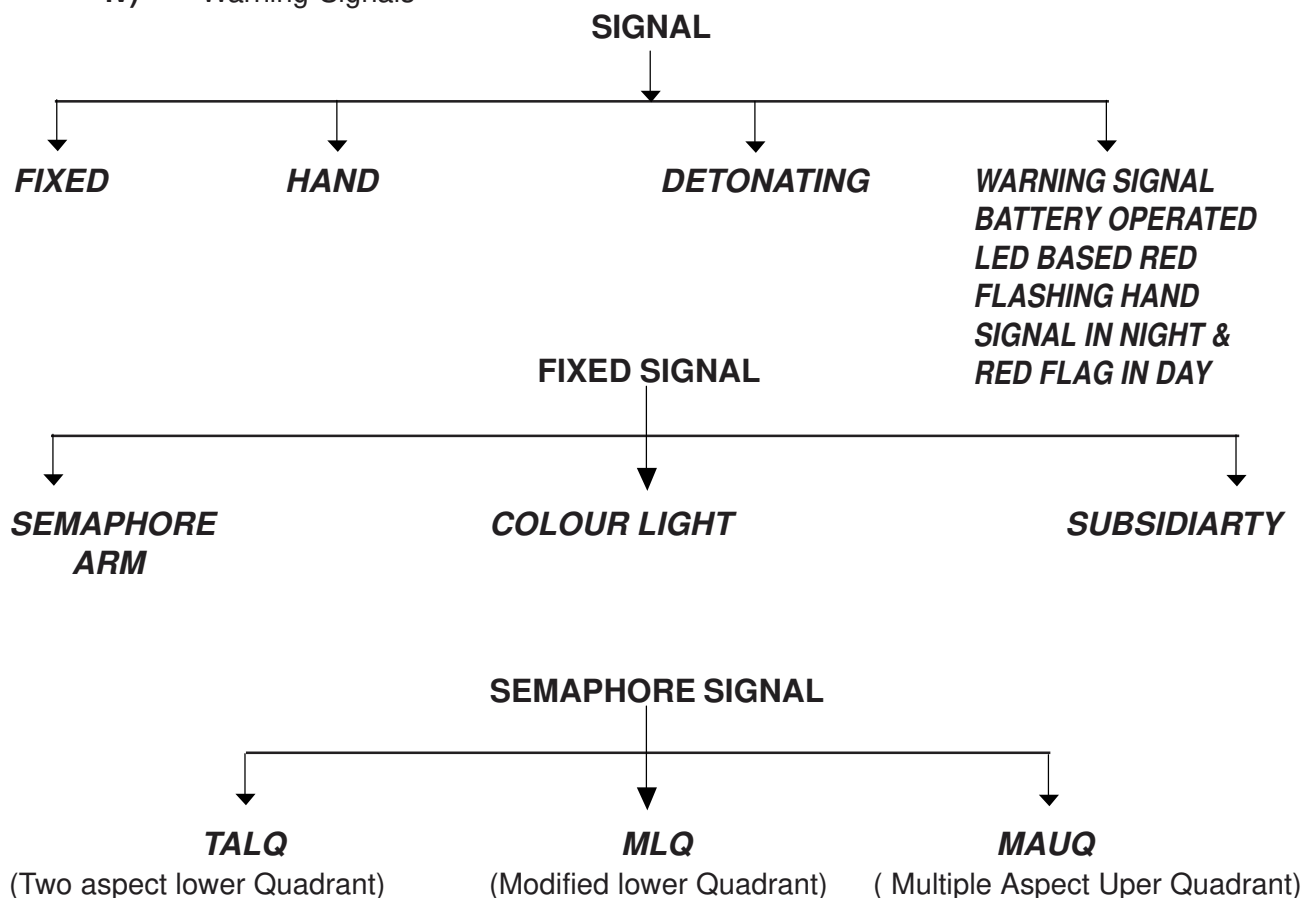
GENERAL USE OF SIGNAL (GR 3.01)

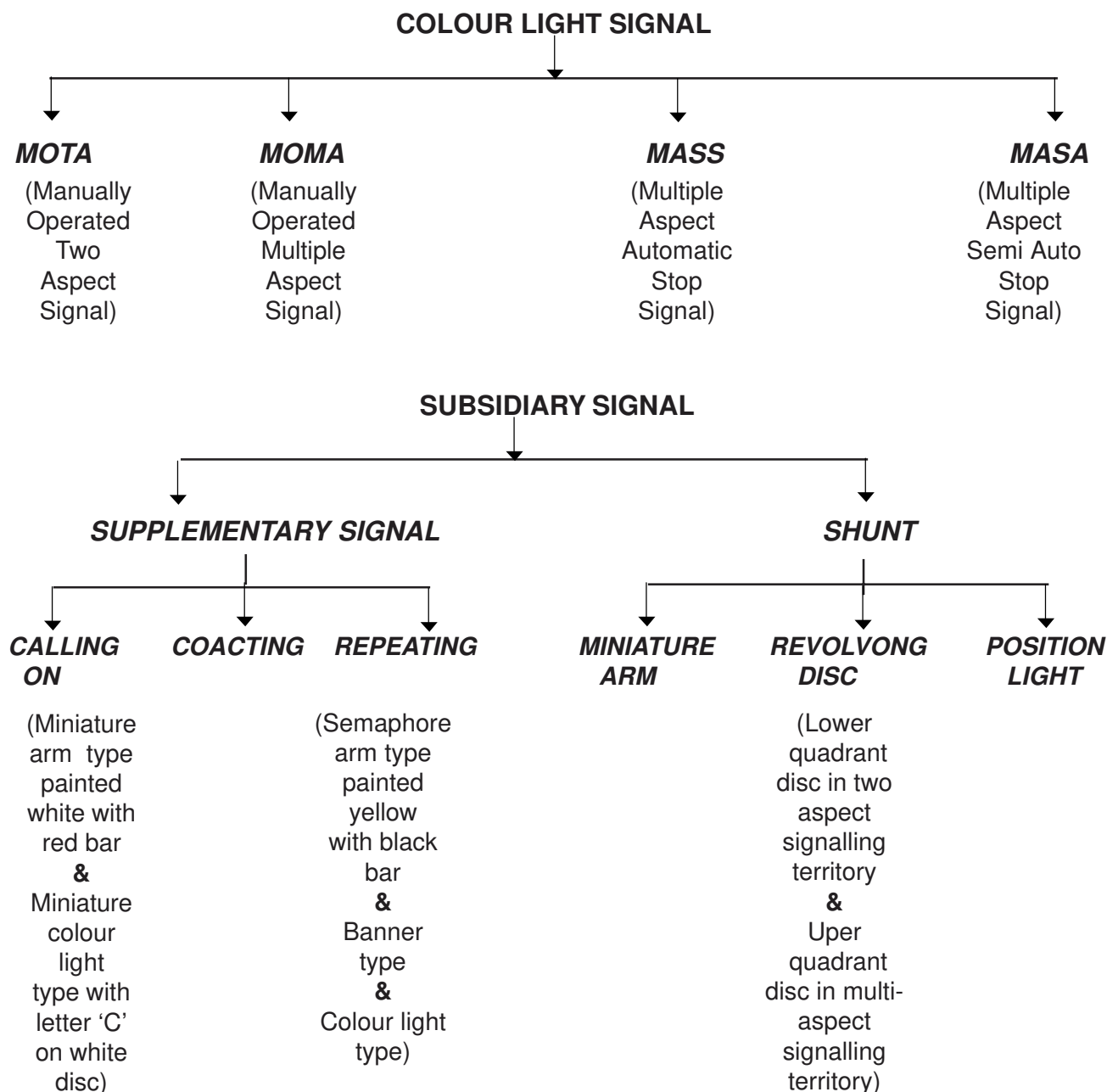
The signals prescribed in these rules shall be used for controlling the movement of trains in all cases in which exceptions are not allowed by approved special instruction.

KINDS OF SIGNALS (GR 3.02)

The signals to be used for controlling the movement of trains shall be

- i) Fixed Signals
- ii) Hand Signals
- iii) Detonating Signals
- iv) Warning Signals





ASPECTS AND INDICATION OF SIGNALS

The "OFF" position of a semaphore signal shall be displayed by day by the inclined position of arms from 45° or 60° below the horizontal position in case of two aspect lower quadrant (TALQ) signal and 45° to 90° above the horizontal position in case of multi aspect upper quadrant (MAUQ) signal. The aspect of colour light signals shall be displayed by the light or lights by the day and by the night.

ASPECT AND INDICATION OF COLOUR LIGHT SIGNALS

LIGHT	ASPECT	INDICATION
Red	STOP	Stop dead
Single Yellow	CAUTION	Be prepared to stop at next stop signal

Double Yellow	ATTENTION	Proceed and be prepared to pass next signal at restricted speed
Green	PROCEED	Proceed with maximum permissible speed.

WARNER SIGNAL (GR. 3.06)

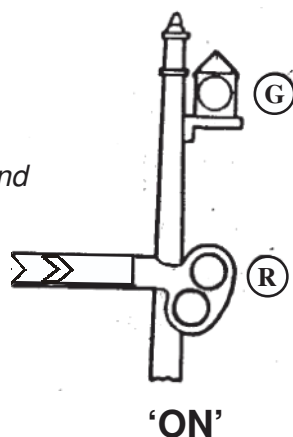
- i) A semaphore Warner signal has a fish tailed arm painted red with white bar
- ii) A Warner signal is intended to warn a Loco Pilot
 - a) About the condition of block section ahead
 - b) He is approaching a stop signal
- iii) When a Warner signal placed below a stop signal, it cannot be taken OFF while the stop signal above it is in ON position.
- iv) A Warner signal on a post by itself shall be located at an adequate distance in the rear of stop signal, the aspect of which it pre-warned provided that when such a Warner signal applies to a gate stop signal, it shall not be display "proceed" aspect, unless there is adequate distance between gate stop signal and the first stop signal of the station ahead and this adequate distance shall not be less than 1200 meter

ASPECT

Proceed with caution

INDICATION

Proceed with caution and be prepared to stop at the next stop signal.



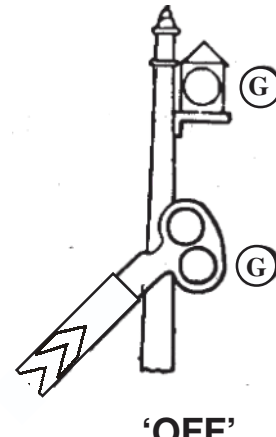
'ON'

ASPECT

Proceed

INDICATION

Proceed



'OFF'

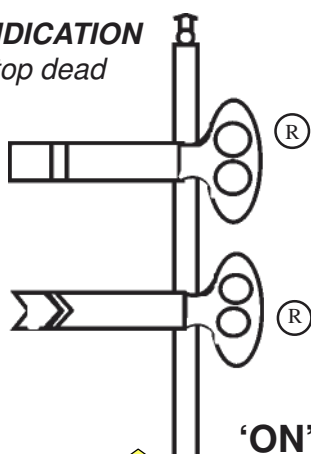
Semaphore Warner signal in TAST - on a post by itself

ASPECT

Stop

INDICATION

Stop dead



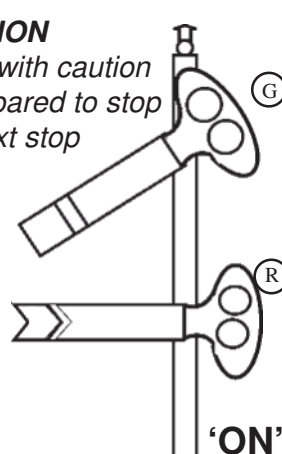
'ON'

ASPECT

Proceed with caution

INDICATION

Proceed with caution & be prepared to stop at the next stop signal



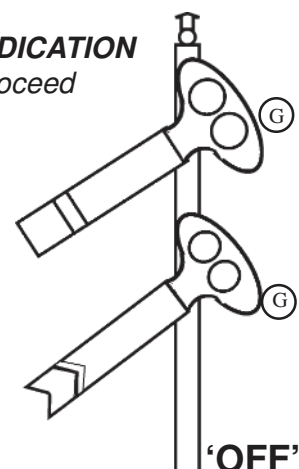
'ON'

ASPECT

Proceed

INDICATION

Proceed

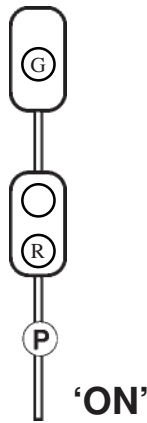


'OFF'

Semaphore Warner signal in TAST - below a stop signal

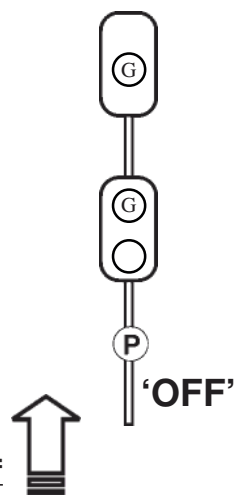
ASPECT
Proceed with caution

INDICATION
Proceed with caution and
be prepared to stop
at the next stop
signal.



ASPECT
Proceed

INDICATION
Proceed



Colour light Warner signal in TAST - on a post by itself

ASPECT
Stop

INDICATION
Stop dead



'ON'

ASPECT
Proceed with caution

INDICATION
Proceed with caution
& be prepared to stop
at the next stop
signal



ASPECT
Proceed

INDICATION
Proceed



'OFF'



Colour light Warner signal in TAST - below a stop signal

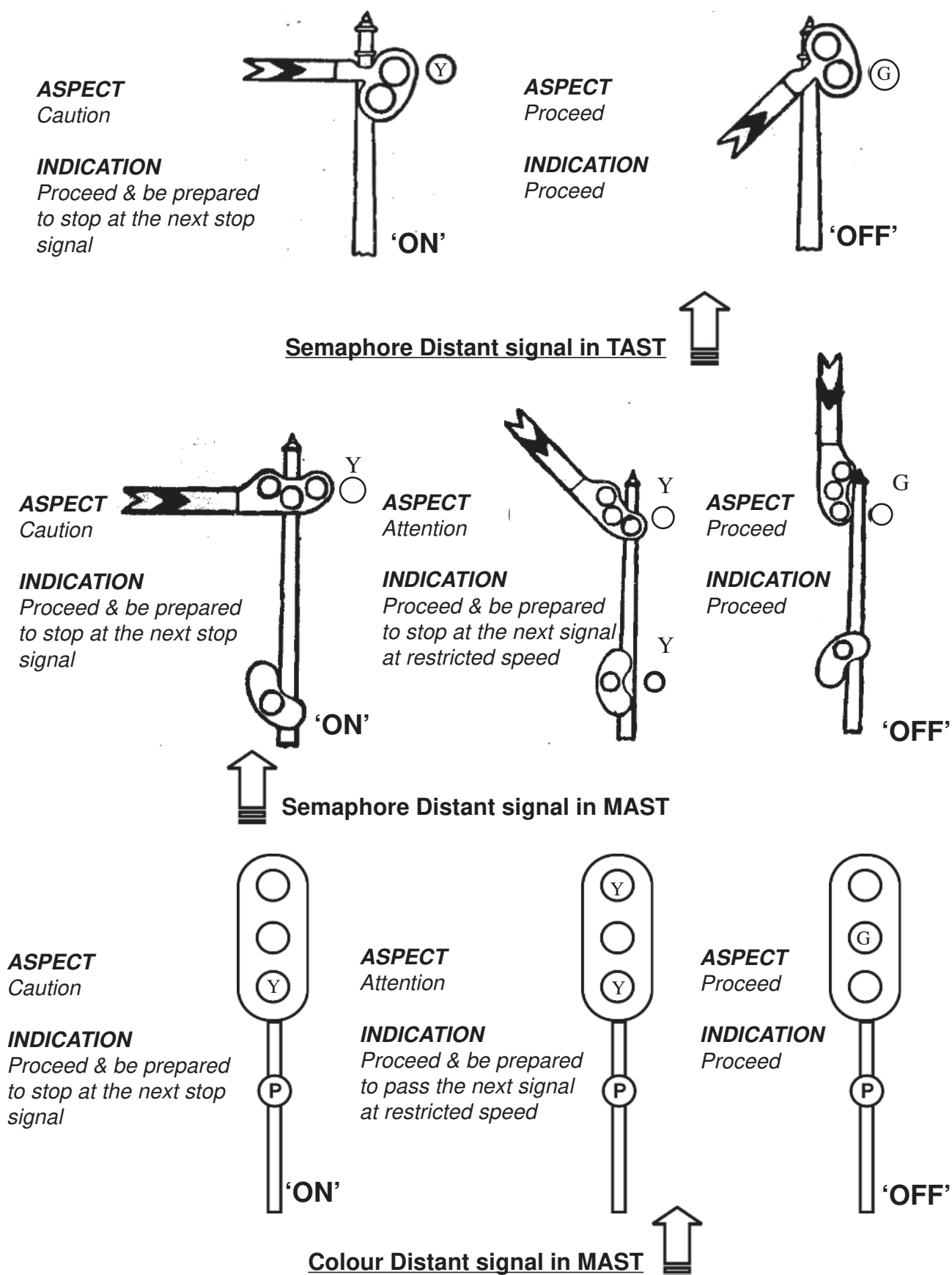
NOTE:

- On a double line "B" class station in TAST, a Warner signal shall be provided below the outer signal. If train runs through stations at 50 kmph without stopping.
- A colour light Warner signal when placed on a post by itself a letter "P" on white circular disc is provided. Letter "P" will be in black colour.
- A Warner signal may be placed either:
 - On a post by itself with a fixed green light 1.5 meters to 2.0 meters above it by night
 - On the same post below the first stop signal or the last stop signal.

DISTANT SIGNAL (G.R. 3.07)

- A distant signal is a permissive signal, which shall be placed at an adequate distance in the rear of stop signal, the aspect of which is pre-warned.
- A distant signal indicates the Loco Pilot
 - The condition of signalling section ahead
 - That he is approaching a stop signal
- The distant signal may be
 - A semaphore arm type is provided in multiple upper quadrants signalling territory
 - Multi-aspect colour light type provided with a letter 'P' in black on white circular disc.
- Where necessary two distant signals may be provided. The outermost signal will be called as distant signal and shall be placed at 2 KMs. from the first stop signal and the second signal

- v) shall be called inner distant and shall be placed at a distance of 1 km from the first stop signal.
A colour light distant signal may be combined with last stop signal or with the gate stop signal.
In such case the "P" marker shall be dispensed with.



STOP SIGNAL (GR 3.08)

- i) Semaphore arm type stop signal shall be square ended arm and is painted in red with white bar.
- ii) Semaphore arm type stop signal shall be provided in TAST & MAST
- iii) Colour light type stop signal shall be provided in TAST and MAST. Multi-aspect colour light stop signal may be in three or four aspect. (See description with illustration in next page)

KINDS OF FIXED STOP SIGNALS FOR APPROACHING TRAIN (GR 3.09)

The stop signal, which controls movement of trains, approaching a station, are of three kinds

- (a) Outer signal (b) Home signal & (c) Routing signal

OUTER SIGNAL

The outer signal where provided is the first stop signal of a station and is located at an adequate distance outside the point up to which line may be obstructed after line clear has been granted to or obtained by the Station in rear.

HOME SIGNAL:

The home signal is the first stop signal of a station at which an outer signal is not provided and the second stop signal of a station at which an outer stop signal is provided. It shall be located outside all connections on the line to which it refers. (Placement/Adequate distance).

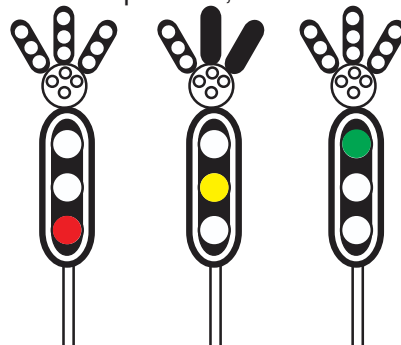
ROUTING SIGNAL:

The routing signal is a signal used to indicate to a Loco Pilot which of two or more diverging route is set for him. When the home signal, is in consequence of its position, inconvenient for this purpose.

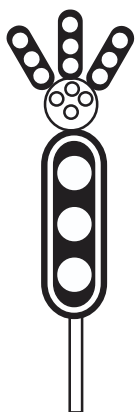
ROUTE INDICATORS:

Route indicators are of three types:

- (a) Junction type
- (b) Multiple Lamp type
- (c) Stencil type



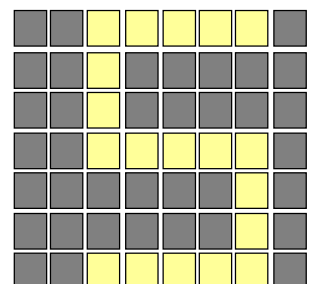
JUNCTION TYPE:



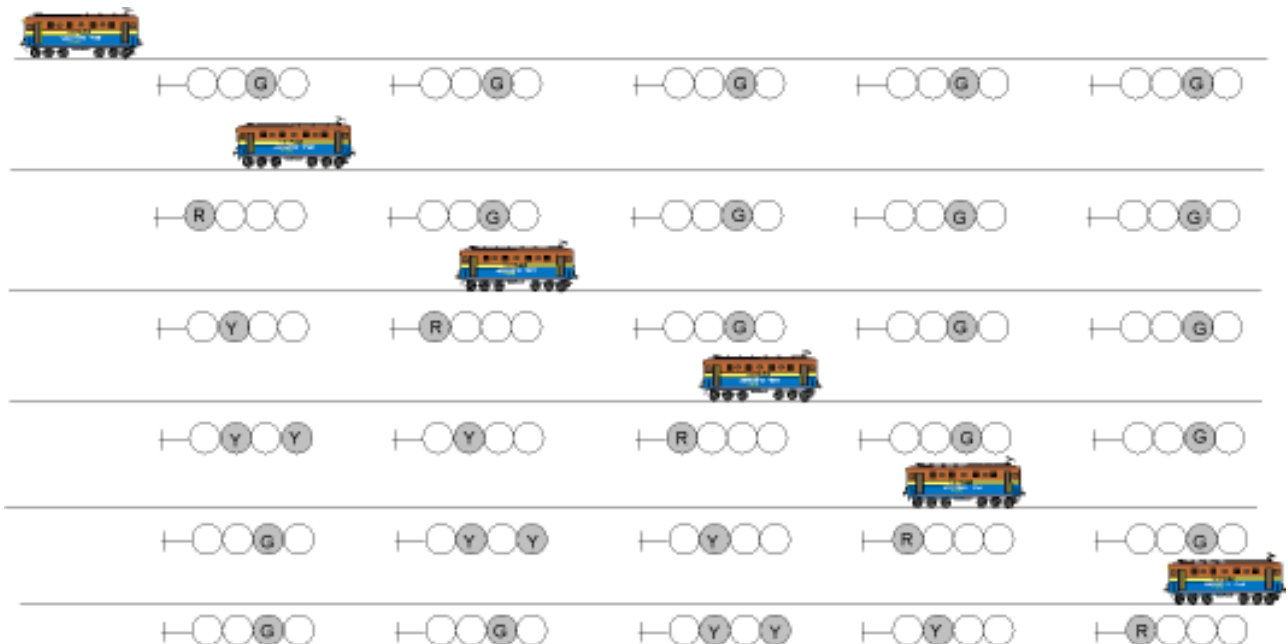
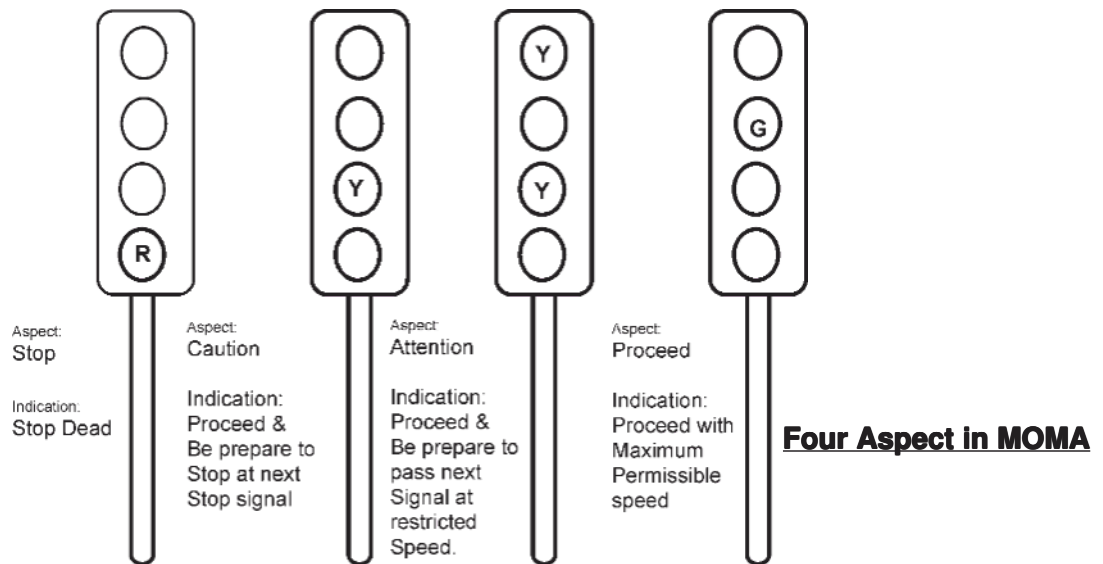
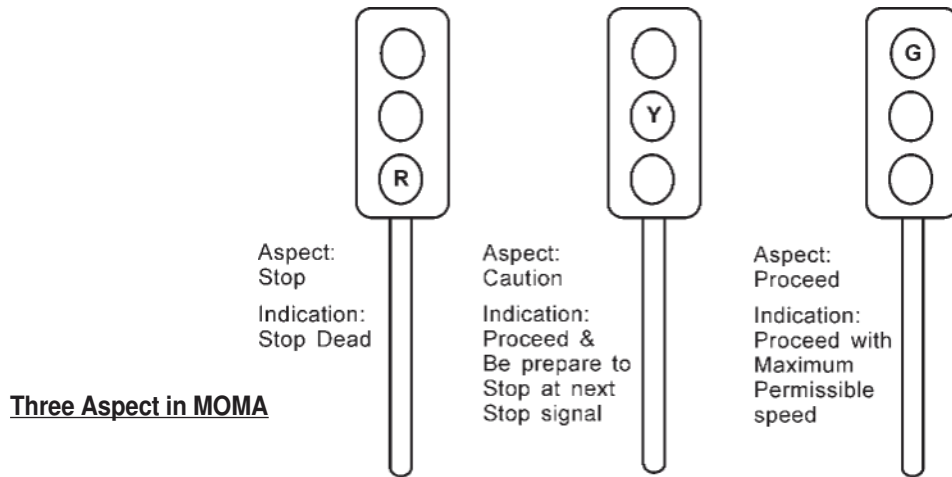
These are provided in colour light signal only. There will be row of five white bulbs, which refer to the line. The row of white bulb will be equal to the number of loop lines (any deviation of the above statement may please refer station working rule of particular station) when the signal is taken OFF, for loop line then particular row of bulb will glow with caution aspect i.e. Yellow. Minimum three bulbs out of five should glow; otherwise that signal shall be treated as defective.

MULTIPLE LAMP TYPE:

It can be provided in both semaphore arm and colour light signal. Horizontal and vertical rows of bulbs are arranged in a box. When a route is set for a particular line and signal is taken OFF, the bulbs will glow to form a number to indicate line.



Aspect of Colour light type stop signal in MAST & AUTO signalling arrangement



STENCIL TYPE:

It may be provided in both semaphore and colour light. It is a box in which various stencils are set for Alphabet letters with lights when the route is set for any line and the signal is taken "OFF", the particular letter will glow.



KINDS OF STOP SIGNALS FOR DEPARTURE OF A TRAIN (GR 3.10)

The stop signal, which controls the movement of a train leaving a station, are called departure stop signal. They are of two kinds:

- (i) STARTER
&
- (II) ADVANCE STARTER

STARTER:

When a train leaving a station is guided by only one starting signal, which is called the last stop signal of the station is known as starter.

ADVANCE STARTER:

When a train leaving a station is guided by more than one starter signal, the outermost starting signal is the last stop signal of the station, and is called the advance starter.

LAST STOP SIGNAL (S.R. 3.10.01)

Where there is no advance starter, the starter is the last stop signal of a station. At "C" class stations, in the absence of a starter, the home signal is the last stop signal.

KINDS OF STOP SIGNAL IN AUTOMATIC BLOCK TERRITORY (GR 3.12)

i) AUTOMATIC STOP SIGNAL

It is the signal, which is not dependent upon manual operation but is controlled automatically by the passage of a train into through and out of the automatic block signalling section.

ii) SEMI-AUTOMATIC STOP SIGNAL

A semi automatic stop signal which is capable of being operated either as an automatic stop signal or as a manually stop signal, as required.

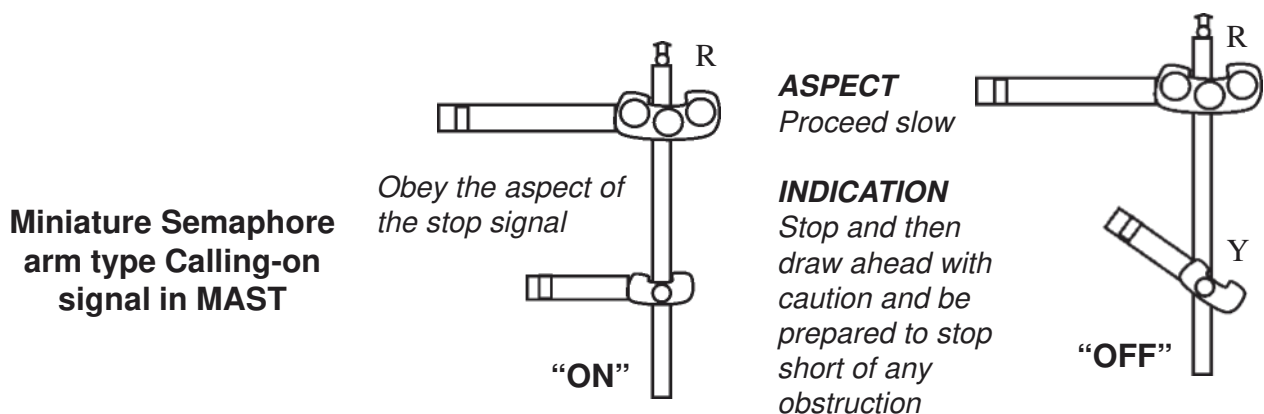
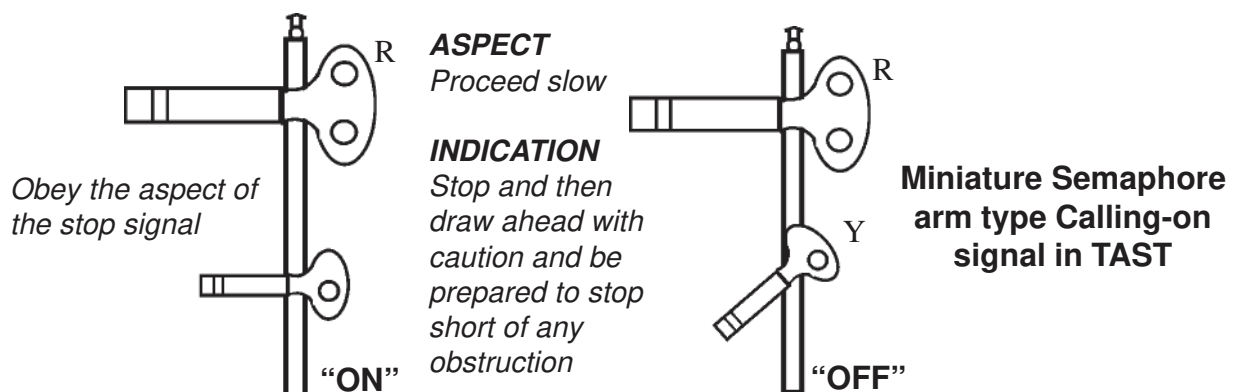
NOTE:

- a) Automatic stop signal shall be provided with a letter "A" in black on white circular disc.
- b) Semi-automatic stop signal shall be provided with illuminated "A" on the black background. When "A" marker is illuminated, it works as an automatic signal, when "A" marker is extinguished; it works as manual stop signal.
- c) When a semi-automatic signal works as an automatic stop signal, it assumes "ON" and "OFF" aspects automatically according to the condition of the automatic block signalling section ahead.
- d) When semi-automatic signal works as a manual stop signal, it assumes "ON" aspect automatically on the occupation of automatic block signalling section ahead but assumes "OFF" aspect when operated manually, provided the relevant automatic block signalling section ahead are clear.
- e) It is provided only in MAST, in colour light signal.

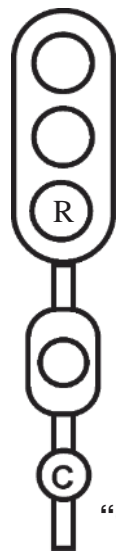
SUBSIDIARY SIGNALS

CALLING ON SIGNAL (GR 3.13)

- i) A Calling ON signal is subsidiary signal, which has no independent aspect in the ON position.
- ii) It shall be –
 - a) A short square ended semaphore arm painted white with red bar.
 - b) A miniature colour light provided with letter “C” in black on a white circular disc
 - c) It shall be fixed below a stop signal governing the approach of a train. Under approved special instructions, it may be provided below any other stop signal except the last stop signal.
 - d) A calling signal when taken “OFF,” calls the Loco Pilot of a train to draw ahead with caution, after the train has been brought to a stop even though the stop signal above it is at “ON” and indicates to the driver that he should be prepared to stop short of any obstruction.
- iii) A Calling ON signal shall show no light in the “ON” position.
- iv) A Calling ON signal shall not be taken “OFF” until the train has been brought to a stand at stop signal below which calling “ON” signal is provided.
- v) The Loco Pilot shall stop his train at the stop signal, if it is at “ON” and if he finds Calling ON signal is taken “OFF,” after bringing his train to stop, then draw ahead with caution and be prepared to stop short of any obstruction.



**Colour Light type
Calling-on
signal in MAST**



*Obey the aspect of
the stop signal*

“ON”



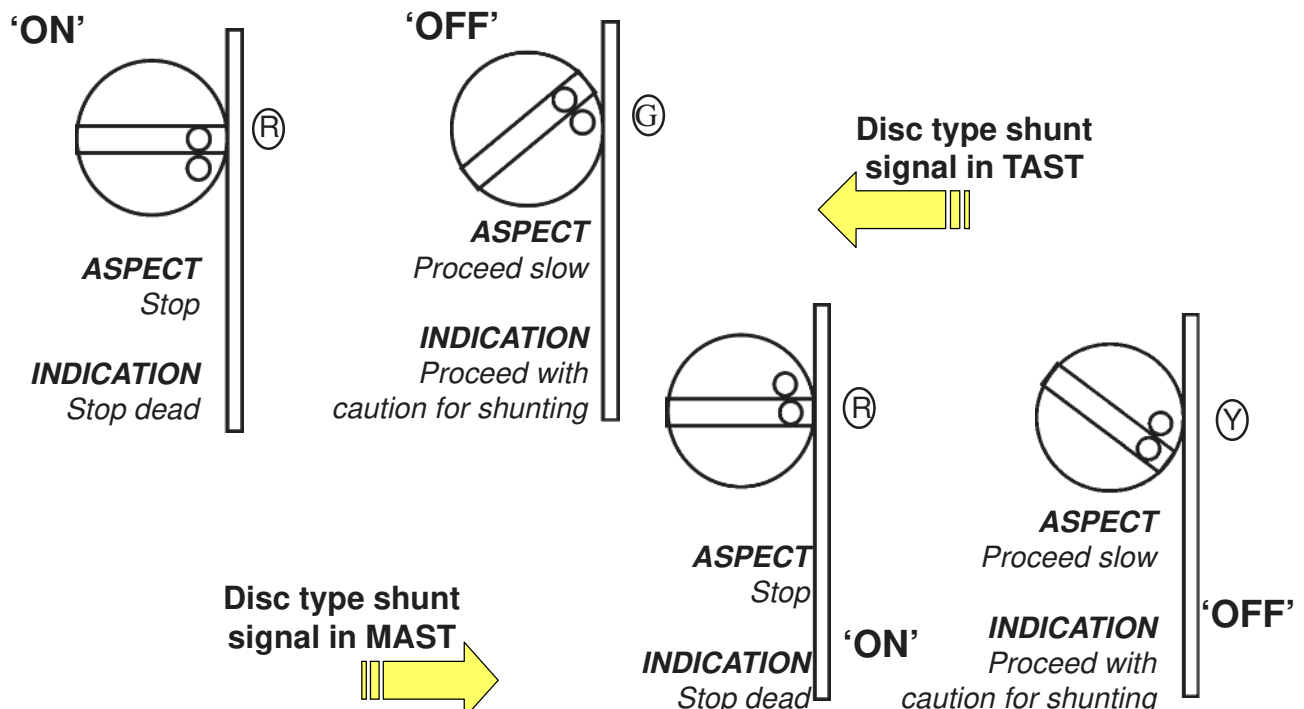
ASPECT
Proceed slow

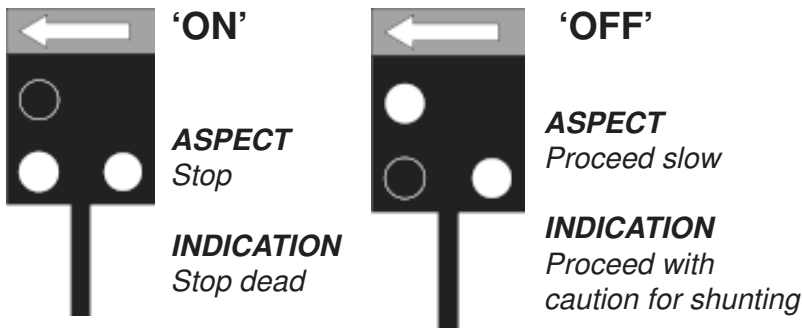
INDICATION
*Stop and then
draw ahead with
caution and be
prepared to stop
short of any
obstruction*

“OFF”

SHUNT SIGNAL (GR 3.14)

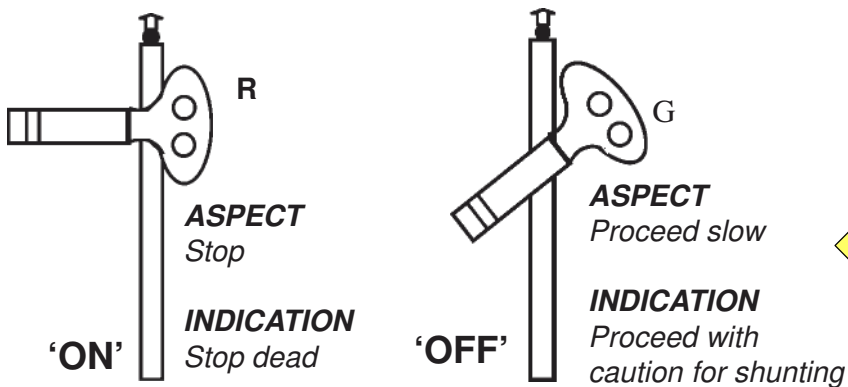
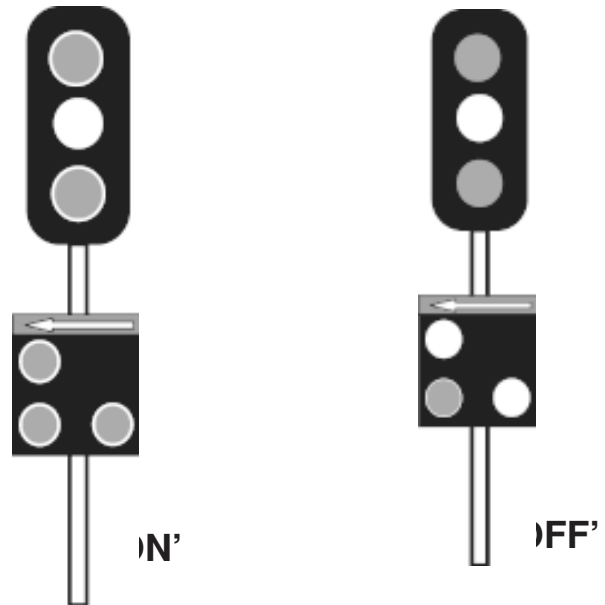
- i) A shunt signal is a subsidiary signal and shall be either –
 - a. A white disc with a read mark across it
 - b. A position light type signal
 - c. Under special instructions, a shunt signal may be a miniature semaphore arm type
- ii) A shunt signal controls the shunting movement
- iii) A shunt signal may be placed on a post by itself or below a stop signal, other than the first stop signal of a station.
- iv) More than one shunt signal may be placed on the same post and in such cases the top most signal shall apply to the extreme left hand line and the second shunt signal from the top shall apply to the next line from the left and so on.
- v) When a shunt signal is taken OFF, it authorises the Loco Pilot to draw ahead with caution for shunting purpose although stop signal, if any, above it is at ON
- vi) When a shunt signal is placed below a stop signal, it shall show no light at ON position.
- vii) In case shunt signals are not provided hand signals may be used for shunting



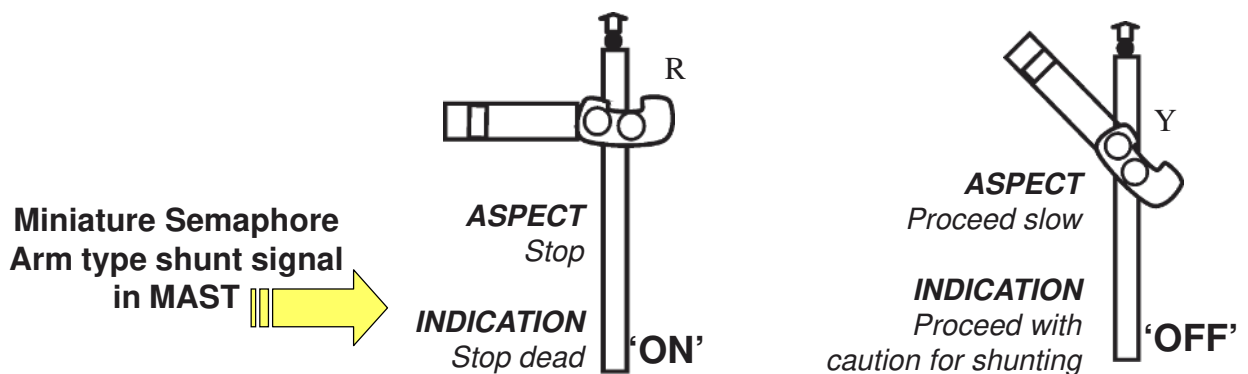


Position Light type shunt signal on a post by itself

Position Light type shunt signal below a stop signal



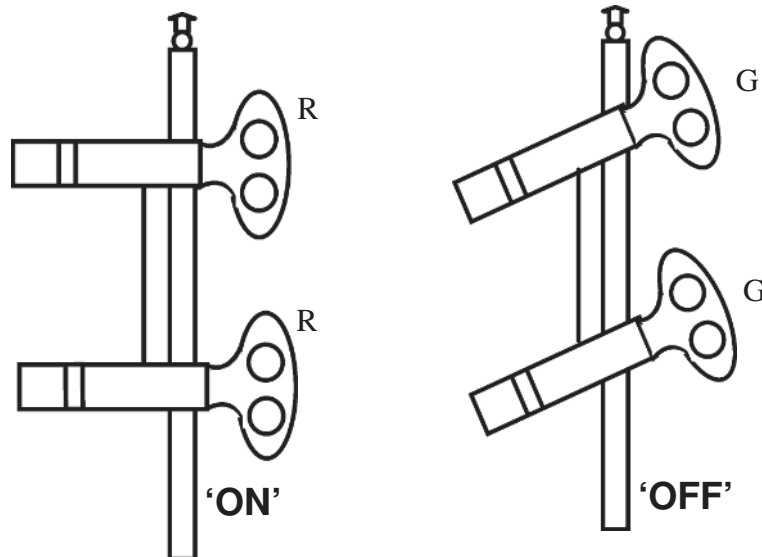
Miniature Semaphore Arm type shunt signal in TAST



Miniature Semaphore Arm type shunt signal in MAST

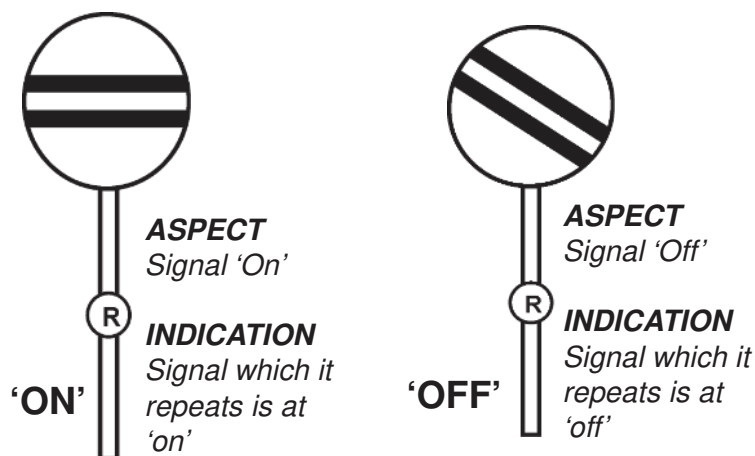
CO-ACTING SIGNAL (GR 3.15)

- i) Co-acting signals are duplicate signals fixed below ordinary signals and are provided where in consequence of height of signal post or of there being an over bridge or any other obstruction, the main arm or light is not in view of the Loco Pilot during the whole time that he is approaching it.
- ii) Co-acting signals shall be fitted at such height either the main arm or light or the co-acting arm or light, is always visible.



REPEATING SIGNAL (GR 3.16)

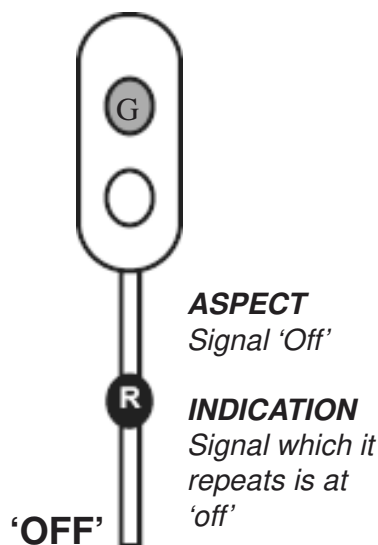
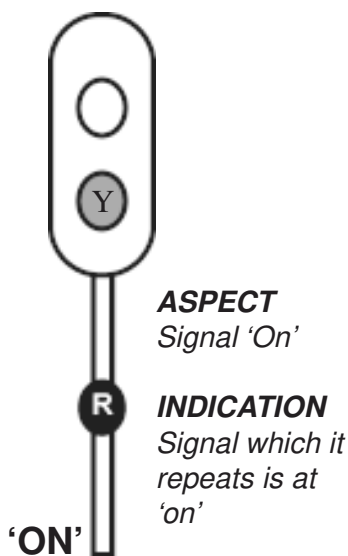
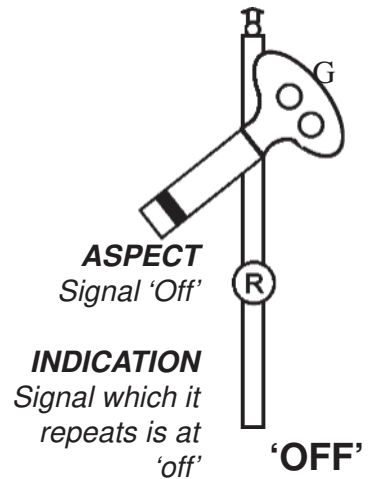
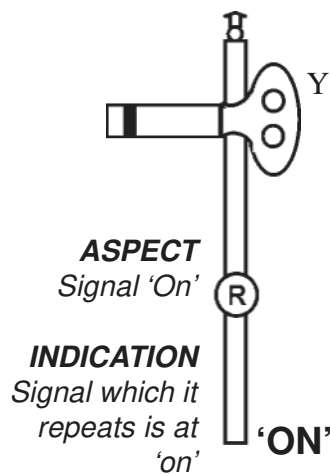
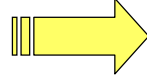
- i) A signal placed in rear of a fixed signal for the purpose of repeating to the Loco Pilot of an approaching train, the aspect of the fixed signal in advance is called repeating signal.
- ii) A repeating signal shall be provided with an "R" marker and shall be of –
 - a. Banner type provided with letter "R" in black on a white circular disc.
 - b. Square ended semaphore arm type with a letter "R" in black on a white circular disc.
 - c. Colour light type provided with white illuminated "R" against the black background.
- iii) A normal aspect of repeating signal is yellow.



Banner type Repeating signal in TAST



Semaphore Arm type
Repeating signal in TAST



Colour Light type
Repeating signal



VISIBILITY OF SIGNAL (SR 3.16.01)

- i) In two aspect signals including modified lower quadrant (MLQ) signals, the minimum visibility distance for various signals shall be as follows:
 - a. Home signal 400 Metres.
 - b. Main starter signal 400 Metres.
 - c. Loop starter signal 200 Metres
 - d. Outer/Warning signal (in TALQ) 1200 Metres.
- ii) In multi-aspect signals, all signals shall be visible for a minimum period of 5 seconds for maximum permissible speed allowed on the section.

SIGNALS OUT OF USE (GR 3.18)

- i) When a fixed signal is not in use, it shall be distinguished by two cross bars, each bar being not less than 1 Meter long and 10 cm wide.
- ii) A semaphore or disc signal when not in use shall be kept fixed in ON position.
- iii) Signals (Light signals) not in use shall not be lit.

HAND SIGNALS (GR 3.52)

EXHIBITION OF HAND SIGNALS:

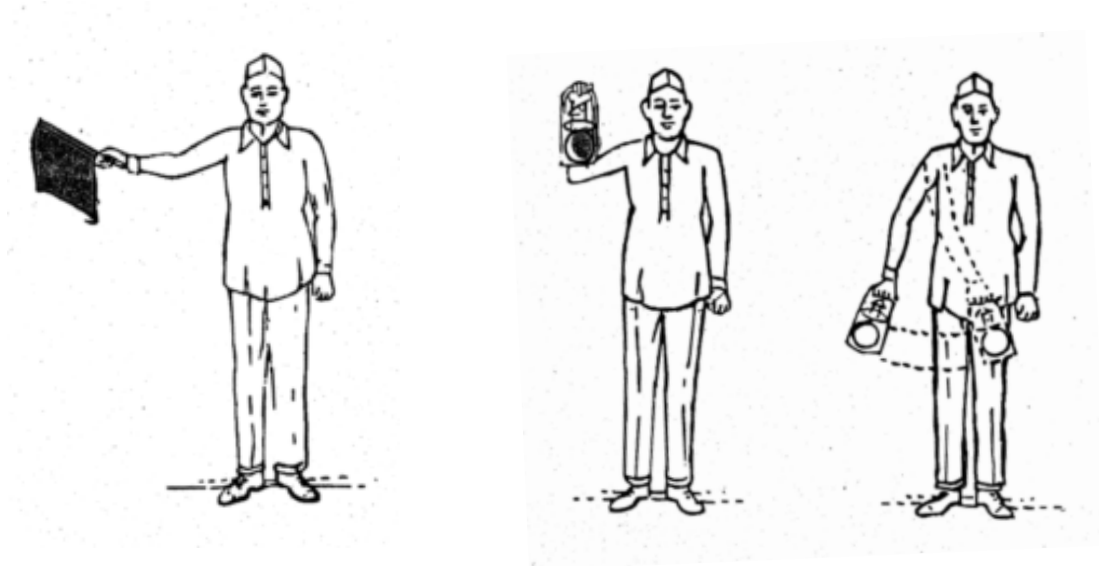
During day flag or flags shall be used as hand signals. Hand shall be used in emergencies only when

flag are not available.

During night a hand signal shall normally be given by showing a red or green light. A white light waved violently shall be used as stop signal only when red light is not available.

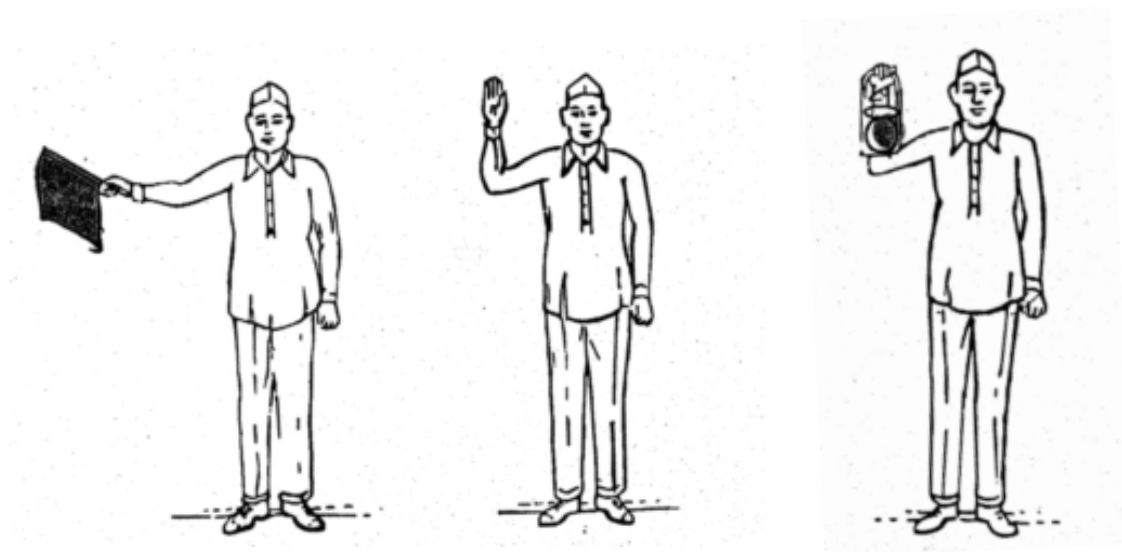
STOP HAND SIGNAL (GR 3.53)

- | | | |
|---------------|---|--|
| Indication | - | Stop dead |
| At Day time | - | By showing a red flag or by raising both arms with hands above the head. |
| At Night Time | - | By showing a red light or by violently waving a white light across the body of the person. |



PROCEED HAND SIGNAL (GR 3.54)

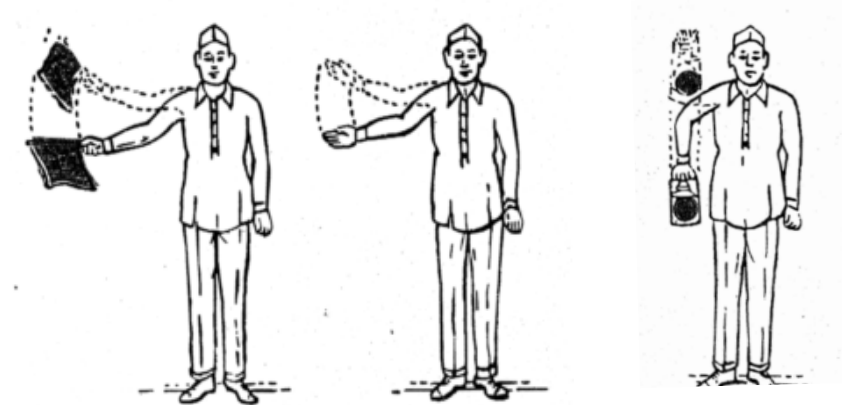
- | | | |
|---------------|---|--|
| Indication | - | Proceed |
| At Day time | - | By holding a green flag or by holding one arm steadily |
| At Night time | - | By holding a green light steadily |



PROCEED WITH CAUTION HAND SIGNAL (GR 3.55)

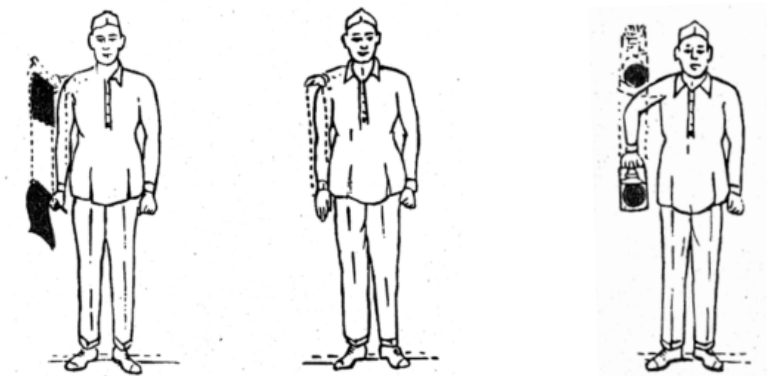
Indication - Proceed slowly reducing speed further if the signal is given at a progressively slower rate.

- At Day time- By waving a green flag vertically up and down or by waving one arm in the similar manner.
At Night time- By waving a green light vertically UP and DOWN.

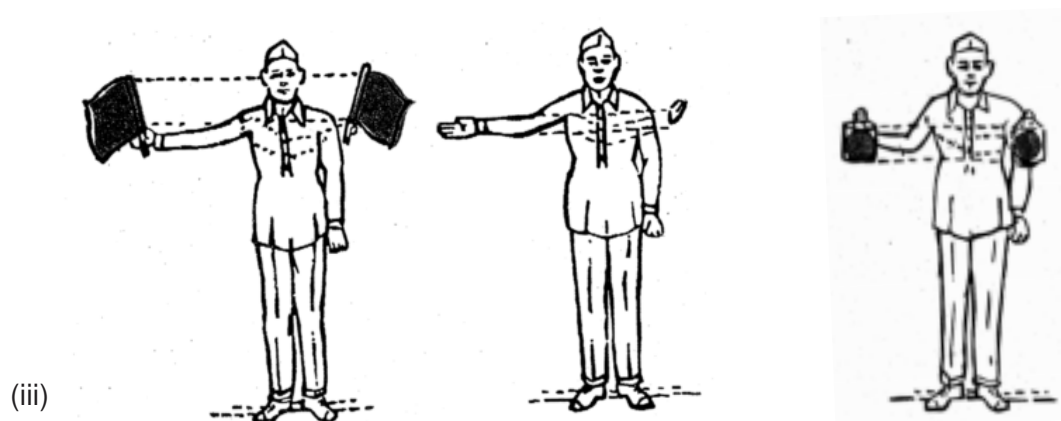


HAND SIGNALS FOR SHUNTING (GR 3.56)

- i) Indication - Move away from the person signalling
In Day time - By a green flag or one arm moved slowly up and down
In night time - By a green light moved slowly up and down



- (ii) Indication - Move towards the person signalling
In day time - By a green flag or one arm moved from side to side across the body
In nighttime - By a green light moved from side to side across the body.



(iii)

In daytime—

By green and red flag held above the head or both hands raised over the head and moved towards and away from each other.

In nighttime —

By a green light held above the head and moved by twisting the wrist



NOTE :- (SR 3.56.01)

Shunting signals are to be given by the station master, guard/brakes man of a train, Yard foreman, Shunting Jamadar or points man or any other duly qualified/authorised person.

(SR 3.56.02)

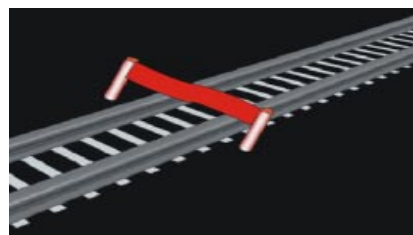
Red or stop hand signal given by any one supersedes all other signals and bars movement beyond such danger signal.

MINIMUM EQUIPMENT OF FIXED SIGNAL (GR 3.27, 28, 29)

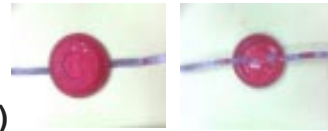
- i) Minimum equipments of fixed signals at a station provided with manually operated multiple aspect signalling (MOMA) -
 - a. At "B" class station – Distant, Home & Starter
 - b. At "C" class station – Distant and Home
- ii) Minimum equipments of fixed signals at stations provided with modified lower Quadrant (MLQ) signalling
 - a. At "B" class station – Distant, Home, A Warner below main home and a starter.
 - b. At "C" class station – Distant and Home
- iii) Minimum equipments of fixed signals at a station provided with two aspect signalling territory (TAST)
 - a. At "A" class station – WARNER, Home and a starter
 - b. At "B" class station –
 - i. On single line – Outer and Home
 - ii. On double line – Outer, Home and Starter
 - c. At "C" class station – Warner and Home

BANNER FLAG (GR 3.57)

A banner flag is a temporary fixed danger signal, consisting of a red cloth supported at each end on a post and stretched across the line to which it refers.



DETONATING SIGNALS



DESCRIPTION OF DETONATING SIGNALS (GR 3.59)

Detonating signals, otherwise known as detonators or fog signals are appliances which are fixed on the rails, when an engine or a vehicle passes over them they explode with a loud sound so as to attract the attention of the Loco Pilot

METHOD OF USING DETONATORS (GR 3.60)

- i) Detonators shall be placed on the rail with label or brand facing upwards and shall be fixed on the rail by bending the clasps around the head of the rail.
- ii) In case of mixed gauge, detonators shall be placed on the common rail or on one rail of each gauge

STOCK OF DETONATORS: 10 Detonators in a box.

USE AND LIFE OF DETONATORS:

When a railway servant placed one or more detonators on the line he shall stand beyond the safety radius of 45 Metres from the detonator. Life span of detonator is 7 years from the year of its manufacture.

PLACING OF DETONATORS IN THICK FOGGY OR TEMPESTUOUS WEATHER IMPAIRING VISIBILITY (GR 3.61)

- i) In thick foggy and tempestuous weather, impairing visibility, the station master shall comply the visibility by "Visibility test object post" which is placed at a distance of 180 Metres from the centre of the station Master's office at each end of the station.
- ii) During thick, foggy and tempestuous weather if a station signal cannot be seen it is necessary to indicate of an approaching train about the locality of signal, the fog signal man will place two detonators on the line about 10 meters apart and at least 275 meters outside the outermost signal. The fog signalman shall show no signal to the approaching train except in case of obstruction he will show a stop hand signal. On single line the fog signalman will show green hand signal to the outgoing train.

NOTE:

- (i) In automatic signalling the placing of detonators (fog signal) shall be Dispensed with.
- (ii) In case of temporary speed restriction, during thick foggy or tempestuous weather by day or night two detonators shall be placed at a distance of 270 meters, 10 meters apart in rear of caution indicators.

PLACING OF DETONATORS IN CASE OF OBSTRUCTION (GR 3.62)

- i) Whenever in consequence of an obstruction of a line, it is necessary for a railway servant to stop an approaching train he shall proceed showing his stop hand signal and place one detonator at 600 meters and 3 detonators at 1200 meters from the obstruction about 10 meters apart from each other.
- ii) If the said railway servant is recalled before the obstruction is removed, he shall leave down 3 detonators and on his way back pick up intermediate detonators.

STOP SIGNAL DEFECTIVE IN OFF POSITION (SR 3.68.02)

- i) In case of reception stop signal is defective in OFF position, the station master shall take measures to put back the signal at ON position either by disconnecting wire or by breaking the green glass.

- ii) Either by disconnecting wire or breaking the green glass if the signal cannot be kept in ON position, then the station master will arrange to protect the signal by placing one detonator at a distance of 90 meters in rear of the signal. A stop hand signal shall be exhibited at the foot of signal. The train shall be piloted by T/369(3b) and proceed hand signal from Engine's footboard.

NOTE:

In case of routing signal (not home) starter or advance starter, the detonator shall not be used.

USE OF ONE DETONATOR:

- i) In case of extreme emergency or where there is no time to place more than one detonator, only one detonator may be placed as far as possible from the obstructed site.
- ii) When approaching stop signal is defective in OFF position, one detonator is placed at a distance of 90 meters in rear of defective signal. The station staff who fix detonator should exhibit the stop hand signal to approaching train Loco Pilot by standing at the foot of the defective signal.

USE OF TWO DETONATORS

- i) In case of thick foggy or tempestuous weather two detonators are placed 10 meters apart at least 275 meters ahead of the outer most signal by the station staff.
- ii) In case of temporary speed restriction during thick foggy or tempestuous weather, two detonators shall be placed 10 meters apart at the distance of 270 meters in rear of caution indicator.

USE OF THREE DETONATORS

- i) Three detonators are used to protect the train in rear in automatic section in case of obstruction or failure. The first detonator is placed at a distance of 90 meters, in rear of obstruction and second at 180 meters and third 10 meters apart (90M–90 M–10M)
- ii) When a train is working with block ticket or with “an authority to proceed without line clear” in both single line (S/L) or double (D/L) absolute block system or with an “Authority to proceed without line clear” in S/L automatic system, three detonators are to be placed in rear of train (250M – 250M – 10M)
- iii) When an engineering protection is required to be done by using banner flag in section when engineering work is in progress, Banner flag is to be placed at a distance of 600 Meters from the point of obstruction and three detonators should be placed at a distance of 1200 meters from the place of obstruction 10 meters apart (600M – 10M – 10M) and a man with red flag hand signal 45 metres away from last detonator. In auto signalling section 90m-10m-10m behind the banner.

USE OF FOUR DETONATORS

Four detonators are to be placed when there is obstruction and train cannot proceed further during normal working in absolute block system and abnormal in double line automatic system from the obstruction in rear and front and adjacent line (600M - 600M – 10M – 10M)

WHEN AN ENGINE EXPLODES A DETONATOR:

- i) Immediately control the train, whistle intermittently & look for any signal and also note down the KM No. & time.
- ii) If no signal is noticed –
 - a. Proceed very cautiously and be prepared to stop short of any obstruction.
 - b. After proceeding 1.5 Kms, if no signal is noticed resume normal speed.
 - c. Assistant driver or guard (in case of EMU) will pilot the train if necessary.
- iii) Report the matter to next station in writing giving KM No. Section and time.
- iv) After controlling the train, if any signal is noticed then act accordingly.
- v) In case guard will have to pilot in EMU, the guard will pilot from the cab only.

The precaution to be observe by loco pilot in case of foggy weather-

1. In absolute block system maximum speed will be 60 kmph.
2. whistle frequently to warn the gate man and road users of an approaching train at level crossings.
3. in automatic section after passing the auto observe the following precaution-
 - a. after passing green,max.speed will be 60 kmph.
 - b. After passing one yellow max speed 30 kmph.
 - c. After passing one yellow be prepare to stop at the next signal.

But in both the system speed will be subject to the judgment of the loco pilot so as to be prepared to stop the train short of any obstruction.

One detonator shall be placed at 285 meters and 275 meter short of the first stop signal at stations where it is necessary.

Now as per GR 9.02 (3)(to pass signal at on) if on account of curvature of the line , fog etc the line ahead cannot be seen clearly, the loco pilot shall proceed at a very slow speed not exceeding 10 kmph.

As per board's letter number 98/safety(A&R)/19/16 dated 02.02.2010: Aftera passing automatic signal at on the loco pilot of a following train shall ensure that minimum distance of 150 metres or two clear OHE masts is maintained between his train(in clear weather) and the preceding train if any, or any obstruction on the line ahead,In the case of EMU the minimum distance of 75 meter or one OHE mast shall be maintained between EMU and a preceding train if any or any obstruction on line ahead.But in case of dence fog after passing an auto signal at on the loco pilot/motorman of the train hauled by any locomotive shall while moving at a speed not exceeding 10 kmph should ensure that he maintains a reasonable distance at which he is able to observe the flashing tail lamp of the train ahead or the obstruction.

I.B. AND GATE SIGNALS

INTRODUCTION

Intermediate block means an arrangement of signalling on double line in which a long block section is split into two portions, each constituting a separate block section by providing an intermediate block post and this may be called a "C" class station which is remotely controlled by the block station in rear .

A fixed stop signal provided before level crossing gate which closes across the line to protect the gate is called gate stop signal when the gates are open for the passage of road traffic this signal shows stop aspect, both the UP and DOWN direction to control the movement of approaching train.

I.B. SIGNAL (GR 3.11)

Intermediate block stop signal is the home signal provided at intermediate block post.

It can be identified by a letter, IB in black on white circular disc provided on the signal post.

PURPOSE: To divide long block section into two short block sections.

UTILITY: To increase the section capacity and to minimise the detention. It is ensured by providing track circuit/axle counter

WORKING: Last stop signal of the rear station shall be taken off on the clearance of the line provided with the track circuit/axle counter between last stop signal in rear and intermediate block stop signal with an adequate distance.

A train may be allowed to proceed up to an intermediate block stop signal even when line clear has not been received from the station in advance. The intermediate block stop signal shall be taken OFF by the station in rear only when line clear has been received on the block instrument.

GATE STOP SIGNAL (GR 3.34)

- i) Except under approved special instructions, every level crossing gate, which is not interlocked with station signal, shall be protected with gate stop signal.
- ii) The gate stop signal shall be placed in rear of level crossing gate at a distance of 400 meters in TAST and 180 meters in MAST/MLQ
- iii) Every gate signal shall be preceded by a Warner/distant/Warning board or both where necessary.
- iv) This signal can be identified by a marker letter "G" in black on yellow circular disc.

I.B. SIGNAL DEFECTIVE (GR 3.75)

PASSING OF IB STOP SIGNAL AT "ON" POSITION –

Sound a long continuous whistle and stop the train at the foot of IB signal and contact with station master in rear over the signal post telephone.

1st Condition:

When telephone is working and station master could be contacted.

- a) Take private number and I. D. No. From the station master in rear, if line clear has been received.
- b) Sound one short one long one short (0-0) whistle. Exchange signal with guard and proceed up to first stop signal of station in advance and act accordingly.

2nd condition:

When signal post telephone out of order/telephone not provided/station master could not be contacted.

- a) Wait for 5 minutes
- b) Give a long whistle and exchange the signal with guard and proceed cautiously up to first stop signal of next station.
- c) Speed not exceeding 15 kmph when view ahead is clear and 8 KMPH when view ahead is not clear or at night.
- d) Stop the train at 1st stop signal of a station, give whistle on hearing whistle signal may be lowered or piloted in.
- e) Report the matter at the next station in writing.
- f) If there is a gate stop signal in between the IB stop signal and the 1st stop stop signal of the station in advance, and even it show green, speed should not be exceeded till the 1st stop signal is reached.

3rd condition:

When telephone is working but station master in rear station unable to obtain line clear due to no communication between two stations (total interruption)

- a) ASM will advice to call the guard by sounding two long and two short (- - 0 0) whistle.
- b) After consulting with station master the guard will give a written authority (memo) with private number received from station master authorising the driver to pass the IB signal at ON.
- c) After handing over memo to Loco Pilot, Guard will go to B/Van. Give one long whistle and exchange signal with guard.
- d) Speed not exceeding 15 KMPH when view ahead is clear and 8 KMPH when view ahead is not clear or at night.
- e) The matter shall be reported in writing at the next station.

4th condition:

If the station master is pre warned of defective IB signal

- a) The train shall be stopped at station.
- b) The Loco Pilot shall be issued T-369(3b) with private and ID No. To pass I.B. Signal at ON without stopping with normal speed.

NOTE: When the train is detaining at the IB Stop signal contact the rear station mast in every five minutes when telephone is working.

BACKING OF TRAIN AFTER PASSING AN I.B. SIGNAL UPTO BACKING LIMIT BOARD (SR 3.75.04)

A Loco Pilot shall sound four short whistle (0 0 0 0). The train shall be protected in rear as per G.R. 6.03 from the place up to which the train is required to back. After taking written permission from the Guard, the Loco Pilot shall back the train upto the backing limit board/point with a great caution.

BACKING OF TRAIN FOULING THE BACKING LIMIT BOARD/POINT/BLOCK OVERLAP.

- i) Guard shall be informed about the fact
- ii) Train shall be protected as per GR 6.03 in rear.
- iii) Guard shall contact the station master in rear over a signal post telephone and confirm that no train has left from rear station support by a private number and he will advise ASM, not to allow train supported with P/No.
- iv) Guard shall hand over a written memo with private number to the Loco Pilot, which has been received from station mast of rear station.
- v) The Loco Pilot shall back the train with great caution. After clear the block section, until the guard has been informed that the block section is clear, no train shall be allowed behind his train from the station in rear.

GATE STOP SIGNAL DEFECTIVE

PASSING GATE STOP SIGNAL IN DEFECTIVE OR AT “ON” POSITION AT ABSOLUTE BLOCK SECTION

- i) Sound one long whistle and stop the train in rear of the gate stop signal.
- ii) Ensure “G” marker on the signal post. Stop one minute by day and two minutes by night. If the signal still remains at ON proceed cautiously up to the level crossing gate after giving long whistle and exchanging signal with guard.
- iii) If the gate is closed against the road traffic and gate man displays hand signal, proceed further and pass the level crossing gate cautiously.
- iv) If the Gateman is not available or is available but not exhibiting hand signals, he shall stop short of the level crossing, where he shall then be hand signalled by the Gateman and pass the gate. If gateman is not available, then-
 - a. Assistant Loco Pilot should close the gate and show proceed hand signal.
 - b. The train shall be stopped again after clearing level crossing by at least two vehicles.
 - c. Assistant Loco Pilot will reopen the gate for road traffic.
 - d. Inform the concerned ASM through gate phone.
 - e. If Gate phone is defective then stop the train at next station and report the matter to ASM in writing.

PASSING GATE STOP SIGNAL AT “ON” IN AUTOMATIC SIGNALLING TERRITORY:

Give one continuous whistle and stop the train in rear of gate stop signal.

Ensure “G” marker and illuminated “A” marker.

i) When “A” marker is illuminated (Gate closed)

- a. Stop one minute by day and two minutes by night
- b. Sound one long whistle, exchange signal with guard.
- c. Proceed cautiously up to the next stop signal not exceeding 15 KMPH when view ahead is clear and 8 KMPH when view ahead is not clear or at night and be prepared to stop within half of the visible distance.
- d. Be guided by aspect of next stop signal only after reaching it.

ii) When “A” marker is extinguished (means gate is open) and section ahead may be occupied

- a. Stop the train in rear of the gate stop signal sound one long continuous whistle and found whistle one long continuous to warn the gateman.
- b. Wait one minute by day and two minutes by night.
- c. If signal still remains at ON, proceed cautiously up to the level crossing.
- d. If the Gate man is available and exhibiting hand signals, proceed further and pass the level crossing gate cautiously and be prepared to stop within half of the visibility distance, at a speed not exceeding 15 KMPH when view ahead is clear and 8 KMPH when view ahead is not clear or at night, up to the next stop signal. Be guided by the aspect of the next stop signal only after reaching it.
- e. If the Gateman is not available or is available but not exhibiting hand signal stop in rear of the level crossing and after ascertaining that the gates are closed against road traffic and on getting hand signals from the Gateman and in his absence from the Assistant Loco Pilot. The Loco Pilot shall sound the prescribed code of whistle and cautiously proceed up to the next stop signal, speed is not exceeding 15 KMPH or 8 KMPH as per visibility. Be guided by the aspect of the next stop signal only after reaching it.

WORKING OF SIGNALS AND POINTS

INTRODUCTION

The working of signals and points are so interconnected mechanically or electrically or by both, that their operation must take place in proper sequence to ensure safety. A signal cannot be lowered (OFF) unless the concerning points are correctly set and locked and points cannot be reversed unless the signal is put back to “ON”. Every fixed signal is so constructed that in case of failure of any part of the connection it shall remain at “ON” or return to its most restrictive aspect.

POINTS AFFECTING THE MOVEMENT OF A TRAIN (GR 3.38)

The station master shall not give permission to take signal off for a train until:

- a) All facing points over which the train will pass are correctly set and locked
- b) All trailing points over which the train passes are correctly set
- c) The line over which the train is to pass is clear and free from obstruction.

FACING AND TRAILING POINT

Points are facing or trailing in accordance with the direction a train or vehicles moves over them. Points are said to be facing points when by their operation a train approaching them can be directly diverted from the line upon which it is running.

LOCKING OF FACING POINTS (GR 3.39):

Facing points when neither interlocked nor key locked shall be locked for the passage of a train either by a clamp or by a through bolt, with a pad lock. It is not sufficient to lock the lever working the points.

INTERLOCKING, ITS KINDS & STANDARDS

INTERLOCKING:

Means an arrangement of signals, points and other appliances, operated from a panel or lever frame so interconnected by mechanical locking or electrical locking or both that their operation must take place in proper sequence to ensure safety.

DIRECT INTER LOCKING

It means all levers i.e. points and lock lever and the signal lever are connected in one lever frame operated by mechanical and electrical means.

INDIRECT INTER LOCKING

It means that the points are set and locked from one frame and signals are operated from another place by other lever frame. Interlocking is affected by means of keys or transmitted from one place to another place.

BASIC PRINCIPAL OF INTERLOCKING

- 1) It shall not be possible to take 'OFF' conflicting signals at once at the same time.
- 2) It shall not be possible to take 'OFF' signal for a running line unless-
 - a) All points on the running line are directly set and facing points are locked.
 - b) All points towards the siding and root line are set against the running line.
 - c) Interlocked level crossing gates are locked across the roadway.
 - d) The running line referred to above shall include adequate distance also.
- 3) Once a signal lever is pulled to take "OFF" a signal, it must be locked or back-locked as necessary.
- 4) Once a signal is in "ON" position all points which would be locked for taking it "OFF" must be free for shunting purpose.
- 5) It must be impossible to take "OFF" a Warner signal until all the relevant stop signal in advance have been taken "OFF" first and when "OFF" it must be back-locked.

STANDARD OF INTERLOCKING

The interlocking of facing points of a station are divided into 3 standard and according to the standard of interlocking the speed of a train over that interlocked facing points is fixed.

i) Standard- I Interlocking

- a) Speed over facing point – Not exceeding 50 KMPH
- b) Isolation - Isolation of main line is recommended but not essential.
- c) Points - The facing point should be provided with key locks.
- d) Interlocking - Interlocking between points and signals may be carried out indirectly by means of key locks.

ii) Standard- II Interlocking

- a) Speed over facing point – Not exceeding 75 KMPH
- b) Isolation - Isolation of main line is essential.
- c) Points - The facing point should be provided with plunger type lock.
- d) Interlocking-Interlocking between points and signals may be direct or indirect.

iii) Standard- III Interlocking

- a) Speed over facing point – Maximum permissible speed in that section.
- b) Isolation - Isolation of main line is essential.
- c) Points - The facing point should be provided with plunger type lock.
- d) Interlocking - Interlocking between points and signals must be direct.

ISOLATION OF POINTS

It is an arrangement, secured by the setting of points or other approved means to protect the line so isolated from the danger of obstruction from other connected line or lines.

TRACK CIRCUIT AND AXLE COUNTER

TRACK CIRCUIT

It means an electrical circuit provided to detect the presence of a vehicle on a portion of track, the rails of the track forming part of the circuit.

AXLE COUNTER

It means an electrical device which when provided at two given points on the track, proves by counting axle in and counting axles out, whether the section of the track between the said two points is clear or occupied.

DEFECTIVE SIGNAL WORKING:

INTRODUCTION:

When a Loco Pilot becomes aware that any signal has become defective or has ceased to work properly on any occasion he shall pass such type of signal after receiving the correct authority to proceed and as per the instructions laid down in the GR/SR and a special instruction if any.

DUTIES OF ENGINE CREW IN RESPECT OF SIGNAL (GR 3.78)

- (1)(a) The Loco Pilot shall pay immediate attention to and obey every signal whether the cause of the signal being shown is known to him or not
- (b) the Loco Pilot shall not, however, trust entirely to signal, but always be vigilant and cautious.
- (2)(a) The Loco Pilot shall whistle intermittently when his engine explodes detonator(s) and take every possible caution including reduction of speed as necessary, so as to have the train well under his control and be able to stop short of any obstruction on the line;
- (b) after proceeding 1.5 kilometres from the place where his engine exploded detonator(s), if his engine does not explode any more detonator(s), he may then resume authorized speed; and
- (c) report the incident to the next station or cabin.
- (3) If in consequence of fog, storm or any other reason, the view of the signals is obstructed, the Loco Pilot shall take every possible precaution, so as to have the train well under control.
- (4) When the Loco Pilot notices a signal warning of an obstruction, except detonator(s), he shall stop his train immediately and act on advice of the person exhibiting warning signal or on the basis of obstruction noticed by him.
- (5) In case no further details of exhibition of warning signal are noticed, after stopping for one minute by day and two minutes by night to ascertain the location and/or cause of the warning, he shall proceed cautiously upto the next block station, keeping a sharp look out.
- (6) The Loco Pilot shall acquaint himself with the system of working, location of signals and other local conditions affecting the running of trains on a section or sections of the railway over which he is to work and if he is not so acquainted with any portion of the railway over which he is to work, obtain the services of a qualified railway servant who is conversant with it to assist him.

LEARNING ROAD (SR 3.78.01)

- a) Assistant Loco Pilot/Loco Pilot/motorman shall operate a minimum number of three trips initially for learning the load (LR) over which he has to work.
- b) In case a staff has not operated on a section for more than 6 months he shall be given “Road Learning Trips” to refresh his knowledge as per the schedule given below:

Sl.No.	Duration of absence	No. of trips
1	6 months to 2 years	one trip
2	Over 2 years	Two trips
3	For sidings –	6 months and above One trip

ASSISTANCE OF ENGINE CREW REGARDING SIGNALS (GR 3.83)

- i) A Loco Pilot and Assistant Loco Pilot shall identify each signal as soon as it becomes visible. They shall call out the aspect of the signals to each other.
- ii) The Assistant Loco Pilot shall assist the driver in exchanging signals when he is not otherwise engaged.
- iii) He shall call out the signals such as “MAIN HOME”, “LOOP HOME”, “MAIN STARTER”, “LOOP STARTER” etc. and also the speed limit as indicated in engineering caution indicators and speed indicator boards and repeated all the caution orders and other engineering boards.

PASSING OF DEFECTIVE APPROACHING SIGNAL AT “ON”

S/No	Circumstances	Authority and action	Speed
1.	When approaching stop signal (Outer, Home, Routing) is defective or at ON in absolute block system	a) Stop dead in rear of defective signal and proceed with great caution after taking OFF the calling on signal (if provided) up to next stop signal or	a) Proceed cautiously and be prepared to stop short of any obstructions.
		b) By obtaining private No. from SM over signal post telephone (if provided) and proceed hand signal given from cabin/station platform or c) By obtaining a written authority (Pilot Memo) T-369(3b) and proceed hand signal from engine foot board by Jamadar or d) by obtaining advance pilot memo T-369(1) from SM in rear or stopping station (Only for Mail/express/ Passenger train) and proceed hand signal from the foot of the defective signal.	b) Not exceeding 15 KMPH c) Not exceeding 15 KMPH d) Not exceeding 15 KMPH

2.	Home signal defective or at ON in auto block section	a) If “A” marker is not glowing, the authority and speed will be same as item No. (a) (b) and (c) above. (b) If “A” marker is glowing (i) Stop the train at signal for one minute by day and 2 minutes by night (ii) Give long whistle and exchange signal with guard and proceed cautiously and be prepared to stop within the half of the visible distance upto the next stop signal. Speed as above in (a) (b) & (c)	Speed as above in (a), (b) & (c) (b) Speed not exceeding 15 KMPH when view ahead is clear and 8 KMPH when view ahead is not clear or at night.
3.	No light in distant signal	Stop the train at the foot of the signal and ensure “P” marker Proceed cautiously and be prepared to stop at the next stop signal and act accordingly to the aspect of stop signal, after reaching to it.	Proceed cautiously and be prepare to stop at the next stop signal and act accordingly to the aspect of stop signal after reaching it.

NOTE:

- (i) For coaching train the advance pilot memo **T-369(1)** may be issued from last stopping station in rear.
- (ii) If the station in rear is a block hut “C” class station **T-369(1)** may be issued from the station in rear of block hut.
- (iii) If signal is defective in OFF position or in auto section, advance pilot memo **T-369(1)** shall not be issued.

PASSING OF DEFECTIVE DEPARTURE SIGNAL

S/No	Circumstances	Authority and action	Speed
1.	Starter other than last stop signal is defective or at ON in absolute block system	i) Stop dead and proceed with great caution upto the next stop signal after taking OFF calling on signal if provided. OR ii) By obtaining a written authority T-369 (3b) and proceed hand signal at the foot of the defective signal i.e. STARTER. If it is not a last stop signal	i) Proceed very cautiously and be prepared to stop short of any obstructions. ii) Not exceeding 15 KMPH upto trailing point of the yard.
2.	Starter other than last stop signal is defective or at ON in Automatic block system	i) If A marker is glowing – stop the train one minute by day and two minutes by night. Give a long whistle, exchange signal with guard and proceed cautiously and be prepared to stop within the half of the visible distance up to the next stop signal. Be guided by the aspect of the next stop signal only after reaching it. ii) If “A” marker is not glowing – authorities will be same as item No. 1 & 2 of absolute block system	i) Not exceeding 15 KMPH when view ahead is clear and 8 KMPH when view ahead is not clear or at night. ii) Same as of item No.(i) and (ii) of absolute block system
3.	Advance starter/ Last stop signal is defective in absolute block system	i) Double line – Pilot memo T-369(3b) with Private Number. ii) Single line – (a) When block token instrument is provided – token and T-369(3b) (b) When block token is not provided or defective – Paper line clear (T/C-1425 for UP & T/D -1425 for DN trains) which includes authority to pass last stop signal at ‘ON.’	Normal speed if there is no point beyond last stop signal. Hand signal to be given, if there is any point beyond last stop signal.
4.	Advance starter/ last stop signal is defective or at ON in Auto block system (Double Line)	i) When the last stop signal working as manual stop signal or semi-auto stop signal as manual stop signal (A marker not glowing) is defective or at ON driver shall receive - T/A-912	Not exceeding 15 KMPH where view ahead is clear and 8 KMPH where view ahead is not clear and at night upto the next stop signal.

Advance starter/last stop signal is defective or at ON in auto block system (D/L) but there is no intervening signal between two consecutive stations.

- A) **T/D-912**
B) Caution Order If Required.

25 KMPH for the first train.
Subsequent trains may run at normal speed. The driver should exercise great caution while approaching level crossing.

On S/L auto system when direction of traffic is established but last stop signal is defective

- i) Starting order **T- 512**
ii) **T/A-912**
iii) Proceed hand signal from foot of the defective last stop signal if there is point beyond last stop signal

15/8 KMPH as per visibility upto the next auto stop signal, then obey the aspect of auto signal ahead.

CONFLICTING SIGNAL

SI/ No.	Nature of defective signal	Action in absolute block system	Action in auto block system
1	Warner is showing OFF while outer or Home is ON	Stop the train and shall be piloted in	Not provided
2	Distant signal shows proceed aspect but home signal shows Yellow with root indicator	Stop the train at the first stop signal and shall be piloted in	Not provided.

BOBBING OR FLICKERING SIGNAL

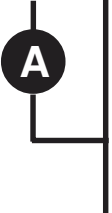
SI/ No.	Nature of defective signal	Action in absolute block system	Action in auto block system
1.	In case of any colour light signal is found flickering or bobbing	i) Treat the signal to be showing its most restrictive aspect ii) Stop the train and obey the steady aspect which remains steady for 60 seconds iii) If so then obey that aspect otherwise the train shall be piloted.	Stop the train and observed the aspect which remain steady for 60 seconds, if so, then obey that aspect, otherwise obey the rule to pass the auto stop signal at danger or at ON
2.	If colour light signal shows more than one aspect at a time	Stop the train and treat the signal to be showing, it's most restrictive aspect. Don't pass it unless piloted with proper authority	Obey most restrictive aspect out of the aspects displayed
3.	If colour light signal showing no light or white light	Stop the train and obey most restrictive aspect of that signal post	Stop the train and obey the most restrictive aspect of the signal post.

DEFECTIVE STOP SIGNAL IN AUTOMATIC BLOCK SYSTEM/DUTIES OF DRIVER TO PASS AUTOMATIC STOP SIGNAL AT “ON” (GR 9.02)

- i) When a Loco Pilot finds an auto stop signal with an “A” marker at ON, he shall draw his train to stop in rear of the signal as close as possible.
- ii) After stopping one minute by day and two minutes by night if the signal continue to be at ON, the Loco Pilot shall sound one long whistle and exchange signal with guards, then proceed ahead as far as line is clear towards the next stop signal in advance, exercising great caution so as to stop short of any obstruction.
- iii) After the signal has been passed at ON the Loco Pilot shall so regulate his speed so as to be able to stop within half of the visible distance. The speed shall not exceed 15 KMPH even in normal conditions of visibility until he reaches the next stop signal. When the view ahead is not clear due to curvature or fog or rain or dust or storm, engine working the train by pushing or other causes the line cannot be seen clearly the speed shall not exceed 8 KMPH.
- iv) After proceeding cautiously upto the foot of the next signal only the Loco Pilot shall act upon its indication, even if the signal is found to be at “OFF” from a distance.
- v) After passing an auto stop signal at ON, the Loco Pilot of the following train shall ensure that a minimum distance of 150 M or two OHE spans is maintained between his train and the preceding train or any obstruction. In case of EMU, the distance may be reduced to 75 Mtrs or one OHE span.

NOTE: On S/L auto block system in addition to the above instruction, if the telephone communication is provided near the signal, the driver shall contact with station master.

DISTINGUISHING MARKS AND SIGNS FOR SIGNALS

APPEARANCE	PROVIDED ON	DESCRIPTION
	Automatic stop signal	Letter 'A' in black on white circular disc
	Semi-Automatic stop signal	White illuminated latter 'A' against black back ground when working as an auto signal & letter 'A' extinguished when working manually.
	Colour light distant or warner signal on a post by itself	Letter 'P' in black on white circular disc
	IB stop signal	Letter 'IB' in black on white circular disc
	Calling-on signal	Letter 'C' in black on white circular disc
	Repeating signal in semaphore signaling territory	Letter 'R' in black on white circular disc
	Repeating signal in colour light signaling territory	White illuminated latter 'R' against black back ground.
	Gate stop signal	Letter 'G' in black on yellow circular disc.
	Gate stop signal in auto block signaling territory	Letter 'G' in black on yellow circular disc & White illuminated latter 'A' against black back ground.

STATION WORKING

STATION AND TRACK WORKING

INTRODUCTION

It is quite essential that movement of a train on a particular track should be safe and for this purpose various working systems are adopted for reception and dispatch of trains at or from a station. The stationmaster shall be responsible to see that working of a station is carried out in accordance with rules and regulations in force.

During working of a train every driver & assistant driver should know about the working system of the station, signals and any special and also station working rules of stations.

CLASSIFICATION OF STATION (GR 1.03)

BLOCK STATION:

Block stations are those at which the Loco Pilot must obtain an authority to proceed under the system of working to enter into the block section with his train. It is of three classes:

i) **“A” CLASS STATION:**

Where line clear may not be given for a train unless the line on which it is intended to receive the train is clear for at least 400 meters beyond the home signal or upto the starter.

ii) **“B” CLASS STATION:**

Where the line clear may be given for a train before the line has been cleared for the reception of the train within the station section.

iii) **“C” CLASS STATION (BLOCK HUT)**

Class “C” station are block huts where line clear may not be given for a train, unless the whole of the last preceding train has passed complete at least 400 meters beyond the home signal and is continuing its journey. This will also include an intermediate block post (I.B.P).

NON-BLOCK STATION OR CLASS “D” STATION:

It means class D station. These are stopping places which are situated between two consecutive block stations and do not form the boundary of any block section.

NOTE: SIGNAL AT CLASS “D” STATION:

At class “D” station a train may be stopped in such a manner as may be authorized by special instruction.

At a class D station a PH marker post shall be provided by the side of the track on either approach to the station, at a distance not less than 400 meters from the centre of the platform.

SPECIAL CLASS STATION:

Any station, which cannot be worked under A, B, C & D class conditions is termed called Special Class station.

NOTE: The classification of each station on S. E.C. Railway shall be given in the working timetable.

DEMARCATON OF STATION LIMIT

STATION LIMIT

It means the portion of a railway which is under the control of stationmaster and is situated between the outmost signals of the station or as may be specified by special instructions.

DEMARCATON-

Section	Signalling Territory	Portion of Railway which lies between
Single line (absolute block system)	TALQ MAST/MLQ/MOMA	Between two outer most signals Between two Distant signals
Single line (Auto signalling system)	Auto signalling section	Between UP & DN first stop signals
Double line (Absolute Block system)	MAST/TAST/MLQ (Single distant) MAST/MLQ/MOMA (Double distant)	Between the outer most signal on each direction separately. Between inner distant and the last stop signal in each direction separately
Double line (Auto Section)	Auto section	Between the first automatic signal in rear of the home signal and the advance starter or Up to the outermost point at the trailing end where no advance starter is provided.

NOTE: At a station where the auto signal falls within the jurisdiction of the Station in rear, the station limit shall be reckoned from the home signal

STATION SECTION:

- Station section is a part of station limit.
- It exists only in B class station
- It is situated between two consecutive block sections
- Under the direct control of stationmaster on which obstruction is freely permitted.

DEMARCATON OF STATION SECTION

	Signalling	Commence from	Ends at
D/L	TAST MAST MLQ	Home signal - At outer most facing point or - At BSLB where there is no point or 1 st point is trailing point	- Last stop signal - Last stop signal - Last stop signal

S/L	TAST	i) Advance starter if provided	i) At the advance starter for opposite direction.
		ii) At the SLB of opposite direction if there is no advance starter or	ii) At the SLB if Advance starter is not provided
		iii) At the home signal if there is no Advance starter or SLB	iii) At the home signal of opposite direction
		iv) Outermost facing point if there is no advance starter or SLB	iv) At the outer most facing point if there is no advance starter or SLB in opposite direction.
S/L	MAST MLQ	i) Advance starter if provided	i) At the advance starter for opposite direction.
		ii) At the SLB of opposite direction if there is no advance starter or	ii) At the SLB if Advance starter is not provided
		iii) Outermost facing point if there is no advance starter or SLB	iii) At the outer most facing point if there is no advance starter or SLB in opposite direction.

DEMARCATON OF BLOCK SECTION:

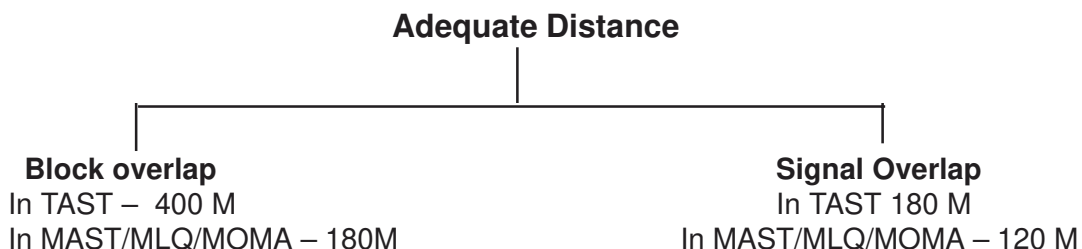
Class of station	Paint at the station in rear Block section commence	Point at the station in advance at which Block section ends
“A” class S/L & D/L section both in TAST	At the last stop signal	At the starter signal
“B” class D/L section in TAST	At the last stop signal	At the home signal
“B” class D/L section in MAST/MLQ	At the last stop signal	At the outermost facing point or at BSLB
“B” class S/L on TAST	i) At the advance starter if provided or ii) At the SLB or iii) At the home signal of the opposite direction if there is no advance starter or SLB iv) At the outermost facing point, if there is no advance starter or SLB or Home signal.	i) At the advance starter of the opposite direction if provided or ii) At the SLB for opposite direction if advanced starter not provided or iii) At the home signal if no advance starter or SLB iv) At the outermost facing point if there is no advance starter or SLB.

Class of station	Paint at the station in rear Block section commence	Point at the station in advance at which Block section ends"
"B" Class S/L in MAST/MLQ	i) At the advance starter if provided or ii) At the SLB if no advance starter or iii) At the outermost facing point if there is no advance starter or SLB	i) At the advance starter for opposite direction or ii) At the SLB for the opposite di- rection if no advance starter or iii) At the outermost facing point if there is no advance starter or SLB for opposite direction.
"C" class in TAST/ MAST/ MLQ on both S/L and D/L	At the home signal	400 M beyond the home signal.

ADEQUATE DISTANCE, BLOCK BACK AND BLOCK FORWARD

ADEQUATE DISTANCE:

It means the distance sufficient to ensure safety. There are different kinds of adequate distances kept for different purposes. The adequate distance is kept for granting line clear for a train is "BLOCK OVERLAP," the adequate distance which is kept for lowering home signal is "SIGNAL OVERLAP," the adequate distance which is required to bring the train to a stop is "BRAKING DISTANCE." In auto section the portion of line which is required to be clear for change of aspect of the previous auto signal is called adequate distance. All the engineering warning boards and indicators also are placed at an adequate distance.



BLOCK OVERLAP:

It means adequate distance beyond the 1st stop signal of a station. This portion is to be kept clear for granting the line clear. This adequate distance shall not be less than 180M in MAST/MLQ/MOMA and 400 M in TAST.

NOTE: In MOMA at "C" class station it will be 400M

SIGNAL OVER LAP

It means the adequate distance kept beyond the starter signal on double line and the last trailing point in S/L for lowering of home signal of the station. This adequate distance shall not be less than 180M in TAST and 120 M in MAST/MLQ/MOMA.

BLOCK BACK:

It means to dispatch a message from a block station intimating to the block station immedi-

ately in rear on a double line, or to the next block station on either side on a single line that the block section is obstructed or is to be obstructed.

BLOCK FORWARD

Means to dispatch a message from block station of double line intimating to the block station immediately in advance the fact that the block section in advance is obstructed or is to be obstructed.

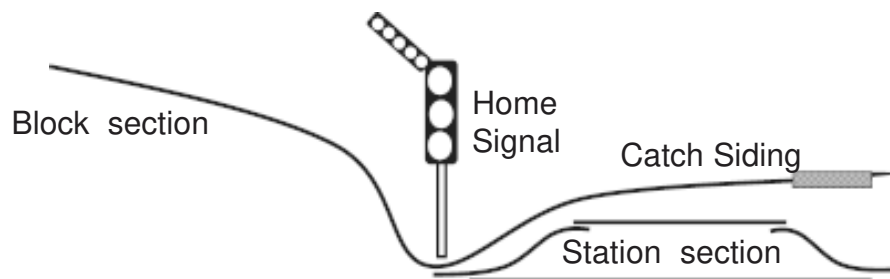
SIDINGS (GR 3.50)

For the purpose of reception and departure of the train in safe manner and to keep sick wagon and to ensure safety of block section and station section, different siding are provided at a station.

Except when the setting of switch, points is indicated by a fixed signal, a driver shall not pass over these points whether facing or trailing unless these are manned and proper hand signal exhibited.

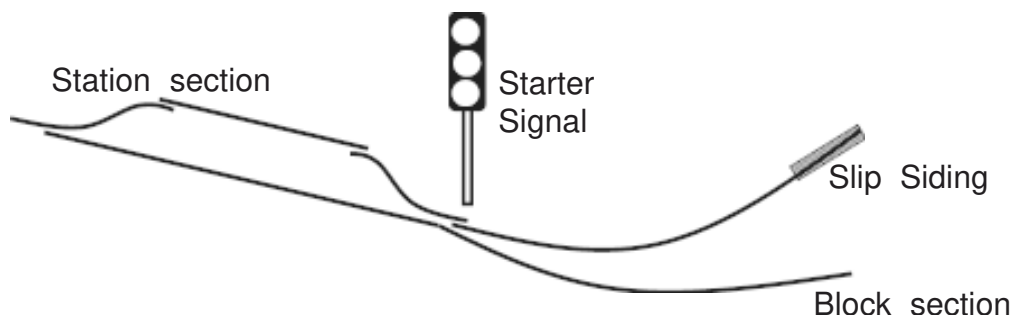
CATCH SIDING:

- (i) This is a safety siding provided to protect the station section/yard.
- (ii) It prevents unauthorized movements from the block section.
- (iii) It is provided where the station yard is in falling down gradient.
- (iv) It is interlocked with home signal in the approaching end and spring loaded in trailing end.
- (v) Normal setting of the point is always for the siding.



SLIP SIDING

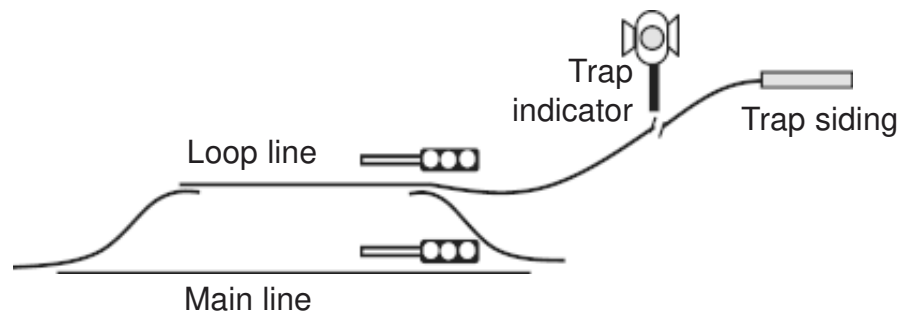
- i) This is a safety siding provided to protect the block section
- ii) It prevents unauthorized movements of vehicles from station yard.
- iii) It is provided where block section is in falling down gradient
- iv) It is interlocked with starter signal and the trailing end, is spring-loaded.
- v) The normal setting of the point is always for the siding.



NOTE: Backing is not permitted after passing spring-loaded point unless Authorized to do so by stationmaster through a written memo.

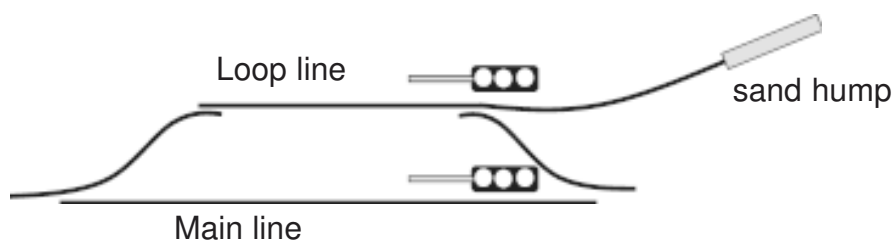
TRAP SIDING

- i) This is an extension of running line, ends in buffer stop.
- ii) It is provided to prevent the accidental escape of vehicle from the siding and fouling the running line.



SAND HUMP

- i) It is an effective substitute for signal overlap
- ii) It is provided to facilitate simultaneous reception and dispatch of a train from the station.



PERMANENT WAY

INTRODUCTION

The finished or the completed track of a railway line over which trains move are commonly known as permanent way. It is a combination of the facing and trailing points, fouling marks, interlocking, isolation, track circuit, axle counter and different types of indicator boards. It essentially consists of the following three parts.

- (i) RAILS
- (ii) SLEEPER
- (iii) BALLAST

These are fixed with each other by means of fishplates; now days rails are welded to increase the length. Rails are rested on sleeper, in turn rest on ballast, which spread over the formation ground preparation for the railway track.

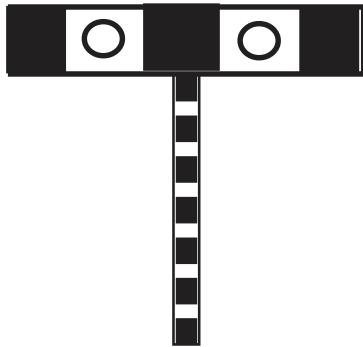
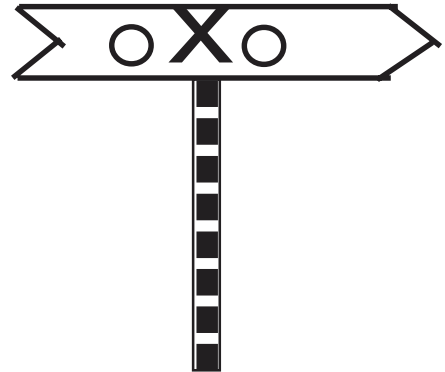
Inspector of way or works is responsible for the construction, maintenance of permanent way, points, bridges or other works connected there with.

BOARDS IN BLOCK SECTION

To guide the Loco Pilot to take action as the conditions deserve, different kinds of boards and engineering indicators are provided in the block section. The appearance, the indications that they provide, where and why are they provided are given below:

CAUTION INDICATOR BOARD

- a) This board indicates the Loco Pilot for permanent, semi-permanent and temporary speed restrictions.
- b) It is a horizontal board fish tailed at left and pointed at right, painted in yellow and black and fixed on a post painted black and white alternately.
- c) This board shall be fixed at a distance of 1200M from the point of restriction.
- d) Two yellow lights will be displayed for temporary restriction and no light for permanent restriction.

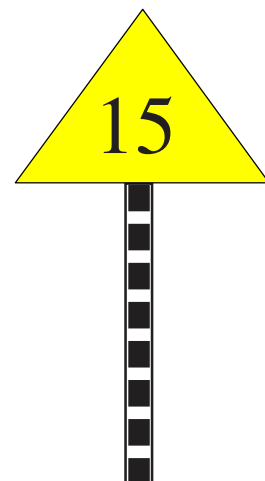


STOP INDICATOR

- i) This board indicates to the Loco Pilot to stop the train.
- ii) This is horizontal board painted in red and white strips, fixed on a post-painted black and white alternately.
- iii) This board shall be provided at a distance of 30M from the point of restriction.
- iv) A signalman is posted for hand signalling the train to pass the indicator.
- v) As soon as the train stops the signalman shall take the signature of the Loco Pilot on the stop order book (ED 9.39 for UP train and ED 9.40 for Dn trains) and allow the train to proceed by showing a green hand signal.
- vi) The Loco Pilot must obey the green hand signal and proceed accordingly unless the full train is passes the obstruction.
- vii) This board shall display two red lights horizontally.

SPEED INDICATOR

- i) This is a yellow equilateral triangular board.
- ii) This board indicates to the Loco Pilot about the speed to be observed.
- iii) This board shall be placed on both permanent and temporary restriction and will be illuminated at night for temporary restriction only.
- iv) It will be placed 30 M before the site.

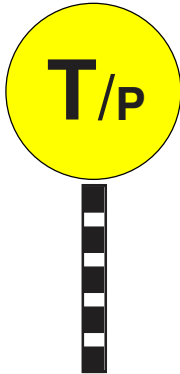
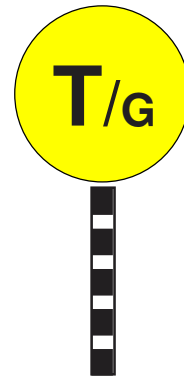


TERMINATION INDICATORS

These indicators consist of yellow disc and bearing letters "T/P" and "T/G" in black.

“T/G” BOARD

- i) This board indicates the Loco Pilot of goods train a point from where normal speed may be resumed.
- ii) This is yellow circular board with letter “TG” in black is fixed on a post painted in black and white.
- iii) This shall be provided at a distance of 700M beyond the point of restriction or length of the longest goods train running on the section.
- iv) For temporary restriction it shall be illuminated at night by fixing a hand signal lamp.



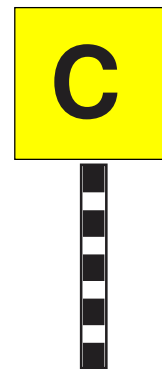
“T/P” BOARD

- i) This board indicates the Loco Pilot of passenger train, a point from where the normal speed may be resumed.
- ii) It is a yellow circular board with letter “TP” in black is fixed on a post-painted in black and white alternately.
- iii) This board shall be provided at a distance of 550 M beyond the point of restriction or length of the longest passenger train running on the section.
- iv) For temporary restriction it shall be illuminated by night by fixing a hand signal lamp.

CAUTION & TERMINATION BOARD AT VULNERABLE LOCATION

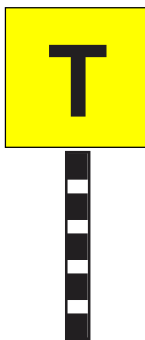
“C” BOARD –

- i) A special type “C” board shall be provided in such of the bridges where during monsoon water level may touch the danger level mark or at other vulnerable locations, danger may lark suddenly during monsoon.
- ii) It is a square board painted in yellow with letter “C” in black fixed on a post-painted black and white alternately.
- iii) It shall be provided at a distance of 1200M in respect of the point from which the vulnerable bridges or locations starts. It need not be lit-up at night.
- iv) A series of “C” boards shall be provided at an interval of 800 M to remind the driver that the vulnerable length has not been passed completely, where the distance between “C” & “T” boards exceeds 1.6kms.



A termination board of same type as “C” board but bearing the letter “T” instead of “C” shall be fixed at a distance of the longest train running in the section, when vulnerable location is over to indicate the driver for resuming the normal speed.

On observing “C” board Loco Pilot shall be extra vigilant and be prepared at any moment to reduce the speed or to stop when ever any signal is received from any stationary watchmen, patrolman or gang man or he himself observes any danger to the track.



“T” BOARD

A termination board of the same description as the “C” board but bearing a letter “T” in place of “C” shall be fixed at a distance beyond the vulnerable bridges and locations equal to the length of the longest train running on the section.

On single line sections, the 1st and last boards may have the letter “C” or “T” painted on opposite faces so that no separate “T” board need be provided. Intermediate boards if any shall have the letter “C” painted on both faces.

WHISTLE BOARDS (SR 4.50.02)

Whistle boards are provided with a view to indicating to Loco Pilot of an approaching train to sound the engine whistle to warn gateman, gang man and staff working on track or road users at the level crossing or site of work or curve or cuttings.

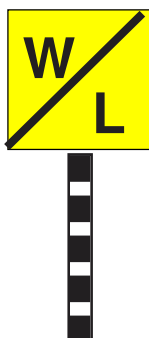
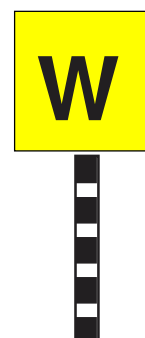
The driver shall sound the whistle/Horn continuously upto level crossing/site of work.

Whistle boards are of two types:

- i) “W” board
- (ii) “W/L” board

“W” BOARD:

“W” board is provided before curve, cuttings and when visibility of line ahead is limited up to **800M** or less. It shall also be provided on approaches to every tunnel at a distance of 800M from it in broad gauge.



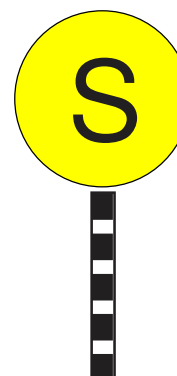
“W/L” BOARD

It shall be provided on the approaches to all unmanned “C” class level crossings and also to those manned level crossings where the view not clear on either side from a distance

600m and those which have normal position open to road traffic without interlocking and protection by signals. This board shall be provided at a distance of 600m from the level crossings.

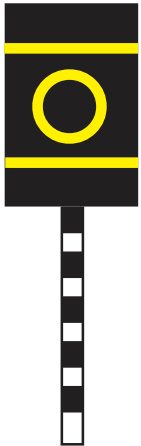
“S” MARKER

- i) It is an indicator to indicate the Loco Pilot for points at the outlying sidings out side the station limit.
- ii) It shall be placed 30 m in rear of the facing points of the outline siding.
- iii) A speed indicator if any shall be fixed below the “S” marker to indicate the speed restriction.
- iv) It shall be preceded by a caution indicator fixed at 800 M in rear of “S” marker, in addition, the speed shall be given on the caution indicator board also.
- v) A “TP” board at a distance of 550 M and a “TG” board at a distance of 700 M shall be placed beyond the outlaying siding point to indicate the driver for resuming the normal speed.



WARNING BOARDS(OPM-1.13)

Warning boards are provided with a view to warn a Loco Pilot of an approaching train that he is approaching the first stop signal, so as to enable him to bring his train under control and be prepared to stop at the signal, unless the "OFF" aspect of the signal has been sighted by him. There are two kinds of warning boards-1) Goods warning board & 2) Passenger warning board.



GOODS WARNING BOARD

This board is provided at a distance of 1.4 KM in rear of the first stop signal or a Gate or IB stop signal which is not prewarned by a Warner/Distant signal located at a distance of not less than 1.4 KM. It is applicable to all goods trains and also to all passenger carrying trains on high speed section. (i.e. where the maximum permissible speed is above 100 KMPH).

Note- In sections provided with double distant and in auto sections no warning board is provided.

PASSENGER WARNING BOARD

Passenger warning boards are provided at a distance of 1 KM in rear of the first stop signal I.B. stop signal or a Gate stop signal which is not prewarned by the provision of a Warner/Distant signal located at a distance not less than 1 KM from it. where the maximum permissible speed is upto 100 KMPH. Where the sectional speed ranges from 50 to 72 KMPH on broad gauge, passenger warning boards to be provided only at those stations where the first stop signal is not visible from an adequate distance of 800 m.



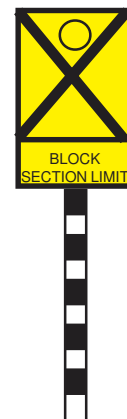
BOARDS IN STATION SECTION



Mainly 2 kinds of boards can be seen in station section to demarcate the block section and station section. They are BSLB and SLB.

SHUNTING LIMIT BOARD

- i) It is a board to demarcate the area of block section and station section in "B" class single line section under the absolute block system.
- ii) It shall be placed at an adequate distance not less than 400 meters from the 1st stop signal in TAST and 180 M in MAST/MLQ where there is no advance starter.
- iii) It shall bear the word "SHUNTING LIMIT" on the side which faces this station and shall be fitted with a lamp.
- iv) It shall mark the limit up to which shunting may be permitted.



BLOCK SECTION LIMIT BOARD (BSLB)

- i) It shall be provided on D/L "B" class station under absolute block system equipped with MAST/MLQ signalling system where there is no points or 1st point is trailing point at the approaching end.

- ii) It shall be placed at a distance not less than 180 M in advance of the home signal and protect the fouling mark of outermost trailing point.
- iii) It shall bear the word "BLOCK SECTION LIMIT" on the side, which faces the station and provided to demarcate the block section.

SPEED RESTRICTION

Speed restrictions are classified as permanent, semi-permanent and temporary.

PERMANENT SPEED RESTRICTION

The speed restriction which is likely to continue over a long period with no prospect of early removal is known as **permanent speed restriction**. This restriction shall be notified in working time table to indicate the Loco Pilot about the location of such restrictions. A caution and a speed indicator shall be provided at 1200m & 30m in rear of such restrictions. T/P & T/G boards also shall be provided at 550m & 700m respectively from the point at which the restriction ends. The engineering indicators provided at such restriction shall not be eliminated at night.

SEMI-PERMANENT SPEED RESTRICTION

The kind of speed restriction which although of limited duration, are likely to extend for the full period of the working time table are termed as 'semi permanent speed restriction.' This shall also be notified in 'working time table'. The rules relating to permanent speed restriction shall equally apply to semipermanent speed restrictions also.

TEMPORARY SPEED RESTRICTION:

This restriction is purely short term in nature. Normally the duration of such restriction is from sunrise to sunset or for a few days only. This restriction shall not be mentioned in working time table but shall be indicated in the caution order. The indicators, provided for temporary restrictions shall be illuminated at night.

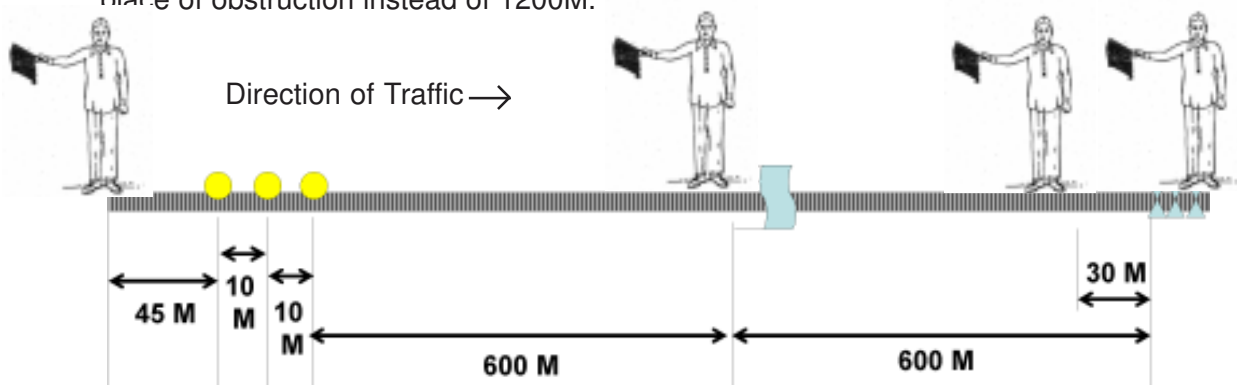
ENGINEERING PROTECTION (GR 15.09)

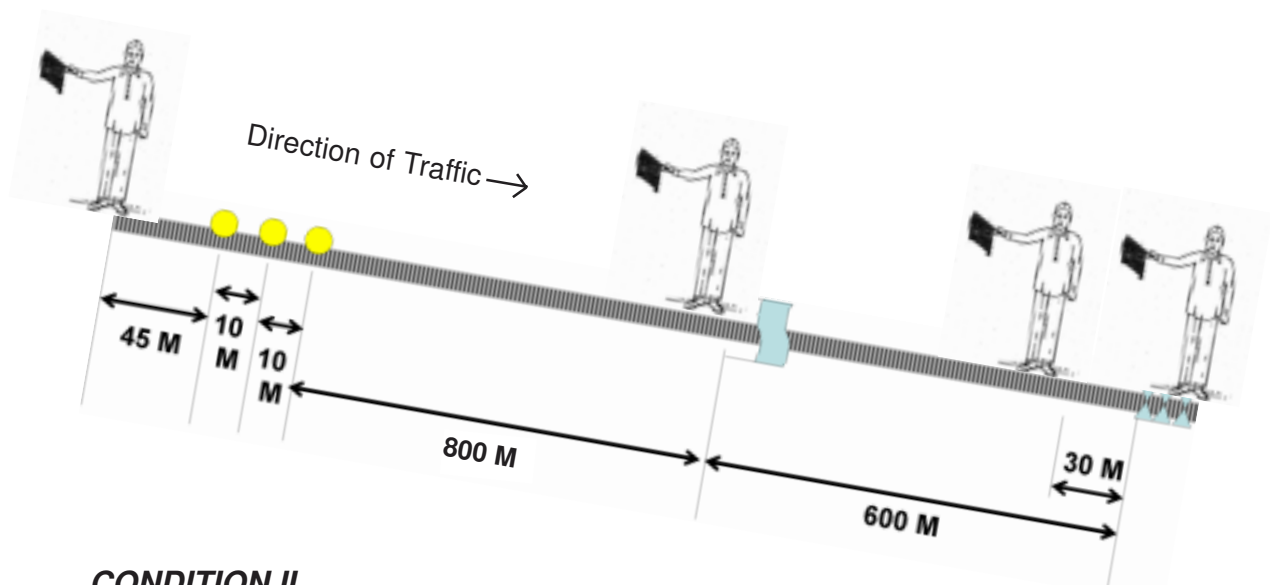
CONDITION (I)

WHEN THE TRAIN IS REQUIRED TO STOP AND THE RESTRICTION IS LIKELY TO LAST FOR A DAY OR LESS (ABSOLUTE SYSTEM)

- a) A banner flag shall be placed at a distance of 600 M on the B/G and 3 detonators shall be placed at 10M apart at a distance of 1200M from the place of obstruction i.e. 600M from the banner flag.
- b) A stop hand signal shall be exhibited at a distance of 30M from the place of obstruction at the banner flag and at a distance of 45M from the 3 detonators.
- c) A proceed hand signal shall be exhibited to the driver for resuming the normal speed after the train has passed the place of obstruction.

NOTE: In 1 in 300 descending gradient 1st detonator should be placed at 1400M from the place of obstruction instead of 1200M.

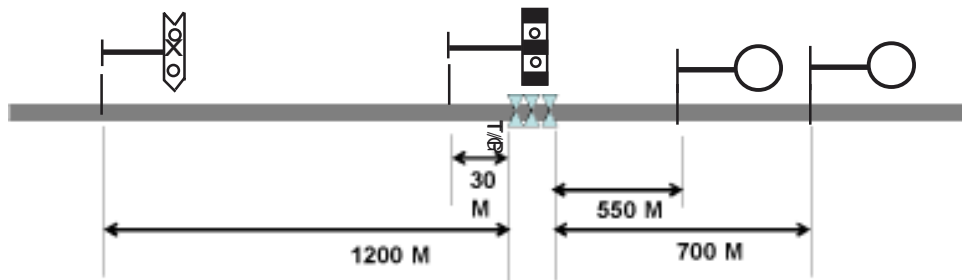




CONDITION II

WHEN THE TRAIN IS REQUIRED TO STOP AND THE RESTRICTION IS LIKELY TO LAST FOR MORE THAN A DAY

a) A stop indicator shall be provided at a distance of 30M from the place of obstruction. In addition termination indicator (i.e. TP or TG) shall be provided at the place from where the Loco Pilot may resume normal speed.

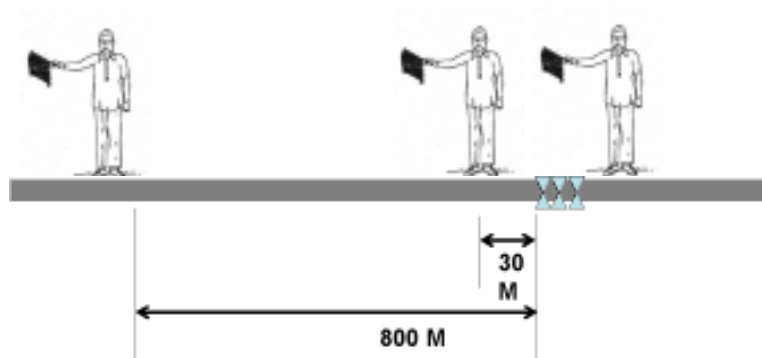


CONDITION III

WHEN A TRAIN IS NOT REQUIRED TO STOP AND THE RESTRICTION IS LIKELY TO LAST ONLY FOR A DAY OR LESS:

- Proceed with caution hand signal shall be exhibited at distance of 30M and again at a distance of at least 800M from the place of obstruction. This distance of 800 M shall be suitably increased if required.
- The railway servant at the place of obstruction shall give proceed hand signal to indicate the driver when he may resume normal speed after the train has passed the place of obstruction.

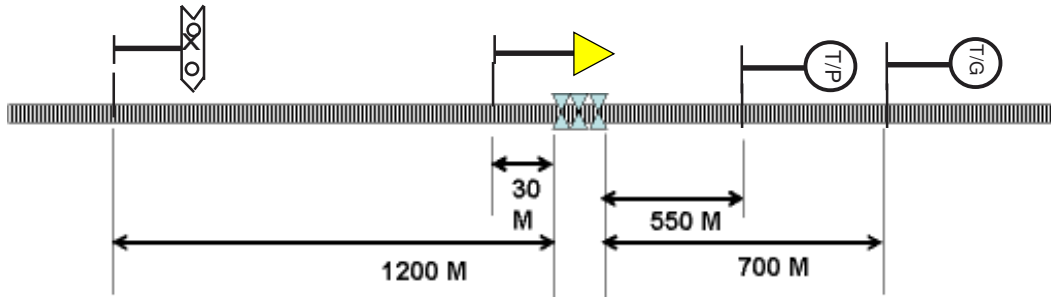
NOTE: In 1 in 300 descending gradient 800M shall be increased to 1200M.



CONDITION IV

WHEN THE TRAIN IS NOT REQUIRED TO STOP AND THE RESTRICTION IS LIKELY TO LAST FOR MORE THAN A DAY

- a) A speed indicator shall be provided at a distance of 30M from the place of obstruction and again a caution indicator, at a distance of 1200M from the place of obstruction.



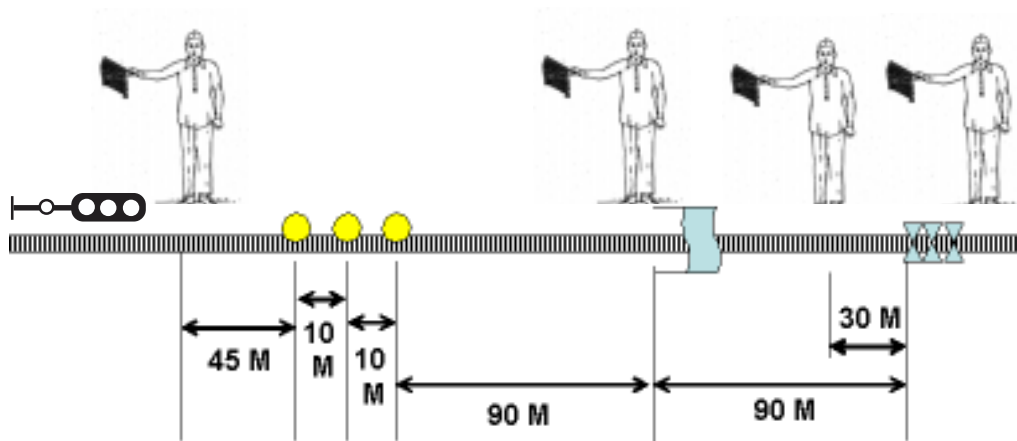
- b) In addition termination indicators shall be provided at the place from where the Loco Pilot may resume normal speed.

CONDITION V

WHEN THE TRAIN IS REQUIRED TO STOP IN AUTOMATIC BLOCK SYSTEM

- a) An automatic stop signal shall be put in ON position.
b) Banner flag shall be placed at a distance of 90M from the place of obstruction and 3 detonators each 10M apart at a distance of 180M from the place of obstruction.
c) A stop hand signal shall be shown at a distance of 30M from the place of obstruction after the banner flag at a distance of 45M from the 3 detonators.
d) A railway servant at the place of obstruction shall give proceed hand signal to indicate the Loco Pilot when he may resume normal speed after the train has passed the place of obstruction.

NOTE: If auto signal is not secured at ON or the distance is more than 1200M, the protection shall be like item No.1 in absolute block system.



BOARDS PROVIDED AT SPEED RESTRICTION ZONE AND THEIR INDICATIONS

The following kinds of boards are provided at speed restriction zones:-

A speed restriction zone is preceded by a **CAUTION INDICATOR** board at 1200 meters and by a **SPEED INDICATOR** board at 30 meters in rear of the zone and the zone is again complemented by the provision of "T/P" and "T/G" boards located at 550 meter and 700

meters beyond the point at which the restriction zone terminates. **STOP INDICATOR** board is provided in place of speed indicator if train is required to be stopped.

INDICATION

The caution indicator board indicates to the Loco Pilot of an approaching train that he is approaching a speed restriction zone so as to enable him to bring his train under control to observe the speed restriction which is forthcoming. The restriction of speed which is to be observed at the zone also is indicated on the caution indicator board.

The speed indicator board indicates to the Loco Pilot what speed he has to observe at the zone and "T/P" boards indicates to the Loco Pilot of passenger train and "T/G" to the Loco Pilot of goods train from where they can resume their normal speed.

GREEN NOTICE(SR 15.08.03)

It is an advance information to all concerned on the ground of safety so as to arrange the train services to suit the block or speed restrictions required for engineering or signalling work.

In case of other works affecting running lines and involving speed restriction block shall not be imposed unless all concerned have been notified by a circular issued on green paper known as 'GREEN NOTICE'

DURATION OF GREEN NOTICE (CURRENCY OF GREEN NOTICE)

Validity of Green Notice - 7 days to 3 months from the date of its issue.

CAUTION ORDER

A caution order is the guideline issued from a block station, in the form of a list or chart in a prescribed form solely meant for the purpose to the Loco Pilot and Guard of a train, where in consequence of the line being under repair or for any other reason, special precaution are necessary to be observed between stations.

A Loco Pilot must not start his train from a originating station or a notice station without a caution order. If no caution is required to be observed in section, a 'NIL' caution order is to be issued.

CONDITION FOR ISSUING CAUTION ORDER

Given below are few circumstances under which a caution order shall be issued.

- i) When in consequence of the line being under repair or for any other reason special precaution is necessary.
- ii) To look out for a train which is overdue.
- iii) When any interlocked points, derails etc. go out of order or become defective in any way.
- iv) When a material train works between stations etc.

TYPES OF CAUTION ORDER

Caution orders are of three types, namely -

- i) Caution order-T/409
- ii) Nil caution order -T/A 409
- iii) Reminder caution order-T/B 409

CAUTION ORDER-T/409

This is issued either from a train originating station or from a notice station. In this caution order, all speed restrictions are written serially which are to be observed through out the journey.

NIL CAUTION ORDER -T/A 409

This caution order is issued either from a train originating station or from a notice station when there is no caution to be observed through out the journey.

REMINDER CAUTION ORDER-T/B 409

This caution order can be issued from any block station in case of any emergency when a restriction or special precaution which is to be observed is not written on T/409 or as a reminder caution order to attract the attention of a Loco Pilot.

TERMINOLOGY OF A CAUTION ORDER

Following are the terminology used in a caution order with their meanings:

Sl. No.	Abbreviation	Explanation
1.	O.E.S	OBEY ENGINEERING SIGNAL
2.	C.T.R	COMPLETE TRACK RENEWAL
3.	T.R.R.	THROUGH RAIL RENEWAL
4.	L.W.R.	LONG WELDED RAIL
5.	C.W.R	CONTINUOUS WELDED RAIL
6.	T.S.R.	THROUGH SLEEPER RENEWAL
7.	S.W.R	SHORT WELDED RAIL
8.	U/S	UNSERVISABLE SLEEPER
9.	S.E.J.	SWITCH EXPANSION JOINT
10.	U.C.	UNCONSOLIDATED (LESS BALLAST)
11.	S.D.I.R	STOP DEAD IF REQUIRED
12.	P.B.R.	PULLING BACK OF RAILS
13.	R.A.T	RAILWAY AFFECTED TANK
14.	T.F.R	THROUGH FITTINGS RENEWAL
15.	T.B.T.R	THROUGH BRIDGE TIMBER RENEWAL
16.	T.W.R	THROUGH WELD RENEWAL
17.	T.T.R	THROUGH TURNOUT RENEWAL
18.	T.B.R	THROUGH BALLAST RENEWAL
19.	B.C.M	BALLAST CLEANING MACHINE
20.	D/S	DEEP SCREENING
21.	PQRS	PLASSER QUICK RELAYING SYSTEM
22.	T.R.T	TRACK RELAYING TRAIN
23.	R.O.B	RAIL OVER BRIDGE
24.	I.M.R	IMMEDIATE REMOVAL(WITHIN 2.3 DAYS) FOR WHEEL BURNT RAILS

PRESERVATION OF CAUTION ORDER

Caution orders are prepared in triplicates. The first foil is issued to the Loco Pilot, the 2nd to the Guard and the 3rd foil is preserved as record. The record foils of the caution orders shall be preserved for a period of 12 months after issue.

DISPOSAL OF CAUTION ORDER

The Loco Pilot and the Guard shall hand over the caution order to the Loco/Traction Foreman (RSO) and station master respectively at the end of the journey along with other train papers.

SYSTEM OF WORKING

INTRODUCTION

The system under which trains are worked between stations is called system of working. All trains working between stations shall be worked on one of the following systems :-

- i) The absolute block system
- ii) The automatic block system
- iii) The following train system
- iv) The pilot Guard System
- v) The train staff and ticket system
- vi) The one train only system

The absolute block system and the Automatic block system normally shall be used on every railway, except any railway or portion of a railway on which the use of any other system of working mentioned above may be sanctioned under special instruction subject to the conditions applicable to each system.

In S.E.C. Railway the Absolute Block System is in force in all sections, except the following

S/No.	Section	System of working in force
1.	Abhanpur – Rajim	One train only system
2.	Abhanpur – Dhamtari	One train only system
3.	Nainpur- Mandla fort	One train only system
4.	Raipur-Telibandha	One train only system

THE ABSOLUTE BLOCK SYSTEM

ESSENTIALS OF THE ABSOLUTE BLOCK SYSTEM (GR 8.01)

- i) Where trains are worked on the absolute block system
 - a. No train shall be allowed to leave a block station unless line clear has been received from the block station in advance.
 - b. On D/L such line clear shall not be given unless the line is, clear not only up to the 1st stop signal at the block station at which such line clear is given but also for an adequate distance beyond it.
 - c. On single line such line clear shall not be given unless the line is clear of trains running in the same direction, not only up to the 1st stop signal at the block station at which such line clear is given but also for an adequate distance beyond it and is clear of trains running in the direction towards the block station to which such line clear is given.
- ii) The adequate distance beyond the first stop signal shall not be less Than 400 m, in TAST and 180 m, in MAST/MLQ.

AUTOMATIC BLOCK SYSTEM

ESSENTIALS OF AUTO-BLOCK SYSTEM ON DOUBLE LINE (G.R.9.01)

- i) Where, trains on D/L are worked on the automatic block system.
 - a. The line shall be provided with continuous track circuiting or Axle counters.
 - b. The lines between two adjacent block stations may, when required be divided into a series of automatic block signalling sections, each of which is the portion of the running line between two consecutive stop signals and the entry into each of which is governed by a stop signal.

- c. The track circuit or axle counters shall so control the stop Signal governing the entry into an automatic block signalling section i.e.
 - i) The signal shall not assume an 'OFF' aspect unless the line clear is not only up to the next stop signal in advance but also for an adequate distance beyond it.
 - ii) The signal is automatically placed to "ON" as soon as the train passes it.
- II) The adequate distance beyond the next stop signal shall not be less than 120 Mtrs.

ESSENTIALS OF AUTO BLOCK SYSTEM ON SINGLE LINE (GR 9.03)

- i) The line shall be provided with continuous track circuiting or axle counters
- ii) The direction of traffic shall be established only after line clear has been obtained from the block station in advance.
- iii) The train shall be started from one block station to another only after the direction of traffic has been established.
- iv) It shall not be possible to obtain line clear unless the line is clear, at the block station from which line clear is obtained, not only up to the 1st stop signal but also for an adequate distance .
- v) The line between two adjacent block station may where required be divided into two or more automatic block signalling section by providing stop signal.
- vi) After the direction of traffic has been established the movement of trains into, through and out of each automatic signalling section shall be controlled by the concerned auto stop signal and the said auto stop signal shall not assume OFF position unless the line clear is up to next automatic stop signal, if the next stop signal is a manual stop signal, then line is clear from an adequate distance beyond it.
- vii) All stop signals against the direction of traffic shall be at "ON"
- viii) The adequate distance refer to shall not be less than 180 m.

MINIMUM EQUIPMENTS OF SIGNAL IN AUTOMATIC BLOCK TERRITORY ON SINGLE LINE (GR 9.04)

- a) Manual stop signals at a station -i) a Home ii) a Starter.
- b) An automatic signal in rear of the home signal of a station.

NOTE Under approved special instruction the automatic stop signal may be Dispensed with.

PROTECTION OF TRAIN STOPPED IN AN AUTOMATIC BLOCK SECTION (GR 9.10)

In case of accident, failure or obstruction, if the train cannot proceed in auto section, driver shall

- i) Sound four short whistle (o o o o)
- ii) Use flasher light to stop an approaching train
- iii) Display danger hand signal
- iv) Use flare fusee if provided to stop an approaching train
- v) Use the short circuiting clip on the adjacent track to control the movement of approaching train if required.
- vi) When a train stops in automatic block signalling section, the guard shall immediately exhibit stop hand signal towards the rear end and check up that the tail light is correctly exhibited.
- vii) The train shall be protected immediately as per (GR 6.03) i.e. 600M – 600M-10M-10M-45M danger hand signal, except that for the protection of the occupied line, one detonator shall be placed at 90M from the train on the way out and similarly two detonators 10M apart, not less than 180M from the train and a stop hand signal shall be exhibited at a distance of 45M from the last detonator i.e. protection in rear from the last vehicle on the

- occupied line as 90 – 90 – 10 – 45M and danger hand signal.
- viii) Protection in front on the occupied line from engine as 600M-600M-10M-10M-45M and danger hand signal if any relief is required.

LOCO PILOT TO REPORT FAILURE IN AUTO SIGNAL (GR 9.11)

When a driver shall pass an auto stop signal at ON he shall stop his train at the next reporting station or cabin and report particulars of auto stop signals passed at ON by him, at the reporting station.

EXAMINATION OF DRIVER/ASSISTANT DRIVER FOR WORKING IN AUTO SECTION (SR 3.78.02)

All Loco Pilots /Assistant Loco Pilots who are required to work train on section having auto block system of working shall be imparted one day's intensive course once in every six months about the rules pertaining to this system and they found qualified, a certificate of competency shall be issued to each Loco Pilot/Assistant Loco Pilot.

ONE TRAIN ONLY SYSTEM

USE OF THE ONE TRAIN ONLY SYSTEM (GR 13.01)

Where trains are worked on the one train only system, only one train shall be on the section on which this system is in force at one and the same time.

Trains may be worked on the one train only system only on short terminal branches on the Single line.

AUTHORITY TO ENTER IN TO THE SECTION (GR 13.03)

A Loco Pilot shall not take his train into the section unless he is in possession of authority to proceed as prescribed by special instructions.

The authority to proceed shall be metallic badge.

NOTE: In case the loss of metallic badge a paper line clear ticket shall be issued, under the signature of the stationmaster on duty.

PROTECTION: (As per GR 6.03)

If a train becomes disable in the mid section and is unable to proceed the guard shall convey a written advice through Assistant Loco Pilot to the Stationmaster of the station from which assistant can be obtained. Disable train shall be secured properly and protected as per GR 6.03.

If no Assistant Loco Pilot, the Guard shall proceed to station and instruct to the Loco Pilot in a written to keep the train stationary until he is return and obtain written acknowledgement.

TRAIN-STAFF AND TICKET SYSTEM

Essentials of the Train-staff and Ticket System.

Where trains are worked between two stations on the Train-staff and Ticket System, -

- (a) a single Train-staff shall be kept at one of such stations, and
- (b) no train shall start from either of such stations to the other unless the said Train-staff is at the station from which the train starts and has either been handed to or shown to the Loco Pilot by the Station Master when giving such permission.

System where applicable .

Trains may be worked on the Train-staff and Ticket System only when the line is single and only between such stations as have been declared by special instructions to be Train-staff stations.

Conditions to be observed for following trains on the Train-staff and Ticket System.

Trains shall not follow one another in the same direction between Train staff stations, unless -

- (a) the Loco Pilot has been properly warned of the time of departure of the preceding train and of the place at which it will next stop;
- (b) all the trains are timed to run at the same speed, and such speed shall not exceed 25 kilometers an hour except under special instructions; and
- (c) an interval of fifteen minutes has elapsed since the departure of the preceding train.

Loco Pilot to have Train-staff or Train-staff Ticket.

No train shall be started from a station unless the Loco Pilot has in his possession to be carried with him on the journey, either the Train-staff or a Train-staff Ticket, for the section of the line over which the train is about to travel.

Note -

The Loco Pilot shall not accept a Train-staff Ticket unless he sees the Train-staff at the same time in the possession of the person who delivers the Train-staff Ticket to him.

AUTHORITY FOR WORKING

Means the authority for working of trains between stations. A Loco Pilot requires authority for working of trains between stations. The authority which is given to the driver for working of trains between stations under the system of working is called the "Authority for working."

AUTHORITY TO PROCEED

Means the authority given to the driver of a train, under the system of working, to enter the block section with his train.

AUTHORITY TO RECEIVE A TRAIN

It means the authority which a Loco Pilot receives for getting into a station with his train. The normal authority which a driver may receive while getting into a station is taking "OFF" of the reception signal.

AUTHORITY TO DESPATCH A TRAIN

It means the authority which a Loco Pilot gets from a block station, for getting into the next block section with his train. The normal authority for a Loco Pilot to get into the next block section with his train is the taking "OFF" of departure stop signal.

AUTHORITY FOR SHUNTING

Means the authority which a Loco Pilot may receive during shunting operations. It may either be the taking "OFF" of fixed signals or exhibition of hand signals or even the verbal order.

When there is no fixed signals for controlling the shunting movement, even though shunting is being performed within the station section, shunting authority T-806 shall be issued. T-806 shall also be issued while shunting is being performed by entering into the block section.

LOCOMOTIVE AND TRAIN WORKING

INTRODUCTION

Loco Pilot/Motorman must be on the alert and be ready to start his train on receiving the guards signal after having satisfied that correct departure signal if any have been taken OFF, and correct authority to proceed under the system of working has been received. No passenger train must be allowed to start before the departure time, notified in public time table.

IMPORTANT TERMS

STANDARD TIME (GR 4.01)

The working of trains between stations shall be regulated by the standard time prescribed by the Government of India which shall be transmitted daily to all Principal stations of the railway at 16.00 hours in the manner prescribed.

On receipt of correct standard time daily at 14.10 hours from the Telecommunication Inspector of each division test room, the Chief controller or in his absence the Deputy chief controller of each divisional control office shall check the time of the Divisional Control Office. He shall then use common ring to communicate at 16.00 hrs to all the Section controllers. He shall also transmit the correct time to other Sub/Area control office in the same manner.

ADHERENCE TO ADVERTISED TIME (GR 4.02)

No passenger train or mixed train shall be despatched from a station before the advertised time.

PUNCTUALITY OF TRAINS (SR 4.02.01)

Loco Pilot/ Motorman must be on the alert and be ready to start his train immediately on receipt of the Guard's starting signal, but only after satisfying themselves that the correct departure signal, if any, have been taken "OFF" and they are in receipt of " Authority to proceed", but no train carrying passenger must be allowed to start before the departure time notified in the public time table.

SETTING OF WATCH (GR 4.03)

Before a train starts from terminal or crew changing station, the guard shall set his watch by the station clock, and communicate the time to the Loco Pilot who shall set his watch accordingly.

SUPPLY OF WORKING TIME TABLE (GR 4.07)

A copy of working time table and (appendices) if any shall be supplied to the Loco Pilot.

TIME OF ATTENDANCE FOR TRAIN CREW (GR 4.04)

Every guard, Loco Pilot, Assistant Loco Pilot shall report for duty at such place as may be ordered by the DRM or by an Officer on his behalf.

LIMITS OF SPEED GENERALLY (GR 4.08)

The driver shall regulate and control the running of the train according to the working timetable so as to avoid either excessive speed or loss of time. He shall not make up more time between any two stations than allowed in this behalf in the working timetable observing the speed restrictions in force.

RIDING ON ENGINE:(SR 4.22.01)

In addition to the engine crew the following persons may ride on the engine or driving cab of EMU/Train , but not more than 3 (three) excluding Engine crew.

- i) An Officer holding a metal pass
- ii) Guard in case of emergency
- iii) Traffic staff when performing shunting or when piloting in.
- iv) Staff proceeding to attend an accident.

ENGINE PUSHING (GR 4.12)

- i) No engine shall push a train outside the station limit except in accordance with special instructions and at a speed not exceeding 25 kmph
- ii) This rule shall not apply to an assisting/banking engine in rear of a train, which may be permitted under a special instruction to run without being coupled or coupled to a train.
- iii) A "Patrol" or "Search Light" special with one or more vehicles attached in front of the engine may be permitted to run at a maximum speed of 40 KMPH.

DIMMING OF HEAD LIGHT (R 4.14.02)

The Loco Pilot shall dim the headlight on their engines in the following cases.

- i) When remains stationary at a station.
- ii) While crossing a train in opposite direction in D/L section
- iii) While performing shunting
- iv) When required to avoid running into dazzled cattle or human being
- v) To spot the light indication of a signal.

BACKING OF TRAIN FROM BLOCK SECTION TO THE STATION FROM WHICH IT LEFT (SR 4.12.01)

Whenever due to unavoidable circumstances the Loco Pilot wants to push back the train to the station from which it has come, the Loco Pilot shall inform the fact to the guard then

- i) Guard will authorise the Loco Pilot to do so by issuing a written memo after obtaining permission from the station in rear to push back ,by any available means.
- ii) If the view ahead is not clear or at night, assistant Loco Pilot or other railway servant shall pilot the train with hand signal at an adequate distance in advance of the leading vehicle.
- iii) The guard shall travel in the leading brake van/vehicle while pushing back and shall be prepared to apply brakes if necessary.
- iv) The driver shall push back the train at walking pace up to the 1st stop signal on single line or 1st stop signal of the correct line/last stop signal of the wrong line, which ever comes first and from their train shall be piloted in
- v) After the train has been brought to stop at the station, the driver shall sound the engine whistle two long. (- -)

LOCO PILOT/ASSISTANT LOCO PILOT TO KEEP GOOD LOOK OUT (GR 4.40)

The Loco Pilot and Assistant Loco Pilot shall keep a constant look out for obstruction and be prepared for an immediate application of brakes at any moment when the train is in motion.

LOCO PILOT & ASSISTANT LOCO PILOT TO LOOK BACK (GR 4.41)

The Loco Pilot and assistant driver shall look back frequently during journey to ascertain the safe running of train.

EXCHANGE OF SIGNALS BETWEEN LOCO PILOTS, GUARDS & STATION STAFF (GR 4.42)

Exchange of signals by the engine crew, guard and station staff is must, which ensures the alertness.

EXCHANGE OF SIGNALS BETWEEN LOCO PILOT AND STATION STAFF

- i) While passing through a station, signal shall be exchanged with the station staff.
- ii) While entering at station – with 1st cabin
- iii) While starting a station – with departure cabin

NOTE: a) One long continuous whistle is to be sounded while passing
Through a station in addition to exchanging of signals.
b) Loco Pilot and Assistant Loco Pilot should remain in driving cab.

EXCHANGE OF SIGNALS BETWEEN LOCO PILOT AND GUARD (4.42;4.42.01)

- i) As soon as the train starts from a station
- ii) As soon as the train starts after stopping outside the station limit
- iii) When the last vehicle leaving the last trailing point of the yard

NOTE: If after starting the guard signal is not visible the exchange of signal should be done when the last vehicle is quitting the yard.

EXCHANGE OF SIGNALS WITH STATION CABIN STOPPING AND NON-STOPPING TRAINS

The Loco Pilot and the guard shall be on the look out for the signals displayed by the cabin man and shall acknowledge and exchange green hand signal with that person when they are opposite to each other. If the cabin man exhibit a stop hand signal, The guard and the Loco Pilot shall take immediate steps to stop the train and find out the reason for showing danger hand signal.

GUARD DOES NOT EXCHANGE THE SIGNAL:(SR 4.42.05)

In case the Loco Pilot does not receive signal from the guard he shall sound two short, (o o) whistles, if still the guard does give any signal, the train shall be stopped and the cause to be enquired.

SIGNAL NOT EXCHANGED BY STATION STAFF:(SR 4.42.07)

In case the station staff does not exchange signal while passing through a station the Loco Pilot shall exercise extra caution to ensure, all is right for the train to proceed on and record the same in the rough journal.

TRAIN HELD UP AT 1st STOP SIGNAL (GR 4.44)

If the train is standing outside the 1st stop signal without any apparent cause for 5 mins

- i) Give continuous whistle to warn the guard and station staff.
- ii) Assistant Loco Pilot is to be sent to station for information with showing a stop hand signal.
- iii) After 15 min. if the signal is not taken OFF or nobody comes to pilot the train.
 - a. The train shall be protected in rear from last vehicle by the guard as per GR 6.03 i.e. 600-600-10-10-45M with danger hand signal.
 - b. In the mean time, if the signal is taken OFF or if driver is authorised to pass the signal at ON – give one long whistle exchange signal with guard and pick up the Assistant driver on the way.
 - c. Guard is to pick up the intermediate detonator when he will be recalled.
 - d. Report the matter in joint train report of the guard and Loco Pilot (T-34 H&F)

NOTE: i) Before going to protect the train GD shall tie a red hand signal (i.e. flag) to side lamp bracket by day and reverse one of his brake van side lamps where provided as an additional indication to the driver.
 ii) After protection Guard shall continue to show a stop hand signal to stop any approaching train until he is recalled.

ODC AND ITS CLASSIFICATION: OPM 15.03)

ODC (OVER DIMENSIONAL CONSIGNMENT) When a bulky consignment with its packing and lashings is loaded in a wagon and its length, width and height are such that one or more of these infringe standard moving dimensions at any point during the run of the wagon from start to destination, then the consignment is called an ODC. It is also known as out of gauge load.

CLASSIFICATION OF ODC:

Depending on the extent to which the clearance of consignment are short of normal clearance, the loads are classified as:

- i) "A" class load
- (ii) "B" class load
- (iii) "C" class load

Class	Sanc. Auth.	Speed	Day/Night Restriction	Escorting
"A" Class	DRM	40 KMPH or at a speed given in BPC	NIL	NIL
"B" class	COM	40/15 KMPH 40 KMPH for straight lines and 15 KMPH while passing stations, yards, curves and turnouts.	NIL	TXR only in critical sections
"C" class	CRS	25/8 KMPH, 25 for straight line and 8 while passing through stations yards, curves, gradients and turnouts	Night Restriction	TXR, PWI DTI & ATFO

SPEED OF LOCOMOTIVE DURING SHUNTING:

- i) 15 KMPH is the general speed of locomotive during shunting
- ii) Wagons loaded with Petrol, kerosene, liquid fuel, spirit and highly inflammable liquid in bulk or packed in tins or drums – 8 KMPH
- iii) All bogie wagons fitted with roller bearing when shunting a single wagon – 5 KMPH
- iv) Group of five roller bearing fitted with transition coupling (screw coupling at either end – 3 KMPH
- v) When rolling down the hump, the impact speed – 5 KMPH

SPEED OF DIFFERENT TYPES OF TRAINS –

- 1.**
 - a) Every train shall run on each section of the railway with the limit of speed sanctioned on that section, under approved special instruction.
 - b) The sectional speed sanctioned and the permanent speed restriction imposed on that section shall be shown in working time table.
- 2.** The Loco Pilot shall -
 - a) Regulate and control the running of the trains according to the working time table and caution order issued to him so as to avoid either over speeding or loss of time.
 - b) The Loco Pilot should not make up time between any two stations more than it is allowed in this behalf in the working time table and he shall also observe all other speed restrictions in force.
- 3.** In order to indicate to the Loco Pilot where train are to run at restricted speed or where trains have to stop due to the line being under repair or due to any other obstructions, caution orders are issued.

MAXIMUM SPEED

i)	Goods train (4 wheeler)	72 KMPH
ii)	Passenger train (4 wheeler attached)	72 KMPH
iii)	Passenger train	100 KMPH
iv)	Mail/Express train	105 KMPH
v)	Superfast train	105 KMPH

NOTE: Goods trains speed depend on the section, types of wagon and type of locomotive.

MAXIMUM PERMISSIBLE SPEED OF DIFFERENT TYPES OF LOCO

i)	WAM4 /WAP-4	105 KMPH
ii)	WAG-5	75 KMPH
iii)	WAG-7	100 KMPH
iv)	EMU	90 KMPH

MATERIAL TRAIN WORKING

MATERIAL TRAIN

It means a departmental train intended solely or mainly for carriage of railway materials when picked up or put down or for execution of works either between stations or within station limit

WORKING OF MATERIAL TRAIN (GR 4.62)

- a) A materials train shall be worked only with the permission of stationmaster of each side
- b) Except under special circumstances, material train shall work between sunrise to sunset
- c) Authority to proceed for material train
 - i. General authority to proceed as per system of working and a caution order
 - ii. A caution order and block back with private No. When materials train is required to work on wrong line on D/L section and return.

- iii. Block ticket (T/A602) and caution order when material train is required to work on obstructed block section

SPEED OF MATERIAL TRAIN

- i) General speed of a material train working between stations shall not exceed 25 KMPH
- ii) When running through between block stations with the engine leading – the speed of the material train shall not exceed the speed prescribed for goods train of similar weight.
- iii) When engine is pushing and brake van is leading 25 KMPH in straight line and 8 KMPH over a turn out.
- iv) When the engine is pushing but the brake van is not leading not exceeding 8 KMPH.

PRECAUTION TO BE OBSERVED

- i) The train crew shall keep a good look out especially in the direction in which the train is moving and must be ready to stop short of any obstruction.
- ii) The Loco Pilot and Assistant Loco Pilot shall keep a sharp look out to hand signals exhibited by Guard and act accordingly and positively obey in the special instructions and restrictions given in caution order.
- iii) Normally material train shall not be divided nor engine uncoupled from train in section except in emergency.
- iv) A material train shall work between sunrises to sunset except under special circumstances.
- v) Female labour should neither be engaged nor carried on the train during night.
- vi) Ensure that railway materials loaded on any wagon, do not infringe standard moving dimensions.

CRANE WORKING (SR 4.27.02 AND 03)

When running on the main line, the crane shall be placed four vehicles away from the engine but when proceeding to brake down i.e. in case of emergency the crane may be placed anywhere on the train.

SPEED: 40 KMPH over straight line, 15 KMPH over turnouts.

LOCO AND TRAIN OPERATION:

Loco and train operation is controlled in a given area of a railway from centralised control office. Generally this is situated in divisional headquarters. The control office is provided with suitable telecommunication system connected with all systems, Stations, yard, loco shed, traction shed, goods shed and officials of various department. Deputy control circuit is provided in each control office, with telephone communication connected to important stations yards and loco sheds for collecting important particulars such as Yard report, mid night figures, train ordering etc.

FUNCTIONS OF CONTROLS IN TRAIN OPERATION:

This may be described under three needs:

- i) **TRAIN CONTROL:**
 - a) Ensuring the punctual running of mail/express/passenger train.
 - b) Running goods train with least possible detention in route
 - c) Arranging relief for engine crew and guard in time etc.
- ii) **TRAFFIC CONTROL:**
 - a) Allotment and distribution of coaching and goods stock in station,
 - b) Securing maximum utilisation of stock.
 - c) Maintenance of marshalling yard.

iii) **POWER CONTROL**

- a) Requisition of locomotive power i.e. engine, direct from locomotive running shed for all operating requirements for example – train working, shunting and banking.
- b) To ensure the most economical use of engine power available.

TIME TABLE (OPERATING MANUAL 5.03)

Railway operating works are divided into two main branches, which are Passenger Train operation, and Goods train Operation. The public judge the efficiency of railway by punctuality and safe running of passenger services. To maintain the punctuality of trains timetables are made and published twice a year, once in the 1st April for the summer season and again on the 1st October for the winter season.

Following types of timetables are published on this railway and issued in every six months to come into force:

1) Public timetable-

This is handy volume for use by the public and are printed in English and regional language. In addition to the time schedule they contain a variety of useful information for passengers including fare-tables.

2) Suburban timetable

This is printed in English and Bengali and is of pocket size. This contains in details the timings of all suburban services as well as of other passengers services running over the sub urban sections and also abstract timings of other main line passenger trains.

3) Sheet timetable

These are exhibited on the notice board of station platform and in passenger waiting halls. They are printed in English and also in regional language and contain the schedules of various coaching services, tabulated and brought together on a large sheet for reading and ready for public reference.

4) Staff copy of the timetable

This is copy of time table meant for use by railway staff.

5) Working timetable

These are issued separately for each division for the information and guidance of all railway staff, specially running staff, station and control staff. These time table contains the schedule of all coaching trains, as well as of goods trains and other valuable information, such as load table of goods train, headquarters of divisional staff, permanent and semipermanent speed restrictions and special working instructions.

TYPES OF SPEED AND TIME.

MAXIMUM PERMISSIBLE SPEED:

The maximum permissible speed over different sections is approved by the commissioner of railway Safety (CRS) taking into consideration the conditions of track, bridges, curvatures, gradient. This speed limit may not be exceeded under any circumstances.

Maximum permissible speed for each type of locomotive is also fixed for different sections and notified in the working timetable.

BOOKED SPEED:

The booked speed means the speed at which a train is booked to run. The booked speed of mail/express/Passenger train and other trains varies according to the classes of train. The booked speed is generally fixed at least 10% less than the maximum permissible speed.

In case of the late running a driver may run up to maximum permissible speed subjected to other restriction in force.

MINIMUM PERMISSIBLE RUNNING TIME (MPRT)

The minimum permissible running time between stations is fixed on the basis of maximum permissible speed. A Loco Pilot should not cover the run between two stations in less than the minimum running time published in the working timetable.

NORMAL RUNNING TIME:

The normal running time for a particular train between stations is fixed on the basis of booked speed of the particular train. While fixing the normal running time, due considerations are given to the permissible track speed, load of the train, type of locomotive, acceleration, deceleration, restriction over facing points and turnouts and engineering restrictions, when a train is running late the driver can exceed the normal running time up to the limit of maximum permissible speed subjected to other speed.

TIME ALLOWANCES.

The time allowances indicated in the working timetable are the following.

TIME ALLOWANCE FOR STOP AND START

Train	Start (In Mts.)	Stop (In Mts.)	Total (In Mts.)
Mail/Express	2	1	3
Suburban	½	½	1
Goods	3	2	5

TIME ALLOWANCE TO COVER TEMPORARY ENGINEERING RESTRICTION

The time allowance to cover temporary engineering speed restrictions is provided in each section based on the work proposed, to be carried out by the engineering department.

RECOVERY TIME

The time allowances are provided to compensate for time lost by a train due to ACP, points and signal failure, fog storms etc. Generally recovery time is provided at the last lap of run of passenger train.

142.03**CLASSIFICATION OF TRAINS AND TYPES OF TRAINS:**

According to the railway operating work trains are classified into two main branches.

- (i) Passenger Train Operation
- (ii) Goods Train Operation

TYPES OF TRAINS:

i) PASSENGER TRAIN

- a. Mail/Express Train - These are high speed trains which carry passenger and postal mail over long distance –stopping only at important stations.
- b. Local Passenger Train – These trains cover relatively short distances, generally stops at every station.
- c. Mixed Train – It carries passenger and goods both and runs on unimportant branch line section such train stops on all stations and are slower in speed.
- d. Suburban Train – these are light trains of greater frequency serving the suburban area of metropolitan city.

ii) GOODS TRAIN

iii) MATERIAL TRAIN

Except these trains special trains are run under the authority of passenger trains such as (a) Special troop train & (b) VIP special

CLEARING LOAD FROM A STATION/SIDING WHERE NO TXR STAFF ARE POSTED (SR 4.31.03)

At stations where no train examining staff are posted or at wayside stations (enroute) while clearing a stable load, the Guard and the Loco Pilot shall jointly responsible for following the procedure described bellow:

1. While starting a train from a station/siding where no Train Examination staff are posted, the condition under which a BPC shall remain valid or become invalid are as under:-

1.1 The BPC shall remain valid:-

- a) When an intensively examined empty rake has been loaded at the station/siding
- b) When a close circuit (CC) empty rake has been loaded or back loaded at a station/siding, provided the validity of the CC rake has not expired.
- c) When a CC rake has been unloaded or back loaded at a station/siding provided the validity of the close circuit rake has not expired.
- d) When a train with valid BPC, loaded or empty, is stabled at a way side station for not more than 96 hours.
- e) When the integrity of a train with valid BPC has not been disturbed by more than 10 FWUs.

1.2 The BPC should be considered invalid:-

- a) When an intensively examined loaded train has been unloaded or back loaded at a station/siding .
- b) When the validity period of a CC rake has expired.
- c) When a train with valid BPC, loaded or empty, is stabled at a way side station for more than 96 hours.
- d) When the integrity of a train with valid BPC has been disturbed by more than 10FWUs.

2. While starting a train after loading/unloading/back loading operation(s) from a station where no train examining staff are posted, the Guard and Loco Pilot shall be jointly responsible for the following actions:

2.1 In case of valid BPC-

a) After the locomotive(s) is/are attached, the Driver will create not less than 46cms. of vacuum or 5kg/cm² of air pr. in the engine and 38cms of vacuum or 4.8 kg/cm² in the rear/brakevan as the case maybe, provided that:

I) in case of an air braked train consisting of more than 56 wagons the pressure in the rear brake van shall not be less than 4.7 kg/cm², and

II) in case of a train consisting of 40 or more BCX/BOX wagons the minimum vacuum level in the engine and the rear brake van shall not be less than 44cms.and 34cms. respectively.

b) On noticing the required amount of vacuum/air pressure in the rear brake van the Guard and the Loco Pilot shall communicate with each other and shall thereafter jointly check the train to ensure the following:

I) All couplings and vacuum hoses, brake pipes are properly coupled.

II) In case of air brake stock, all the angle cocks are open condition and the last angle cock is in closed condition.

III) In case of vacuum stock, rear most end hose pipe of the last vehicle is kept in the dummy plug.

IV) Empty load device handle is in proper position.

V) There is no loose fittings/hanging parts like push rod, brake beam, safety brackets, brake blocks etc. which may endanger safe running of the train. The Loco Pilot shall check the train from the station side and the Guard shall check it from the off side.

VI) All the hand brakes are released.

VII) Doors of wagons are closed and locked/secured.

c) Thereafter, Guard shall prepare a memo in triplicate jointly with the Driver on a plain sheet in the following format and shall append their signatures in each copy.

I) Date

II) Train description and No.

III) From- To-

IV) Engine No.

V) Engine attached at-

VI) Total load-

Guard shall obtain SM/YM's endorsement on two of these copies of the joint memo, out of which one would be handed over to the Loco Pilot and the other would be retained by him. He would hand over the third copy to the SM/YM.

d) The SM shall not permit the train it start until he has received a copy of this joint memo.

2.2. In case of invalid BPC:-

- a) The Loco Pilot and the Guard, in addition to taking all actions as mentioned in Clause 2.1 above shall also be responsible to ascertain the brake power of the train with the following results.
 - Total number of cylinders.
 - Total number of working cylinders.
 - Percentage of Brake power.
 - Vacuum/ air pressure in engine.
 - Vacuum/ air pressure in brake van.
- b) These information will be recorded in the same joint memo as mentioned in Clause 2.1 (C) itself.
- c) In this case, the memo/certificate shall automatically revalidate the BPC normally upto the next train examination point in the direction of movement of the train where this train will be offered for examination except in case of Para 3(b) mentioned below.

3. While starting a train from a way side station for clearing a stable load, the following checks should be performed by Guard and Loco Pilot.

- a) When a BPC is valid and the load has not been stabled for more than 24 hours, check mentioned in Sub clause 2.1(a) shall only be required.
- b) In case of valid BPC, if the period of stabling is more than 24 hours but not more than 96 hours checks shall be governed by Sub-clause 2.1 and 2.2. However, in this case the BPC shall automatically get revalidated after the checks for the train to travel up to the destination/station.
- c) In case of invalid BPC, irrespective of the period of stabling checks shall be governed by Sub-clause 2.1 and 2.2.

Formula for calculating the brake power

$$\frac{\text{Total No. of working Piston}}{\text{Total No. Of Piston on trains}} \times 100$$

BRAKE POWER CHECKS:

The Loco Pilot has to inspect the train along with the guard if no train examination staffs are available and calculate the brake power percentage. He should find out the number of working pistons and count total number of pistons of the train. The ratio of the number of working pistons and the total number of pistons multiplied by 100 gives the percentage of brake power of the train.

PIPED VEHICLE

A vehicle whose distributor valve (DV) have been kept isolated by putting the handle into the isolating position and does not response to the variation of brake pipe pressure, such vehicle is known as a piped vehicle and is deemed to be having inoperative cylinders.

INOPERATIVE PISTON (IOP)

A vehicle whose distributor valve (DV) does not respond to the variation of brake pipe pressure and the piston remain inside the cylinder is deemed to be having inoperative piston, when the driver reduces the brake pipe pressure, the piston of such a vehicle does not respond and remain inoperative due to the non functioning of the DV.

ROLLING PISTON

Normally after destruction of full brake pipe pressure an active piston should remain in applied condition for a period not less than 30 minutes.

The piston of a brake cylinder, which does not fulfil this condition, is known as rolling piston.

EXCESSIVE PISTON STROKE (EPS)

Any vehicle that has a piston stroke i.e. more than the stipulated length is deemed to be having excessive piston stroke. This results in poor brake power and the Loco Pilot should take it into account while checking the brake power percentage of the train.

PISTON STROKE OF BOX 'N'/BCN WAGON

<u>WAGON</u>	<u>EMPTY</u>	<u>LOADED</u>
BOX 'N'/BCN	85 mm + 10 mm	130 mm + 10 mm

WORKING ON ELECTRIFIED SECTION:

- i) When any defects of the OHE which is likely to interfere with smooth movement of pantograph or cause or damage, it is noticed ahead, the Loco Pilot shall trip DJ and lower pantograph immediately. The train be brought to stop short of such place.
- ii) If the damage of OHE is slight, the Loco Pilot may coast through such damaged section if possible.
- iii) In all cases, the defects shall be reported to TPC (Traction Power controller).

APPLICABILITY OF GENERAL RULES

In electrical section special precaution should be taken by controllers, TPC, Station master and train crew when a section of OHE found faulty.

PRECAUTIONS WHEN A TRAIN IS DETAINED IN SECTION FOR MORE THAN 5(FIVE) MINUTES DUE TO NO TENSION ON LINE:

When ever any train gets held up for more than 5 minutes in the block section on account of no tension, the Loco Pilot of the train shall depute his Assistant Loco Pilot to check the train in order to look for any abnormality and to advise the guard for no tension on line (in OHE) together with Assistant Loco Pilot the guard shall check the entire train.

If in the mean time power supply restored the Loco Pilot shall call back the Assistant Loco Pilot to the locomotive and resume journey

Otherwise after the train is checked the Loco Pilot/Guard inform the section controller/TPC of the details of abnormality if any or otherwise an assistant required to the nearest emergency telephone socket or by other available means.

The Loco Pilot and Guards shall arrange protection of line to the affected line in accordance

with GR 6.03 in auto section as per GR 9.10.

Further the journey will be resumed as per decision regarding recharging of OHE by TPC/ section controller.

SPECIAL DEFINITIONS

POWER BLOCK

It means blocking of section of line to electric traffic only.

FEEDING POST

It means a supply control post where the incoming Feeder line from grid sub-station is terminated.

TOWER WAGON:

It means a self-propelled vehicle which is used for the Maintenance and repair of OHE.

TRACTION POWER CONTROLLER:

It means a competent railway servant who may for the time being be responsible for the control of power supply on the traction distribution system.

OHE(OVERHEAD EQUIPMENTS)

It means the electrical conductor over the track together with their associated fittings insulators and other attachment by means of which they are suspended and registered in position.

EMERGENCY TELEPHONE:

It means a telephone circuit provided for contacting the traction power controller

TRACTION LOCO CONTROLLER (TLC/LC)

It means an official under the control of Sr. DEE (OP)/OP/ADEE (OP) who will be responsible for booking of electric locomotives and running staff to meet the requirement of the traffic.

SECTION INSULATOR:

It means a device for dividing the contact wire into electrical sections while maintaining mechanical continuity and a continuous path for pantographs.

ASSISTANT LOCO PILOT:

Assistant Loco Pilot means duly certified assistant loco pilot of single or multiple unit train or of an electric engine.

RULES FOR ELECTRIFIED SECTION (SR 17.09.01)

- i) In the event of OHE failure, the traction power controller shall immediately locate the faulty section and isolate the same, in case of double line or multiple line section he shall also isolate the healthy section on adjacent tract on the same route length. As the faulty section, then he advises the section controller about the same.
- ii) On receipt of the information from TPC, the section controller shall immediately inform the stationmaster, if the station who are concerned with the working of train in

which healthy OHE temporarily is isolated. They shall not allow any train to leave their stations unless both the drivers guards of the first train have been issued a caution order to proceed at speed of 35 KMPH by day and 20 KMPH by night subjected to observance of other speed restrictions exercising great caution.

ELECTRICAL SAFETY

SAFETY RULES CONCERNING 25 KV AC

- i) Do not approach to 25 kv AC directly or indirectly within a radius of 2 meters.
- ii) On line before going to the roof of the loco take emergency power block first by exchanging private No. With TPC and ground the loco, OHE on both side of loco by earthing poles and cables. After that climb on the roof of the loco.
- iii) Do not enter in the high-tension compartment of loco until loco is grounded.
- iv) Do not perform roof inspection in shed, stations or yards until correct isolating switches are opened and grounding switch is closed. Put your personal pad lock on the isolating switch for more personal safety.
- v) Do not project part of body above the roof of the loco while changing head light bulb.
- vi) Do not project jet of water towards OHE.
- vii) Do not touch any conductor line near the electrified track.
- viii) Do not throw any conductor on the OHE from over bridge.
- ix) Do not walk between track and under OHE.

WHISTLE CODE

INTRODUCTION:

To establish the communication in between or to warn the guard, station staff, train driver, gateman and road users and staff working on track, prescribed whistle codes are sounded from engine cab as per situation and conditions.

TYPES OF WHISTLE CODE.

The whistle codes, which are used in different occasions, are mainly two types.

- | | | |
|------|---------------|----------|
| (i) | Short Whistle | 0 |
| (ii) | Long Whistle | — |

The prescribed code of whistle shall be blown in different conditions in different ways.

SIGNIFICANCE OF DIFFERENT CODES:

The driver shall sound engine whistle as per special instructions.

- i) Before putting an engine on motion
- ii) When entering tunnel
- iii) At such other times and places as may be prescribed by special instruction.

WHISTLE CODES AND THEIR INDICATIONS (SR 4.50.01)

Sl. No.	Code of engine whistle	Indication
1.	0	a) Before starting - i) Indication to driver of assisting/banking engine that the driver of leading engine is ready to start. ii) Acknowledgement by the driver of assisting/banking engine to leading engine. iii) Engine ready to leave loco yard or after completing loco works. iv) Engine ready to go to loco yard. b) On run (i) Assistance of other engine not required (ii) Acknowledgement of driver of assisting/banking
2	00	a) Call for guard signal b) Signal not exchange by guard c) Signal not exchange by station staff
3	— 0	a) Guard to release brake b) Before starting engine, train from station/mid section c) Main line clear after backing into siding
4	0 0 0	a) Guard to apply brake b) Train is out of control guard to assist
5	0 0 0 0	a) Train cannot proceed due to accident, failure, obstruction or other exceptional cases b) Protect the train in rear
6	— — 0 0	a) Call for guard to come to engine
7	0 — 0	a) Token not received or missed b) With wrong "Authority to proceed" c) Passing stop signal at ON with proper authority
8	—	a) Before starting vacuum recreated on Ghat section remove sprang b) Passing an auto stop signal at ON c) Passing an IB stop signal at ON when telephone is out of order d) On run acknowledgement of guard's signal.

9(i)	----- (Intermittent) ----- (Continuous)	Approaching level crossing a) Approaching , tunnels or or curves or cutting, or site of accident or area of restricted visibility due to fog storm or any other reasons the view of signal is obstructed. b) Recall the railway servant protecting the train in rear c) Material train ready to leave d) Running through a station e) Approaching stop signal at ON f) Detained at a stop signal.
10	— 0 — 0	a) Train parting b) Train arriving in complete.
11	0 0 —	a) Alarm chain pulled (ACP) b) Insufficient vacuum in engine c) Guard applies vacuum brake
12	— —	a) Raise pantograph to be acknowledged by the other engine
13	— 0 —	Lower pantograph to be acknowledged by the other engine
14	— 0 0	a) Signal arm lowered but light extinguished b) Signal arm improperly/insufficiently taken OFF c) Defective signal
15	— — —	Fouling not cleared
16	0 0 0 0 0 0 (Frequently)	a) Apprehension of danger b) Danger signal to the driver of approaching train whose path is fouled or obstructed for any reason c) While working on a single line section during total failure of communication or when single line working is introduced on double line section. d) Moving in wrong direction on double line or against the signalled direction in the auto block signalling territory (D/L) or against established direction in the automatic block signalling territory (S/L)

Bell signals (SR 4.51.01)

For single or multiple unit electric trains, the following bell signal codes shall be used between the Motorman and the Guard. In case of failure of electric bell equipment in the cab, the codes shall be sounded by horn.

S/No.	Code of Bell signal	Indication	Acknowledgement
1.	0	Stop train	0
2.	00	Start train	00
3.	000 pause 0	To push back train	000 pause 0
4.	00	During run through	00
5.	000	Guard required by Driver	000
6.	00000	Brakes tested found alright	00000
7.	00000 pause 0	Brakes tested and not found alright	00000 pause 0
8.	0000 pause 0000	Information to Motorman regarding guard leaving the cab.	0000 pause 0000
9.	00 pause 00	Passing automatic signal at on	00 pause 00
10.	0000	Protect train in rear	0000
11.	00 pause 0	Proceed with caution while working under GR 4.21 (2) (a)	00 pause 0
12.	000 pause 000	Guards warning to Motorman if EMU exceeds the prescribed speed after passing an auto signal at 'ON'	000 pause 000
13.	0 pause 0	Zone of speed restriction is over, resume prescribed speed	0 pause 0

ABNORMAL WORKING

ABNORMAL LOCO WORKING

INTRODUCTION

There are the occasions in which defects arises in respect of working of head light, speedometer, engine horn etc. or at times engine's driving cabs gets defective or a Loco Pilot gets incapacitated during run.

In any of the above mentioned circumstances a trainee has to be able to undertake required measures when required after completion of this lesson.

DEFECTIVE HEAD LIGHT

- (i) In case the engine head light goes out of order after leaving engine or crew changing station, the Loco Pilot shall work the train at night or during thick foggy or tempestuous weather or inside a tunnel cautiously at a Speed not exceeding 20kmph after ensuring the proper glowing of marker lights up to the repairing station and the driver shall sound long continuous whistle frequently.
- (ii) The Loco Pilot shall hand over a written memo to the station master of the next station for necessary repair of headlight.

DEFECTIVE DRIVING CAB(SR 4.21.01) –

In case of an electric engine, if the leading driving compartment is defective –

- (i) The Loco Pilot shall inform the TLC.
- (ii) The Asstt shall work the train from the trailing cab.
- (iii) The Loco Pilot shall remain in leading cab and be responsible for the correct operation of the train.
- (iv) SPEED not exceeding 40 kmph up to the repairing station or work till such time relief engine is arranged by the TLC in consultation with deputy chief controller.

DEFECTIVE SPEEDOMETER

No locomotive shall be turned out from Shed with deficient or defective speedometer. If it becomes defective enroute, Loco Pilot shall work the train at a speed of 10% less than the permissible speed by estimating the speed with help of his watch, km. Post and inter section running time given in the working time table.

The following formula will help a Loco Pilot to calculate the speed in this regard.

- i) No. of electric mast passed in one minute $\times 4.3 =$ _____ kmph
- ii) $\frac{60 \times 60}{\text{Time taken to cover one km. In seconds}} =$ _____ kmph

LEADING CAB HORN NOT SOUNDING (SR 4.21.01) –

- (i) In case horn/whistle of the driving cab of an engine becomes defective in the section

- the Loco Pilot will run the train/engine at 40kmph in day time up to the destination or till the defect is rectified and at night he will only clear the section at a speed not exceeding 40kmph.
- (ii) During run the horn/whistle as prescribed should be sounded by Asstt. Loco Pilot from rear cab.

DRIVER IN CAPACITED – (G.R. 4.20. SR. 4.20.01)

- (i) If a driver becomes incapacitate while the engine in motion, the Asstt. Loco Pilot if duly qualified may work the train up to the next station cautiously and where the Asstt. Loco Pilot is not duly qualified he shall bring the train to a stop and send a message to the station master of the nearest station to make arrangements for a Loco Pilot to take over the train and for so doing he may take assistance of the guard.
- (ii) The speed of the train shall not exceed more than 25 kmph up to the next block station and on arrival at next station the fact shall be reported to the stationmaster.
- (iii) If required train shall be shunted into a suitable line and shall remain under the charge of Asstt. Loco Pilot till relief arrives.

FLASHERLIGHT

It is an emergency warning light filled on the roof of The loco by the side of the head light. Flasher light works on 110V DC Battery. When a Loco Pilot senses an emergency, he should switch the Flasher light 'ON' by operating an 'ON' & OFF switch, provided on the Flasher light control box. When the switch is put 'ON' the light starts Flickering in 2 seconds, interval with an amber/orange colour to attract the Attention of an approaching trains driver of the other line to stop for any obstruction.

USE OF FLASHER LIGHT – It is under the following circumstances –

- (i) During sudden jerk on run with drop of vacuum/pressure of the Train.
- (ii) Sudden increase in airflow indicator (gauge).
- (iii) During train parting/derailment of train.
- (iv) During OHE failure or it is required to send information through the opposite line's train Loco Pilot.
- (v) When S/L is introduced on D/L absolute block system and train is running on wrong line.
- (vi) While LE will go to open the communication in S/L total interruption.
- (vii) Coaching trains when stops in block sections out of Coarse

DUTIES OF ADJECENT LINE LOCO PILOT, WHILE HE SEES A FLASHER LIGHT IS GLOWING –

Soon after the Loco Pilot of the train coming from the opposite direction finds that flasher light is glowing, he shall immediately take action by applying emergency brake if required and stop his train, short of the flasher light to avoid any mishap.

REPORTING OCCURANCES ON LINE –

- (i) When a train is stopped for any reason between station, the Loco Pilot shall sound four short whistle (0 0 0 0) and the train shall be protected as per G.R. 6.03.
- (ii) A memo jointly signed by the guard and the Loco Pilot indicating the Site, date, time, cause, nature of failure and kind of relief required shall be send to the nearest station by quickest possible means through a competent railway servant. (Asstt. Loco Pilot, Brakes man, Gagman) quoting the name and designation of such messenger in the

memo. The Guard and Loco Pilot shall try to inform the fact to the controller through portable telephone if provided.

INTRODUCTION

While on duty during run a Loco Pilot may have to face different circumstances like bad riding, wheel slipping, storm, cyclone, breakage of pantograph, no tension on line, defect on OHE line and run over cases etc.

At the end of the lesson the trainee should be able to negotiate different situations, adopt required measures and report the occurrences to the concerned authority.

BAD RIDING

BAD RIDING DUE TO DEFECT IN TRACK (S.R. 2.11.01)

a) DUTY OF THE LOCO PILOT WHO EXPERIENCES BAD RIDING-

- i) The Loco Pilot shall note as accurately as possible the KM No. at which bad riding has occurred and he shall also put on the flasher light and put off the head light if it is night time.
- ii) He shall apply train brakes and look back to see the condition of the train behind and after the train comes to a stop, he shall take account of the extent of damage.
- iii) After going through the condition of track if the Loco Pilot thinks that he can pass the spot safely with his train he will proceed pass the spot carefully and then resume normal speed.
- iv) He shall while approaching the next block station, whistle frequently and bring his train to a stop in such a manner that the engine is in front of the station building or the cabin where block instrument is located without clearing fouling mark. Hand over a written memo to the station master or the switch man as the case may be, specifying the exact location as far as practicable and nature of defect noticed. On single line section the driver shall not hand over the authority to proceed till such time he has issued the memo to the station master /cabin man.
- v) He shall also make a report of the occurrence to the CCC(E) on return to the head quarter.

b) DUTIES OF LOCO PILOT OF SUBSEQUENT TRAIN-

The Loco Pilots of the subsequent trains shall be guided by the caution order issued to them by the station master.

BAD RIDING DUE TO ENGINE DEFECT (SR 2.11.02)-

In such a case the Loco Pilot shall-

- i) Note down the KM. No. and time.
- ii) Stop his train and take account of the extent of damage and proceed cautiously to the next station and report the station master about the fact by giving a written memo.
- iii) Inform LC (loco controller) and SCR (section controller) whether any assistance is required or he will be able to continue the journey up to the next repairing station.

WHEEL SLIPPING (SR 2.11.03)

- i) In the event of wheel slipping of an engine while on run between stations ensure the safety of track and try to stop wheel slipping as quickly as possible.
- ii) Note carefully the kilometrage and the extent of damage. If required ask for necessary assistance by contacting TLC over telephone.

- iii) Inform the SM of the next block station in writing about the full particulars of the incident.
- i) In case the track in his opinion is not safe for passage of trains at the normal speed take necessary action, as per action during bad riding (Point 3 & 4) (SR 2.11.01)

STORMS AND CYCLONE (SR 2.11.04)

In case of a train be caught on the run in a cyclone, storm or strong wind which is likely to an danger the safe running of the train in the opinion of Loco Pilot, he should,

- i) Stop the train avoiding sharp curves, high embankment and bridges
- ii) In coaching train ensure that doors and windows of the coaches are kept open by the passengers to allow the free passage of winds through the coaches.
- iii) The train should be restarted in consultation with the guard only after the cyclone, storm or strong wind abates and it is considered safe for the train to proceed.

DEFECTS ON OHE

It shall be the responsibility of every railway servant to report immediately any abnormality, which may come to his notice on the OHE such as

- i) The whole or part of the OHE or a feeder or a cable falling down and or persons, animals or vehicles coming in contact with or likely to come in contact with live equipment.
- ii) Pantograph of an electric rolling stock getting damaged and entanglement of the same with OHE
- iii) A damaged catenary or contact wire falling on the vehicle
- iv) Damage to the track or structure of OHE
- v) Derailment and any other traffic accident

Railway servant noticing the above shall at once report the same to the TPC either directly or through station master over nearest available phone.

RUN OVER CASES

A) WHEN THE PERSON IS ALIVE

- i) When a railway employee finds an injured person near the railway track he shall "Render first Aid"
- ii) Take the injured person to the nearest station in the direction of run and hand over to the station master for further medical aid with the memo showing time and place of incident, make declaration if the condition is found dangerous with the following items. Name, Age, Sex, father/Husband's name, caste, village, how he or she happened to be on the track, cause of accident, time, KM No. (where the injured was found).
- iii) The statement should be signed by guard/driver and a responsible person as a witness if available and to be handed over to the SM
- iv) If the person is found to be in serious condition, the dying statement should be recorded.

B) WHEN THE PERSON IS DEAD

- (i) When a dead body is found on or near the track by Railway servant he shall keep the evidence intact specially the fingerprint.

- (ii) The dead body should not be removed until the arrival of police. However to clear the line for the movement of subsequent trains, the body may be removed to some minimum requirement.
- (iii) The dead body should be left in charge of a village Chowkidar/Lineman/Gateman/other responsible Person with a memo stating the following particulars –
 - (a) Line (up or down)
 - (b) Km. No., where the body was found.
 - (c) Time
 - (d) Position of the body in relation to the track, blood Stains in the ballast or on the engine.
 - (e) Cause of injury whether by a train or otherwise.
 - (f) Position of clothing found on or near the track.
- (iv) Loco Pilot's name and address to be given in the memo for further disposal.

FOGGY WEATHER WORKING

DUTIES OF LOCO PILOT-

- (i) When due to thick, foggy or tempestuous weather the visibility is impaired , the Loco Pilot shall regulate the speed of his train to a maximum of 60 KMPH in absolute block territory and to 30 KMPH in Automatic Block territory. The lower speed limit shall depend upon the visibility condition and the conscience of the Loco Pilot.
- (ii) In Auto section if green aspect is displayed then the speed of train should not exceed 60 kmph.& if the double yellow aspect is displayed then the speed of train should not exceed 30 kmph.
- (iii) Placement of detonator in foggy weather - In Multiple Aspect Signalling system where single distant signal is provided detonators should be placed at 275 meters short of the first Stop Signal .

ACCIDENT AND UNUSUAL OCCURANCES

INTRODUCTION –

While working on line a Loco Pilot may have to meet different circumstances in the form of an accident/obstruction, single line working on double line, running on wrong line, total interruption and train parting etc. And it requires adopting different measures.

The trainee should be able to have a proper knowledge to negotiate different circumstances and be able to follow the rules at the end of the lesson.

WORKING IN THE EVENT OF ACCIDENT –

In case of accident on the line or to any train or of failure or in an emergency or interruption of communication train shall be worked between stations in accordance with a special instruction.

WORKING OF L/Eng OR TRAIN TO AN ACCIDENT SITE/OBSTRUCTED LINE –

SR 6.02.05-Rules and regulation for working of trains on the obstructed line in case of obstruction or accident on the authority of Block Ticket/Authority to proceed for relief engine/train into an occupied block section in Absolute Block System **T/A-602**, which includes -

- i) Block Ticket to proceed without line clear.
- ii) Authority to pass signal in 'ON' position and
- iii) Caution order

Until obstruction is removed, if required, trains shall be worked on the obstructed line upto the place of obstruction on **Block Ticket (T/A-602)** with clear instruction whether the train is to return, or to wait at the place of obstruction or to proceed to the next station.

After sending a train on **Block Ticket**, a following train shall not be despatched in the same direction unless-

- i) The previous **Block Ticket** is collected and cancelled, or
- ii) Necessary endorsement is made on the previous Block Ticket with the advice to wait at the site for a next train to follow, or
- iii) The previous train has met with an accident or has been disabled, or
- iv) The **Block Ticket** has been collected from the Loco Pilot of the previous train by the official incharge at the site and kept in his personal custody and the fact has been relayed to the S.M. over phone quoting the **Block Ticket** number.

If more than one train accumulates at the site, after completion of work, the traffic official incharge at the site with the concurrence of the Station Master through the exchange private no., shall decide which train to go th which side and if possible he shall arrange to couple the trains together and instruct them to proceed cautiously to the station as directed having due consideration to minimise delay to more important trains.

If it is not possible to couple up trains one train may follow another after an interval of 15 minutes at a speed not exceeding 8 KMPH.

SPEED-

Not exceeding 15 KMPH during day light when view ahead is clear and 10 KMPH during night or whenever clear view for 800 meters is not available.

INSTRUCTIONS-

- a) Check carefully the Block Ticket about the instruction given.
- b) Check the date and time of issue of previous Block Ticket and instructions given there in.
- c) Use engine whistle frequently.
- d) Maintain the speed restriction of 15/10 KMPH.
- e) If required protect your train in rear from the last vehicle by placing detonators at **250M-250M-10M-45M** with danger hand signal (SR 6.02.03) if it gets detained at the block section for more than 10 minutes.
- f) On arrival of the train, hand over the Block Ticket to the S.M. with signature, date, time, etc.

SR 9.12.08

Rules and regulation for working of trains on the obstructed line in case of obstruction or accident on the authority of **Block Ticket/Authority** to proceed for relief engine/train into an Automatic Block Signalling section (**T/C-912**)

When it becomes necessary to work trains on the obstructed line on the authority of Block Ticket/Authority to proceed for Relief engine/train into an automatic Block Signalling section, all the instructions contained in SR 6.02.05 shall be followed except in lieu of prescribed form no. T/A-602, form T/C-912 shall be used.

[SR 6.02.05(g)(i)]
SINGLE LINE WORKING INTRODUCED ON DOUBLE LINE WHEN
ONE LINE IS OBSTRUCTED AND COMMUNICATION AVAILABLE

SR6.02.01 Rules and regulation for single line working on a double line section when one line is obstructed:

Single line working is introduced only after obtaining reliable information in writing, that the line on which single line working is going to be introduced is clear of all obstruction in consultation with the Section Controller.

When for the time being trains are being worked on single line working system, the Driver of each train shall be handed over with an “ **Authority for temporary single line working on double line**” (T/D-602), which includes-

- i) A line clear ticket
- ii) An authority to pass signal in ‘ON’ position and,
- iii) A caution order

The authority indicates-

- i) The line on which the train or the light engine is to run.
- ii) The KM between which the obstruction exists.
- iii) Any speed restriction which may have been imposed by way and works staff.
- iv) An assurance to the effect that any trap points on the line in question have been spiked and clamped.

The caution order portion of the authority issued to the Loco Pilot of the first train shall have an endorsement to inform all Gatemen and Gangmen on the way about the introduction of temporary single line working and to specify the line over which the trains shall run.

SPEED-

First train 25 KMPH and subsequent train shall run at normal speed.

INSTRUCTIONS:

a) **For trains proceeding on the right line-**

The last stop signal shall be kept at ‘ON’. It can be passed at ‘ON’ on the authority issued to the Loco Pilot by the S.M. The train may be admitted on signal or be piloted in at the next station.

b) **For trains proceeding on the wrong line-**

- i) The train shall be piloted out of the station on the written authority (T/D-602) issued by the Station Master to the Driver.
- ii) The train shall be brought to a stop at the first stop signal of the right line or the last stop signal of the wrong line whichever ever comes first and from there the train shall be piloted in by T-509.
- iii) The Loco Pilot should switch on the flasher light of his engine when proceeding via wrong line.

SR 6.02.05 (g)(i)

TRAIN RUNNING IN RIGHT DIRECTION BUT ON WRONG
LINE WITHOUT INTRODUCING S/LINE WORKING

On a double line section, if for any reason other than introducing single line working,

it becomes necessary to run an engine or train against the normal direction of traffic, the line concerned shall be blocked (Block back) indicating the section to which the train /engine is to proceed.

After obtaining acknowledgement from the Station Master of the station at the other end, the Station Master intending to run the train shall issue a Block Ticket (**T/A-602**) which shall include-

- i) Block ticket to proceed without line clear.
- ii) Authority to pass signal in 'ON' position.
- iii) Caution order

The caution order portion of the Block Ticket shall have the following particulars-

- a) Existing speed restrictions.
- b) Instructions to Loco Pilot/Guard-
 - i) To work out for possible obstruction and take action accordingly.
 - ii) To ascertain the condition of the train/obstruction over the adjacent line and to report the same at the next station in writing.

SPEED

15 KMPH at day time or when view ahead is clear and 10 KMPH at night or when view ahead is not clear from a distance of 800 meters.

ACTION

The Loco Pilot should sound the engine whistle frequently and he should observe the items mentioned in the caution order.

PROTECTION

If the train which is proceeding with block ticket is held up in the block section due to accident, failure or obstruction and cannot proceed further the train should be protected in rear by placing 3 detonators at 250 meters, 250 meters, 10 meters and exhibit danger hand signal at a distance of 45 meters from the 3rd detonator.. For the protection in front GR 6.03 to be followed.

NOTE

The procedure is restricted to only one goods train or light engine and not for passenger train.

TOTAL INTERRUPTION OF COMMUNICATIONS

MEANS OF COMMUNICATION –

- (i) Block instrument, Track circuit and Axle counter.
- (ii) Telephone attached to block instruments.
- (iii) Station to station fixed Telephones, where available.
- (iv) Fixed Telephones such as Railway Auto Phones and BSNL Phones(with caller ID facility where feasible)
- (v) Control telephones
- (vi) VHF sets

If any of the above mentioned means of communication is not available then it is called total interruption of communication. In this situation, before any train is allowed to enter into the block section in advance, the train shall be brought to a stop at the station first and the Loco Pilot and guard of the train shall be advised of the circumstances by the S.M. on duty.

TOTAL INTERRUPTION OF ALL COMMUNICATION ON DOUBLE LINE SECTION (S.R. 6.02.03)

AUTHORITY - T/C-602,

An authority for working of trains during total interruption of communication on Double Line section which shall include-

- 1 Authority to proceed without line clear.
- 2 An authority to pass the last stop signal at 'ON'
- 3 A caution order restricting the speed to 25 KMPH for straight line or when view ahead is clear. 10 Kmph when view ahead is not clear due to curve, rain, fog, obstruction or any other cause. If required Asstt. Loco Pilot will pilot the train with proceed hand signal.

ACTION/ INSTRUCTION

- (i) Fixed signals except the last stop signal can be taken off.
- (ii) Engine whistle to be used frequently.
- (iii) A tunnel should be entered only after ensuring that it is clear.
- (iv) An interval of 30 minutes is to be maintained between two trains.
- (v) Train must be stopped at the outside of the first stop signal of station in advance first then it may be taken on signal or piloted in.
- (vi) Loco Pilot must give four short whistles (0 0 0 0) if he is unable to proceed further, in block section due to accident/failure or any other reason.
- (vii) Loco Pilot will hand over the authority for working of trains during total interruption of communication on double line section. "T/C-602" to the S.M. of the station at the other end i.e. next station.

PROTECTION

Protection in rear from last vehicle 250m-250m-10m-45m with danger hand signal. When train stop in block or detained more than 10 minutes at the out side of 1st stop signal of the next station and protection in front or adjacent line as per G.R.6.03.

TOTAL INTERRUPTION OF ALL COMMUNICATION ON SINGLE LINE SECTION (S.R.6.02.04) –

In case of total interruption of all communications the driver and guard will be informed and the engine will be detached from the train to open the communication.

AUTHORITY

Before despatching the Light Engine/Train Engine the Station Master on duty shall hand over the Driver T/B-602, an authority for opening of communication during total interruption of communication on single line section, which include-

- i) An authority to proceed without line clear
- ii) A caution order specifying the limit of speed that has to be observed in the affected section.
- iii) An authority to pass the last stop signal at 'ON.'
- iv) A line clear enquiry message (asking line clear for awaiting train).
- v) A conditional line clear reply message for the light/train engine to return light or attached to a train.

NOTE

- i) The last stop signal shall be kept at 'ON'.
- ii) After the light/train engine has been despatched with "line clear enquiry message "

and conditional line clear reply message, no other engine or train shall be allowed to proceed in the same direction until the return of that engine.

SPEED

15 KMPH during day or when the view ahead is clear and 10 KMPH at night or when the view ahead is not clear.

PRECAUTION/INSTRUCTIONS

- (i) Engine whistle to be used frequently.
- (ii) Switch 'ON' the F/light of LE while going to open the communication.
- (iii) A tunnel should be entered only after ensuring that it is clear.
- (iv) When view ahead is not clear then Asstt.Loco Pilot will pilot the engine with hand signal.
- (v) After the engine brought to a stand at the 1st stop signal of next station it may be admitted on signal or may be piloted in.
- (vi) The Loco Pilot of light engine will hand over the authority for opening communication during total interruption of communication on S/L section containing the "**line clear enquiry message**" and "**condition line clear reply message**" to the S.M. at the next station.
- (vii) In the event of the light engine which is going to open communication, meeting another engine coming from the other end, the distance from the nearest station and the importance of the train for which line clear is required, to be taken into consideration, to which direction they will proceed.

NOTE

In case a light engine which is being despatched to proceed to the next station on the authority **T/B-602**, is not meant for opening the communication and to continue its journey onward, "**Line clear enquiry message**," and "**Conditional line clear message**" shall be struck out. In this case another engine can be sent at an interval of 30 minutes. The Loco Pilot of Light Engine will hand over the authority **T/B-602** to the S.M. on arrival at the next station.

After opening communication, when the engine is returning light or attached to a train

the Loco Pilot shall be handed over with-

- i) Condition line clear ticket **T/G-602** for up train and **T/H-602** for dn train.
- ii) Conditional line clear reply message (**T/F-602**).
- iii) **T/369(3b)** to pass the last stop signal at on, and
- iv) Caution order (**T/B-409**).

SPEED

While returning light or attached to a train after opening communication on the authority of **conditional line clear ticket**, it may run at **normal speed**.

INSTRUCTION

On approaching the next station the Loco Pilot of the light engine/train shall bring his train to a stop at the first stop signal, from where it may be taken on signal or be pilotet-in to the station. On arrival at the station, the Loco Pilot shall hand over the **conditional line clear reply message** to the Station Master.

On the basis of line clear reply message now the S.M. will allow trains into the affected section with the following authorities-

- i) Condition line clear ticket. (**T/G-602** for UP trains & **T/H-602** for DN trains)

- ii) Authority to pass last stop signal at 'ON' **T/369(3b)** and
- iii) Caution order (**T/B 409**).

NOTE

- i) The first train entering into the affected section after opening communication may run at normal speed. A clear interval of 30 minutes shall be kept between the preceding train and the next train to proceed into the affected section
- ii) The second and subsequent train shall maintain a speed restriction of 25 KMPH on straight when view ahead is clear and 10 KMPH when view ahead is not clear due to rain, fog, curve, obstruction etc. which shall be endorsed in the caution order issued to the Loco Pilot of the subsequent trains.
- iii) The line clear ticket shall contain the departure time of the last preceding train.

PROTECTION

When being worked on the authority of "Conditional line clear ticket" issued on the basis of "Conditional line clear reply message", if due to accident, failurer or obstruction a train can not proceed further, the Loco Pilot shall sound "0000" (4 short) whistle, where upon the Guard shall protect the train in rear by placing detonator in rear from the last vehicle at 250M-250M-10M and at 45M with danger hand signal and also the train shall be protected in the same manner when it is being detained outside the signal for more then 10 minutes.

NOTE: No train shall be backed in the block section when trains are following one another, if it becomes necessary under exceptional circumstances, it can be backed only after providing protection as mentioned above.

AUTOMATIC SYSTEM

PROCEDURE TO PASS LAST STOP SIGNAL AT 'ON' ON DOUBLE LINE AUTO BLOCK SECTION (S.R.9.14.01(b))

When the last stop signal which may either be a manual stop signal or semi automatic signal working as manual stop signal ('A' marker is not glowing) is defective or at 'ON', it can be passed on the authority of

- (i) T/A-912
- (ii) Proceed hand signal shall also be exhibited at the foot of the defective signal if there is any point beyond the signal.

SPEED

15 kmph in normal visibility condition up to the next auto stop signal or 8 KMPH when owing to curvature, fog, rain, dust, storm etc. the visibility condition is impaired and be prepared to stop within the half of the visible distance.

PROCEDURE TO PASS LAST STOP SIGNAL AT 'ON' IN AUTO SECTION WHEN THERE IS NO INTERVENING AUTO STOP SIGNAL BETWEEN TWO CONSECUTIVE BLOCK STATION (i.e. no intervening signal)SR9.14.01(b)

AUTHORITY –

- (i) An authority to proceed on Automatic Block System during prolonged failurer of signals, **T/D-912** with private no. .

- (ii) Caution order, **T/B-409**

SPEED – 25 kmph for 1st train and subsequent train with normal speed.

ACTION – Train may be admitted on signal or piloted in at the next station.

ABNORMAL WORKING IN AUTO BLOCK SECTION

PROLONG FAILURE OF AUTO STOP SIGNAL WHEN COMMUNICATION ARE AVAILABLE ON DOUBLE LINE (SR 9.12.01)

AUTHORITY/ACTION –

- (i) An authority to proceed on auto block system during prolonged failure of signals **T/D-912** with private number distinguishing numbers of the departure and gate signals required to be passed, shall also be indicated in this authority authorising the Loco Pilot to pass them.
- (ii) Caution order **T/B-409**
- (iii) On approaching at the next station train may be admitted on signal or piloted in.

SPEED – 25 kmph for 1st train and normal speed of subsequent trains.

AUTO STOP SIGNALS ARE WORKING BUT COMMUNICATION IS NOT AVAILABLE ON DOUBLE LINE SECTION (SR 9.12.05)

AUTHORITY/ACTION

For 1st train

- (i) A caution order (Containing the list of trains if any) indicating the circumstances.
- (ii) A copy of list of trains to be run on that section.
- (iii) Train may be admitted on signal or piloted in only after the train stop at the first stop signal of the next station.

For Subsequent Train –

Train may run as per signal aspect.

SPEED – Follow as per aspect of auto signal for first and also subsequent train.

NOTE – At next station the driver of the first train shall hand over the list of trains to the S.M., which was given by the Station Master of the rear station.

FAILURE OF AUTOMATIC STOP SIGNALS AND ALSO COMMUNICATION IS NOT AVAILABLE ON DOUBLE LINE (Total interruption) (S.R.9.12.02)

AUTHORITY –

Authority to proceed without line clear on auto block signalling territory (**T/B-912**) which shall include-

- i) An authority to proceed without line clear.
- ii) An authority to pass automatic/ semiautomatic/manually operated gate signal at 'ON' with individual distinguishing number of each signal.
- (iii) Caution order.

SPEED – 25/10 kmph as per visibility.

ACTION –

- (i) Engine whistle is to be used frequently.
- (ii) A tunnel should be entered only after ensuring that it is clear.
- (iii) 15 mts. time interval to be maintained in between the two trains.
- (iv) Loco Pilot must stop the train at first stop signal of the next station. The train may be admitted by taking 'OFF' the signal or piloted in.
- (v) In case of accident or failure in block section, give four short whistle (0000) and the train to be protected in rear from last vehicle as G.R.6.03 i.e. 600-600-10-10-45m & danger hand signal .When train is detained for more than 5 mts. at the out side of first stop signal.GR 4.44 to be followed.
- (vi) Hand over the "Authority to proceed without line clear"(T/B912) in Automatic block signalling territory to S.M of the next station.

ON S/L AUTO SECTION WHEN COMMUNICATION ARE NOT AVAILABLE AND DURING SUSPENSION OF PANNEL WORKING OR S/L WORKING INTRODUCED ON D/L WHEN NO MEANS OF COMMUNICATIONS ARE AVAILABLE (S.R.9.12.06 or S.R.9.12.04)

In the above mention two circumstances, trains shall be worked between the stations concerned in the manner as laid down in **SR 6.02.04.** except that in lieu of **T/369 (3b).**

The prescribed form. **T/A-912** shall be issued. **T/A -912** shall indicate the individual number of Automatic/semiautomatic/manually operated gate signals at 'ON'. In case of departure signals, the Loco Pilot shall pass the signals only on being hand signalled past by a railway servant gate signals shall be passed cautiously up to the level crossing , where he shall ascertain that the gates are closed and proceed hand signal is being displayed by the gate man.

SR 9.12.03 Working of trains under Automatic block system (Double line) during obstruction of one line when signals are operative and communications are available:-

1. Single line working will be introduced only on receipt of written information from guard/driver/PWI/Traction Foreman that the line on which single line working is to introduced is clear of obstructions.
- 2 Single line working will be introduced between the nearest station provided with cross - overs between UP and Down lines on either side of the obstruction.

Authority/Action-

- a) **Loco Pilot of all trains running on the wrong line and each first train to run on right line shall be given**
 - i) A paper line clear ticket on prescribed form (**T/C-1425** for UP train and **T/D -1425** for DN trains)
 - ii) On authority on prescribed form (**T/A- 912**) authoring the Loco Pilot to pas the intervening non governing /governing semi automatic and maually operated signal on being hand signal past by a point man uniform. The distinguishing no of each signal shall be indicated in the authority .Semi automatic gate signal shall be passed cautiously only after ascertaining that the gate are closed and proceed hand signal displayed by the gateman.
 - iii) A caution order indicating -
 - a) The line on which single line working has been introduced
 - b) The kilometerage at which obstruction exist on the other line
 - c) Speed restriction if any.

An endorsement shall also be made in the caution order issued to the Driver of the first train after introduction of the single line working to stop and inform all gate men ,

gang mates and key man about the introduction on single line working specifying the line on which trains shall run.

- b) Subsequent trains run in right line shall be allowed to follow to each other on the indication of Automatic signals.

Special instructions-

1 For right line trains - All fixed departure signals can be taken off and the trains also can be admitted on signals at the next station.

2 For wrong line trains- All trains shall be piloted out with **T/A-912** and the trains shall also be piloted in at the next station by **T-509** after it has come to a stop opposite first stop signal of the right line or last stop signal of the wrong line, whichever comes first and whistle the horn.

Speed First train 25 kmph and subsequent train may run normal speed.

ON SINGLE LINE AUTO SECTION PANEL WORKING IS SUSPENDED BUT COMMUNICATIONS ARE AVAILABLE (S.R.9.12.07)

AUTHORITY

- (i) A line clear ticket on prescribed form (**T/C-1425** for up train & **T/D-1425** for DN train)
- (ii) An authority on prescribed form (**T/A -912**) indicating the individual number of Automatic /semiautomatic/manually operated signal to be passed at 'On'.
- (iii) Caution order (**T/B- 409**).

ACTION

- (i) Loco Pilot should be very cautious while passing level crossing gates. He must ensure that gates are closed and locked against the road traffic and proceed hand signal displayed by the gateman.
- (ii) Train may be admitted on signal or piloted in at the next station.

Speed:- 25 kmph for the first train, subsequent trains may run at normal speed.

WORKING OF TRAIN IN AUTO BLOCK SYSTEM ON SINGLE LINE WHEN DIRECTION OF TRAFFIC IS ESTABLISHED BYUT LAST STOP SIGNAL IS DEFECTIVE (B.W.M. 6.23)

AUTHORITY –

- (i) **T/512**
- (ii) **T/A-912.**
- (iii) Displaying hand signal at the foot of the defective signal.

SPEED – Not exceeding 15 or 8 KMPH as per the visibility up to the next Auto Stop Signal.

AUTHORITY TO PROCEED IN AUTO BLOCK SYSTEM –

Under normal working on double line or single line –taking ‘OFF’ of the last stop signal.

TRAIN PARTING, INABILITY OF THE ENGINE TO HAUL THE FULL LOAD

TRAIN PARTING (G.R.6.08)

ACTION –

- (i) If a train is parted on run, the Loco Pilot should keep the first portion in motion if possible, till such time the rear portion comes to a stop to avoid collision between the two portions (but do not pass any stop signal at ‘ON’ for doing this).
- (ii) Loco Pilot should give engine whistle code (– 0 – 0) one long one short one long one short, to inform the guard and put ‘ON’ the flasher light to stop an approaching train on adjacent line if required.
- (iii) Guard will apply hand brake or drop the vacuum/air pressure to stop the second portion and will show a green flag by day. While light by night, up and down vertically as high and as low as possible and protect the train in rear and front portion as per G.R.6.03.
- (iv) When both portions of parted train are brought to a stand, try to coupled-up both portions if possible with due caution under the hand signal from the guard and couple the train.
- (v) If the coupling is not possible then work the train into two portions. The guard will allow 1st portion mentioning last vehicle number advising SM to arrange to clear the 2nd portion left in block section. With written memo on prescribed form no- T/609. On single line section, Loco Pilot shall hand over the token to the guard and obtained a written receipt. He shall bring his train to a stop at the first stop signal at next station and sound one long one short one long one short whistle. The station master to arrange to receive the train on pointing in on convenient line.
If asked for by the guard after dropping the first portion the driver will return to the spot to clear the second portion on guard’s written authority.
- (vi) If first portion already reached the block station in advance the Loco Pilot will give parting whistle code (- 0 - 0) repeatedly to warn the cabin man and station staff and stop the train without clearing the fouling mark (F.M.) of the station.
- (vii) In single line section do not hand over the tangible authority if any to the S.M.
- (viii) If there is a banking engine on train, the Loco Pilot of banking engine shall arrange to stop the rear portion and he will give one long one short one long one short (-0-0) and will arrange to protect the second portion in rear as per G.R.6.03.
- (ix) The guard will protect in front as per GR6.03
- (x) An engine will be sent by ASM with Block ticket to clear 2nd portion, which was left in block section.

Miscellaneous

AIR BRAKE SYSTEM USED IN ROLLING STOCK

INTRODUCTION:

Air brake used in rolling stock utilizes compressed air for brake application. The reserve braking power in the form of compressed air is stored in the reservoir of each vehicle. A control valve or a distributor valve is attached by brake pipe pressure that run through the length of the train and governs the flow of compressed air into and out of the brake cylinder. /The pressure in brake pipe is normally maintained at a predetermined value of 5 Kg/cm² and brake application occurs when the pressure in the brake pipe is reduced either intentionally or automatically.

SALIENT FEATURES

Air brakes are efficient and reliable to run heavy and long train at high speed. Air brake has following advantages:

- i) Shorter braking distance
- ii) High braking force
- iii) Reduced brake power deterioration
- iv) Uniform brake power over train length of the train
- v) Compact and light equipment.

TESTING PROCEDURE

The Loco Pilot and guard shall ensure at the time of taking over charge of a train that their train has the prescribed minimum brake power as follows:

OPM 17.26

(AIR STOCK) SPECIFICATION FOR PERCENTAGE OF BRAKE POWER

Types of rake	Originating	Enroute
CC Rake	95%	75%
Normal Rake	85%	75%
Coaching train	100%	10% IOP of the total No. of cylinders (Max. 02 cylinders only)

OPM 17.04 (VACUUM STOCK)

SPECIFICATION OF PERCENTAGE OF BRAKE POWER

Types of rake	Originating	Enroute
Goods Train	85%	75%
Coaching train	100%	10% IOP of the total No. of cylinders (Max. 3 cylinders only)

OPM 17.24 (b) GAUGE READING (AIR STOCK)

TYPE OF TRAIN	BP ON LOCO	BP ON BRAKE VAN
Coaching train	5 kg/cm ²	4.9 Kg/cm ²
Goods train up to 56 wagons	5 kg/cm ²	4.8 kg/cm ²
Goods train more than 56 wagons	5 kg/cm ²	4.7 kg/cm ²

SPECIFICATION OF VACUUM ENGINE AND BRAKE VAN

Type of train	Engine Gauge	Brake Van gauge
Mail/express	53 cm	47 cm
Passenger Train	50 cm	44 cm
Goods train	46 cm	38 cm
40 or more BCX/Box wagon	44 cm	34 cm
Engine	58 cm	53 cm
	on dummy	on disc

CHECKING OF BP CONTINUITY

• At originating station

1. Ensure BP pressure is 5kg/cm² in engine & 4.8 to 5.0 kg/cm² in brake van.
2. Open rearmost BP angle cock to blow. When BP drops to 4 kg/cm² ,isolate L/T cock & wait till BP comes to '0'.(In this process all the dirt present in the BP pipe line will be exhausted to atmosphere)
3. Again recharge BP in the train.
4. By operating A-9, check brakes are applying & releasing in entire train with indication of air flow indicator.
5. Operate guard's brake valve & drop BP Pressure. Same amount of BP Pressure should be dropped in engine gauge & air flow indicator should shoot up.
6. Now after recreating BP, drop BP from engine by operating A-9 handle. The same drop of BP to be noted on B/V gauge.

• At enroute

1. Operate A-9 to drop BP pressure.
 1. LP & ALP shall check 4 to 5 wagons in front for brake application & shoot up of air flow indicator
 2. Guard shall check brake application of 4 to 5 wagon in rear.
 3. Again operate guard's brake van valve to drop BP & check application of brake in front by LP/ALP & by guard in rear.

Note:

1. Feed Pipe continuity to be done by dropping from rearmost vehicle.
2. Sensitivity of the brake system demands a minimum brake pipe pressure reduction of 0.6kg/cm² in 6 seconds for brakes to apply. If a train is stalled or comes to a halt for any reason with coach brake pipe fully charged, no brakes will apply even if brake pipe pressure reduces to '0' due to normal leakages in the system(i.e; if the rate drop is less than 0.3kg/cm² in 60 seconds. Loco Pilot must therefore ensure that brakes are applied by a positive act of placing A-9 at full service, so as to avoid rolling down of the train on its own on gradients.

RELEASING THE BRAKE BINDING OF AIR STOCK WAGON:

- i) Ensure BP pressure on engine gauge is 5 kg/cm².
- ii) Ensure that there is no air leakage from the affected wagon.
- iii) Ensure that the hand brakes of the wagon are fully released.
- iv) Ensure empty/load handle is on proper position.
- v) If above all items are at right and normal then release the distribution valve (DV) by pulling the releasing valve till brakes are released even after releasing the valve then

- isolate distributor valve by pulling the isolating handle to the isolate position and then release the QRV.
- vi) If still brakes are not released, operate SAB (Slack adjuster Barrel) in anti clockwise direction.
 - vii) If the brake binding still does not release, disconnect the SAB by taking out its connecting pins from the live lever.
 - viii) If still does not release, take out pin from long or short pull rod and secure the loose fittings.
 - ix) If still does not release, remove the brake block of affected wagon.

CHECK LIST ON AIR BRAKE SYSTEM

A Loco Pilot must not work his train without a valid brake power certificate (BPC).

INTRODUCTION:

A BPC is the document through which a Loco Pilot know whether his train is possessing the required percentage of brake power, it is the document by receiving which the Loco Pilot grows in confidence that he can work his train safely and can bring his train under control or to a stop as and when required. A Loco Pilot must not work his train without valid BPC.

IDENTIFICATION OF BPC

A BPC shall be in a prescribed form No. Mech./V-5 RB-1 for vacuum stock trains and Mech/V-5 RB2 for air stock trains used for this purpose and may be of rosy or greenish colour. Now a days it may be in white colour.

ISSUE OF BPC

BPC for a train is issued by the train examiner (TXR) who shall before issuing the same satisfy himself that the train is having prescribed amount of vacuum/air pressure on engine and brake van gauges. The train must also have the required percentage of brake power. He shall also ensure that the relevant columns of BPC have been properly filled up and BPC is duly signed by the train driver guard and finally he himself (TXR) has signed the same.

PLACE OF ISSUE:

Except otherwise required, BPC shall be issued from the train originating station. It may also be required to be issued from a station enroute under certain conditions.

TYPES OF BPC:

BPC caters to two types of rakes i.e.

- (1) NORMAL RAKES
- (2) CLOSE CIRCUIT RAKES (CC RAKE)

NORMAL RAKES:

The BPC of the normal rakes/trains contains the details of the train No./load, percentage of brake power and details of brake system, effects on the vehicles of the train TXR driver and guard should sign the certificate respectively.

CLOSE CIRCUIT RAKES:

The BPC for close circuit rakes is in the form of full scrape sheet (A-4 size) apart from the details as contained in a normal brake power certificate sheet, it contains columns indicating the driver's name, HQ, Train No. Load details, journey from & to, No. of KM run & the cumulative KMs covered by the rake.

The Loco Pilot should fill in the relevant details such that the total KM covered by CC rakes can be monitored and offered for intensive examination by carriage and wagon staff (C&W), which is normally done after running 5500 KMs/6000KMs.

VALIDITY OF BPC:

A BPC normally remains valid from the train originating station (place of issue) to destination station yard. BPC may get invalid under the following conditions:

- i) If the integrity of the rakes is disturbed and the composition is changed by 10 or more 4 wheeler units or its equivalent, in case of other units.
- ii) If the train remains stabled at a station for more than 24 hours.
- iii) The BPC for the close circuit rakes (air brake) is valid for 5500 KMs for Box N and 6000 KMs for BCN/BTPN rakes or 1 month from the date of issue which ever is earlier provided the integrity of the rakes is not altered.

The Loco Pilots are instructed to make entry in the proforma given in the BPC of CC rakes. Loco Pilot/SM/SS should inform to controller if the rake is running with invalid BPC so that the same can be directed for further examination.

CALCULATION OF WAGON IN TERMS OF UNITS

A) COACHING STOCK

- i) A four wheeler or six wheeler wagon/vehicle 1 unit
- ii) A bogie vehicle i.e. 8 wheeler 2 units

B) GOODS STOCK

- i) A four wheeler wagon 1 unit
- ii) A eight wheeler wagon 1 unit
- iii) BRH/BOB/BCN/Box N/Box 1 unit

ACCIDENT:

Any occurrence which does or may affect the safety of the railway/loco/rolling stock/ permanent way/Passenger/railway servant or which does or may cause delay to the train movement or loss to the railway is called the accident.

SERIOUS ACCIDENT

Accident to a train carrying passengers which is attended with loss of life or with grievous hurt to a Passenger or Passengers in the train or with serious damage to railway property of the value exceeding Rs. 2 Crore (2,00,00,000) and any other accident or commissioner of Railway Safety shall also be deemed to be a serious accident. However, the following shall be included :-

- a) Cases of Trespassers run over and injured or killed through their own carelessness or of Passengers injured or killed through their own carelessness.
- b) Cases involving Persons being Railway servant or holding valid Passes/tickers or otherwise who are killed or grievously injured while travelling outside the rolling stock of a passenger train such as on foot board or roof or buffer but excluding the inside of vestibules between Coaches or run over at a Level Crossing or elsewhere on the Railway track by a train; and
- c) Level Crossing accident where no railway servant is killed or grievously hurt unless the Chief Commissioner of Railway Safety or Commissioner of Railway Safety is of the opinion that the accident requires the holding of an inquiry by the Commissioner of Railway Safety.

AVERTED COLLISION

An averted collision is a circumstance under which but for the vigilance shown by any person or person, a collision would have occurred either in the block section or within the station limits between two trains or a train and an obstruction.

Provided, Further that such an occurrence may not be treated as an averted collision:

- i) If outside the station limit, the distance between the two trains or the train and obstruction at the time the train or trains have finally come to a stop in 400 meters or more.

- ii) If within the station limits, there is an intervening stop signal at danger governing the moving train and compliance by the moving train with indication converted by the stop signal averted the collision between the trains, however between the train and obstruction.

CLASSIFICATION OF ACCIDENT:

There are 16 classes of accident i.e. “A” to “R” except “I” and “O”

1	Class “A”	Collision
2	Class “B”	Fire or explosion in train
3	Class “C”	Train running into road traffic or traffic running into train at L/C
4	Class “D”	Derailment
5	Class “E”	Other train accident
6	Class “F”	Averted collision
7	Class “G”	Breach of block rules
8	Class “H”	Train passing signal at danger
9	Class “J”	Failure of engine and rolling stock
10	Class “K”	Failure of permanent way
11	Class “L”	Failure of electric equipment
12	Class “M”	Failure of signal and telecommunication
13	Class “N”	Train wrecking
14	Class “P”	Casualties
15	Class “Q”	Other incidents
16	Class “R”	Miscellaneous

TYPE OF ACCIDENT RELIEF EQUIPMENT:

Following types of accident relief equipments are provided in this railway.

- i) Accident relief medical equipment (ARME) ii) Accident relief train (ART)

SOUNDING OF HOOTER FOR TURNING OUT OF RELIEF TRAIN/ARME VAN IS AS FOLLOWS:

Equipments	No. of blast	Duration of Each blast	Gap between two consecutive blasts
Relief Train with ARME	5	60 seconds	10 seconds
Relief train without ARME	3	60 seconds	10 seconds

TARGET TIME FOR THE TURN OUT OF ARME/ART

- i) The ARME van must be dispatched to the site of accident within 15 minutes from the base station, after sounding the hooters with double exit siding and 25 minutes in case of single exist siding with the first available engine.
- ii) The ART be turned out/dispatched from the base station to the site of accident within 30 minutes by day and 45 minutes by night after sounding the hooters.

MISCELLANEOUS:

TRAIN HELD UP AT 1ST STOP SIGNAL (GR 4.44)

The Loco Pilot shall sound one long continuous whistle after 5 minutes Assistant Loco Pilot will proceed to cabin or stationmaster’s office showing a danger hand signal towards the station. If the signal still remains at ON

Circumstances	Absolute Block System	Auto Block System
Normal case	After 15 minutes if neither the signal is OFF nor anybody comes to pilot the train, It shall be protected in rear as per GR 6.03 (600-600-10-10-45M danger hand signal)	After stopping for 15 minutes the train must be protected as per GR 6.03.
During total interruption of all communication	After 10 minutes the train shall be protected in rear as per SR 6.02.03 on D/L as 250M-250M-10M-45M with danger hand signal and as per SR 6.02.04 on S/L 250M-250M-10M-45M with danger hand signal	ON DOUBLE LINE – after 5 minutes train shall be protected in rear as per GR 6.03 i.e. 600M-600M-10M-10M-45M danger hand signal ON SINGLE LINE – After 10 minutes train shall be protected in rear as per SR 9.12.06 i.e. 250M-250M-10M-45M with danger hand signal.

AUTHORITY OF SHUNTING:

Sl.No.	Rule No.	Circumstances	Authority
1	GR 8.15 SR 5.13.01	When fixed signals are provided for shunting	i) Written permission for shunting ii) By taking OFF shunt signal, calling on signal, starter (other than last stop signal)
2	SR 5.13.02	When fixed signals are not provided or defective	i) T-806. ii) Hand signal
3	SR 5.16.01 GR 8.06 (iii)	Infringing block section during shunting purpose	i) T-806 with Private No. for block back or block forward or a written advice to perform shunting in rear of travelling away train ii) Hand signal
4	SR 8.15.01 (b)	When shunting is required by obstructing block section on S/L token block system	i) T-806 with private No. for block back or written advice to do shunting in rear of a travelling train
5	SR 5.14.06	Shunting from UP line to DN line or vice versa	i) T806 with Private No. and block back/block forward. ii) Hand signal

FIXED SIGNAL TO CONTROL SHUNTING(SR 5.13.01)

1. Shunt Signal
2. Calling On signal
3. Starter signal (other than last stop signal)

SHUNTING AUTHORITY(SR 5.13.02)

T-806 (When shunting operation is not controlled by fixed signal)

NOTE: T-806 shall be returned after the completion of shunting.

SHUNTING ON THROUGH GOODS TRAIN(SR 5.14.02):

Whenever shunting on a through goods train is to be performed at a station, person in charge of shunting operation shall satisfy himself that at least 15% of hand brakes are applied for the wagons which remains stationary i.e. which are not involved in shunting operation (SR 5.14.02).

LOOSE SHUNTING

It is a shunt movement in which vehicles which have to be moved are given an impetus by an engine (with or without other vehicles attached) and then after allowed to run forward unattached (SR 5.21.01 (a))

FLY SHUNTING (SR 5.21.01(b))

It is a shunt movement in which two or more vehicles to be moved after being given an impetus by an engine (with or without vehicle attached) are separated at the points, by the points being reversed smartly between “cuts” in order to send them on to different lines.

LOOSE /FLY SHUNTING IS NOT PERMITTED IN FOLLOWING CASES:

- i) The wagon on which hand brake is defective
- ii) Loose/fly shunting is prohibited on the face of gradient or dead end siding or an open derauling switch
- iii) During storm or cyclonic weather
- iv) With the wagon loaded with inflammable liquids, explosives, articles/coaching bogies/empty oil tank or the wagons stencilled or labelled “(NOT TO BE LOOSE SHUNTED)” and ODC wagons also single consignment loaded on more than one wagon.

Responsibility of Loco pilots/Shunters during shunting.

For commencing shunting:-

- 1) Check the shunting order (Property filled and you understood the instructions there in). Do not start shunting unless the shunting movement is clearly understood.
- 2) Check the signature of SM & Shunting supervisor on the shunting MEMO.
- 3) Ensure that shunting supervisor (Guard/SM/Brakes man /YM/AYM/YF/Shunting Master/shunting Jamadar/Points man) is present during shunting and is personally checking the setting & clamping of points as when required (SR 5.13.03).
- 4) The Shunting order shall be retained till the shunting is completed (SR 5.03.02).

During Shunting:-

- 1) Check the brake power of the loco before starting shunting (ACTM 30644).
- 2) Ensure brake power after every cab changing (ACTM 30635 (7)).
- 3) Always perform shunting from leading cab, avoid back cab shunting (GR 4.21)
- 4) While EOT on coaching train stop the Engine/coach 20mts before the train and move with great caution to avoid any bump (SR 5.13.04).

- 5) While passing over points the LP/ALPs to be vigilant & cautious, check visually points are set & locked clamped (GR 5.13).
- 6) Shunting speed should not exceed 15 Kmph if not other wise restricted (GR 5.13).

After Completion of shunting:-

- 1) Handover the shunting order duly signed with time to the shunting supervisor (SR 5.13.02).
- 2) Check the continuity if attaching/detaching has been done (GR 4.32).
- 3) On run do not forget to do brake feel test if attach/detach has been done, the brake feel test should be conducted between 20 to 30 Kmph.

RAIL FRACTURE (SR 15.17.01)

If a rail is badly fractured, no train shall be allowed to pass over the affected portion of track until the fractured rail is replaced or otherwise made safe by an official not below the rank of PWI and certified fit by him.

A gap of more than 75mm (25mm in the case of outer rail of a curve) or there is more than one fracture in a length of 1 meter along the rail head are known as badly fractured rail.

SR 15.17.02 IN CASE OF RAIL FRACTURE LESS SERIOUS IN NATURE

In this case permanent way Mistry/gangmate/Keyman may allow the train at the speed not exceeding 20 KMPH but only after making the rail safe for purpose by using special/joggled fishplate if available for the passage of train until the track has been made safe as above. No train shall be allowed to pass over the affected portion.

COMMON LOOP

The loop line in which both UP and DOWN trains may be taken and dispatched easily on signal is called common loop, speed not exceeding 10 KMPH.

PASSENGER TRAIN ADMITTING IN GOODS YARD (SR 3.40.03)

If for any reason a train carrying passengers is required to be admitted in goods loop or in goods yard then.

- i) If possible a caution order is to be issued to the Loco Pilot and Guard from station in rear stating that the train shall be received on goods yard or goods loop at next station.
- ii) In above both cases the train shall be stopped at the 1st stop signal and then the signal shall be taken Off.
- iii) In case the station in rear cannot be advised then the SM of the concerning station shall send a memo in duplicate to the driver informing the fact. Speed 10 KMPH.

DIFFERENT FORMS/AUTHORITIES USED IN CONNECTION FOR TRAIN MOVEMENT

T/369(1) _

1. Advance authority to pass defective reception signals of the station in advance.
2. This is used in absolute block system only.
3. For goods trains it is issued from the station immediately in rear of the block station at which the signal become defective
4. If the station immediately in rear happens to be a block hut, T/369(1) shall be issued from the station in rear of the block hut.
5. For mail express trains this authority shall be issued from the last stopping station of the train.

T/369(3b)

It is a written authority

- i) For Piloting a train into the station when outer, home or routing signal at ON or defective.
- ii) For piloting a train into the station where conflicting signal occurs.
- iii) For piloting in a train into the station when approaching stop signal remains flickering/bobbling or showing more than one aspect or no light or white light.
- iv) For passing IB signal at ON or defective when rear station is aware of the defective of IB signal.
- v) For dispatching a train when lock and block instrument is suspended.
- vi) For piloting out a train from the station during non-interlocking work.
- vii) For piloting out a train when departure stop signals at ON or defective.

T/509- Authority to receive a train on an obstructed line or non signalled line.

T/511

It is a written authority to start a train from a non signalling line.

NOTE:

- a) Tangible authority and T/511 will be given when more than one train is waiting for departure for the same direction in single line sections.
- b) If there is only one train T/511 will not be given, only tangible authority will be given
- c) During S/L working for wrong line if more than one train is waiting for same direction then T/511 is given.

T/512- Authority to start a train from a line with common starter.

Working of a train in auto block system (S/L) when direction of traffic is established but last stop signal is defective T512 and T/A-912 is given.

T/806- Shunting order

This authority is given -

- 1) When there is no fixed signals to control shunting movements.
 - 2) When fixed signals provided for shunting movement has become defective.
 - 3) When it is required to perform shunting movement by entering in to the block section.
- Note- When shunting movements are being performed by entering in to the block section after 'Blocking back' or "Blocking forward" the authority shall contain Pvt. No .

T/A-912

It is a written authority

- i) To pass governing and non governing auto stop signal/semi-auto stop signal, working as manual stop signal/gate stop signal in auto block territory at ON or defective without private No.
- ii) To pass last stop signal at ON or defective on D/L auto section

T/B 912

Authority to proceed with out line clear on Automatic block signalling territory.

T/C-912

Authority to proceed for relief engine/train into on automatic block signalling section

T/D-912

- i) Authority to proceed on Automatic block system during prolong failure of all or series of automatic stop signals.
- ii) To pass the defective last stop signal when there is no intervening stop signals provided between two block stations (consecutive) in auto signalling territory.

T/A - 602 - Block ticket

- i) Authority to proceed for relief engine /train on occupied line during heavy engineering work.
- ii) With out interduceing single line working train to run on right direction but on wrong line

T/409-Caution order.

T/A-409- Nil Caution order.

- i) From train originating station
- ii) From the noticed station.

T/B -409- Remainder caution order- Form intermediate stations.

T/C-1425 Paper line clear ticket for UP train

T/D-1425 Paper line clear ticket for DN train

- i) During S/L working introduced on D/L auto block system.
- ii) To pass last stop signal at ON or defective on token less system (S/L).
- iii) When block token instrument is out of order or not provided.(S/L).

ED 9.39 (UP) /ED 9.40 (DN)

Engineering stop order for UP and Down trains.

OCC key or SKB

Shunting authority to the station provided with token less instrument on S/L section.

PRECAUTION AND PROCEDURE WHEN TRAINS STOP IN 1 IN 150 (1/150) OR STEEPER GRADIENT IN SECTION FOR MORE THAN 15 MINUTES DUE TO ANY REASON (SR 6.04.02 d(i))

- (i) Apply hand brakes of the loco in addition to vacuum/air brake.
- (ii) In case of roller bearing stock on a section 1/150 gradient or plane bearing stock on a stock 1/100 gradient, the following additional precaution to be taken.
 - (a) **PASSENGER TRAIN** – Guard will apply hand break of the brake van and put sprags and wedges or scotch block in the wheels of two vehicles nearer to the down gradient.
 - (b) **GOODS TRAIN** –
 - (I) Asstt. Loco Pilot will apply hand brakes of 10 wagons in rear of loco and guard will apply hand brakes of brake vans and 5 wagons from the brake van side or 1/3 of the total wagon which ever is more.
 - (ii) When train is ready to start create/charge normal vacuum/air pressure and again apply the brake.
 - (iii) Remove the sprags/wedges, release the hand brakes of wagon.
 - (iv) Recreate or recharge vacuum/air pressure and then start the train after ex changing signal with guard.

NOTE

If the stock is air stock then before applying the hand brake of loco, Loco Pilot must operate DAB handle of loco to emergency position.

SPEED RESTRICTION ON DIFFERENT CIRCUMSTANCES:

Sl.	Rule No.	Speed	Circumstances
1	SR3.40.03	10 KMPH	Passenger train admitting in goods yard
2	SR 3.70.01	15 KMPH	Piloting in and piloting out
3	GR 4.10	15 KMPH	Non interlocked points
4	SR 4.10.01	15 KMPH	Passing through the loop line
5	SR 2.11.03	20 KMPH	When wheel slipping on electric loco then subsequent trains speed
6	SR 4.10.02	10 KMPH	1 in 8½ turnout
7	SR 4.11.01	50 KMPH 75 KMPH	Standard I interlocking Standard II interlocking Max. Perm. Standard III interlocking
8	GR 4.12	25 KMPH	Engine pushing
9	SR 4.20.01(a)	25 KMPH	Driver becomes incapacitated, Asst. driver to take the train up to the next station
10	GR 4.12	40 KMPH	Search light special
11	SR 4.27.03	40/15 KMPH	Steam crane attached to a train
12	SR 4.62.05	25/8 KMPH	Material train pushing
13	GR 5.13	15 KMPH	Shunting speed generally
14	SR 5.14.07 (c)(i)	8 KMPH	Shunting with inflammable goods
15	SR 5.14.07(c)(ii)	5 KMPH	All bogie wagons fitted with roller bearing
16	SR 5.14.07(c)(iii)	3 KMPH	When shunting a group of 5 wagon fitted with roller bearing and transition coupling (T.C.)
17	SR 5.14.07(d)(iii)	5 KMPH	The impact speed with rolling down hump yard
18	SR 6.02.01	25 KMPH	S/L working introduced on D/L speed for the first train
19	SR 6.02.03 (g)(ii)	15/10 KMPH	Train to run on wrong line without introducing S/L working on D/L with an authority OPT 462
20	SR 6.02.02 (a) (ii)	15/10 KMPH	Train run with block ticket

Sl.	Rule No.	Speed	Circumstances
21	SR 6.02.03	25 /10 KMPH	Total interruption on D/L absolute block system
22	SR 6.02.04 and SR 9.12.06	15/10 KMPH	During total interruption on S/L section on absolute and auto block system
23	SR 9.12.02	25/10 KMPH	Total interruption on D/L auto block system
24	SR 9.12.01	25 KMPH for 1 st train and booked speed for subsequent trains	Failure of auto signal when communications are available on D/L
25	SR 9.12.03(m)	25 KMPH for 1 st train and booked speed for subsequent trains	Single line working introduced on D/L section
26	SR 15.05.01©	40/15 KMPH	When a patrolman not returned within 15 minutes of his schedule time
27	SR 15.17.02	20 KMPH	Rail fracture less serious in nature
28	SR 15.17.04	15 KMPH	Rail closure shorter than 5.5m.
29	SR 4.10.01	15 KMPH	1 in 12 turnout not provided with curve switch.
30	SR 4.62.06(e)(i)	8 KMPH	Material train pushing when brake van is not leading
31	SR 4.62.06(d)(i)	25/8 KMPH	Engine pushing and brake van is leading of 25 for straight material 8 for curve
32	SR 9.02.01	15/8 KMPH	Passing auto stop signal at ON
33	SR 17.09.01	35/20 KMPH 35 kmph by day& 20 kmph by night	On resumption of tension in OHE, train wants to move after experiencing of no tension on line for 5minutes or more and TPC could not be contacted
34		Not exceeding 60 KMPH	Working in thick and foggy weather in absolute block system

35		Not exceeding 30 KMPH	Working in thick and foggy weather in auto block system
36	OPM 17.15/17.37	32 KMPH	Non vacuum goods train or non air goods train
37	OPM 17.15/17.37	40 KMPH	Non-vacuum passenger train or non-air passenger train.
38	SR 4.10.01	30 KMPH	Over 1/12 turn out provided with curve switch
39`	SR4.10.02	30KMPH	Over 1/8 ^{1/2} turn out provided with curve switches and laid as symmetrical splits

NOTE:

- i) Sanctioning authority for non-vacuum goods train is DRM
- ii) Sanctioning authority for non-vacuum passenger train is COM.

AUTHORITIES

S/No.	Description	Form No.
1	Advance authority to pass defective signal	T/369(i) (Blue)
2	Authority to pass signals at ON or defective	T/369(3b) (Blue)
3	Caution order	T/409 (Green)
4	NIL caution order	T/A409
5	Reminder caution order	T/B-409
6	Authority to receive a train on obstructed line and also on non signalled line.	T/509(Black)
7	Authority to start from a non signalled line	T-511 (Blue)
8	Authority to start from a line with common starter	T-512 (Blue)
9	Authority to proceed for relief engine train into an occupied block section	T/A602 (Red)
10	Authority for opening communication during total interruption of communication on a S/L section	T/B 602 (Red)
11	Authority for working of trains during total interruption of working on D/L	T/C 602 (Red)
12	Authority for temporary S/L working on D/L section	T/D 602 (Red)
13	Line clear enquiry message	T/E 602 (Red)
14	Conditional line clear message	T/F 602 (Red)
15	Conditional line clear ticket for UP train	T/G 602 (Red)
16	Conditional line clear ticket for DN train	T/H 602 (Red)
17	Written permission by guard to driver to proceed to next station from mid section	T/609(Blue)
18	Shunting order	T/806 (Blue)
19	Authority to pass automatic /semi auto/ manually operated gate stop signal at ON	T/A 912 (Blue)

20	Authority to proceed without line clear Auto section	T/B 912 (Red)
21	Authority to proceed for relief engine train into an auto block signalling territory	T/C 912 (Red)
22	Authority to proceed on auto block system during prolong failure of signals	T/D 912 (Red)
23	Line clear enquiry message (Outwards /inwards) in auto block signalling section	T/A 1425 (Black) & T/B-1425
24	Paper line clear ticket for UP train in auto section	T/C 1425 (Blue)
25	Paper line clear ticket for DN train in auto section	T/D 1425 (Blue)
26	Train intact register for arrival	T/1410

Extracts of Some Correction Slips

Addendum & Corrigendum No. 2 to G & SR Book (2008) of S.E.C.Railway.

In the General and Subsidiary Rule book 2008:-

1. The existing **G. Flare Signals** may be replaced by "**G. Signals to warn incoming train of danger ahead.**"

2. The existing GR 3.65 may be replaced by the following:-

3.65 – The signals to be used to warn the incoming train of an obstruction shall be a red flashing hand signal lamp at night or a red flag during day.

3. The SR 3.65.01 may be replaced by the following:

SR 3.65.01 –Battery operated LED based Red Flashing Hand signal lamp will be used at night or Red Flag during day time for this purpose.

4. The existing GR 3.66 may be replaced by the following:-

3.66. Use of warning signals.- When it becomes necessary to protect an obstruction in a block section, a signal may be used, as prescribed by special instructions under rule 3.65, while the railway servant proceeds to place detonators.

5. The SR 3.66.01 may be replaced by the following:

3.66.01 – Exhibition of warning signal:

The warning signal shall be exhibited at a suitable place in the vicinity of obstruction so that the warning signal could be seen by the Loco Pilot of an approaching train from as great a distance as possible.

6. The existing GR 3.67 may be replaced by the following:-

3.67. Knowledge and possession of warning signals.-

(1)(a) All concerned railway servant on whom this duty is laid by the Railway Administration shall keep a stock of such signal as may be prescribed by special instructions under rule 3.65;

(b) The Railway Administration shall be responsible for the supply, renewal and safe custody of such signals as may be prescribed by special instructions under rule 3.65 and for ensuring that their use is properly understood;

(c) The Railway Administration shall supply every Guard, Loco Pilot, Patrolman and Gateman working on the Double or Multiple line, Ghat, Suburban or Automatic Block Territories with such signal as may be prescribed by special instructions under rule 3.65.

(2) Every railway servant concerned with the use of signals as prescribed by special instructions under rule 3.65, shall have a correct knowledge of their use and keep them ready for immediate use.

(3) Every railway servant shall see that the railway servants in his charge concerned with the use of warning signals as prescribed by special instructions under rule 3.65, have a correct knowledge of their use."

7. The SR3.67.01 (a) may be replaced by the following:

3.67.01 – Bonafide users -

(i) Guards, Loco Pilots, Patrolmen working on Ghat, Suburban, Automatic signaling, Double or Multiple line sections shall be supplied with such signals, which shall form the part of his personal equipment. Gatemen working in these sections shall also be supplied with such signals.

(ii) The Loco Pilot/Assistant Loco Pilot shall give a continuous series of short sharp whistle which he shall continue to sound until the approaching train on the adjacent line has been brought to a stop.

8. The existing SR 3.67.01(b), (c), (d) & (e) may be deleted.

9. The existing GR 3.78 may be replaced by the following:-

3.78 – Duties of engine crew in respect of signals:-

1.(a) The Loco Pilot shall pay immediate attention to and obey every signal whether the cause of the signal being shown is known to him or not,

(b) the Loco Pilot shall not, however, trust entirely to signal, but always be vigilant and cautious.

(2)(a) The Loco Pilot shall whistle intermittently when his engine explodes detonator(s) and take every possible caution including reduction of speed as necessary, so as to have the train well under his control and be able to stop short of any obstruction on the line;

(b) after proceeding 1.5 kilometres from the place where his engine exploded detonator(s), if his engine does not explode any more detonator(s), he may then resume authorized speed; and

(c) report the incident to the next station or cabin.

(3) If in consequence of fog, storm or any other reason, the view of the signals is obstructed, the Loco Pilot shall take every possible precaution, so as to have the train well under control.

(4) When the Loco Pilot notices a signal warning of an obstruction, except detonator(s), he shall stop his train immediately and act on advice of the person exhibiting warning signal or on the basis of obstruction noticed by him.

(5) In case no further details of exhibition of warning signal are noticed, after stopping for one minute by day and two minutes by night to ascertain the location and/or cause of the warning, he shall proceed cautiously upto the next block station, keeping a sharp look out.

(6) The Loco Pilot shall acquaint himself with the system of working, location of signals and other local conditions affecting the running of trains on a section or sections of the railway over which he is to work and if he is not so acquainted with any portion of the railway over which he is to work, obtain the services of a qualified railway servant who is conversant with it to assist him.

10. The existing clause (b) of sub-rule (1) of rule 4.16 may be replaced by the following:-

“(b) by night, as well as in thick, foggy or tempestuous weather impairing visibility during day, a red tail lamp of approved design displaying a flashing red light to indicate last vehicle check device, or.”

**Addendum & Corrigendum No. 3 to General and Subsidiary
Rule Book 2008 (3rd edition) of S. E. C. Railway.**

In the General & Subsidiary Rule Book-2008,

The existing SR 3.36.04 (C) may be replaced by the following-

- (c) In case of suburban/Passenger trains (conventional, MEMU & DEMU) where The stoppage of the train at a station is two minutes or less, starting signals may be taken 'off' as soon as 'Line Clear' has been received from the station in advance and even before the train has come to a stop at the station. It shall be the responsibility of the Loco Pilot/Motorman to stop the train at the station at which it is booked to stop according to Working Time Table although the starting signals at the station may have been taken 'off' earlier.

**Addendum & Corrigendum No. 5 to General and Subsidiary
Rule Book 2008 (3rd edition) of S. E. C. Railway.**

In the General & Subsidiary Rule Book-2008,

1. The existing SR 5.16.01 may be replaced by the following:

5.16.01- Shunting during reception/dispatch of trains- When signals have been taken 'off' for an incoming/outgoing train on/from a line which is not isolated, no shunting movement shall be carried out towards the points over which the incoming/outgoing train is to pass except on stations where frequent shunting movement take place and where such points are protected by Stop signal or by a Shunt Board with the precautions to be observed while performing shunting that:-

- (a) shunting shall be carried out under the supervision of authorized competent railway servant.
- (b) Rake/Load should be fully vacuum/air brake.
- (c) The maximum speed during shunting operations shall not exceed 15 kmph.

**Addendum & Corrigendum No. 6 to General and Subsidiary
Rule Book 2008 (3rd edition) of S. E. C. Railway.**

In the General & Subsidiary Rule Book-2008,

1. The existing item 15 of SR 4.50.01 may be replaced by the following:
15. _____ Fouling not cleared.
2. The existing item 9 of SR 4.50.01 may be replaced by the following:
9 (i). _____ (intermittent) Approaching level crossing.
9 (ii). _____ (continuous) (a) Approaching tunnel or area of restricted visibility or curves or cutting or site of accident or when in consequence of fog, storm or any other reason the view of signal is obstructed.

**Addendum & Corrigendum No. 8 to General and Subsidiary
Rule Book 2008 (3rd edition) of S. E. C. Railway.**

At page no. 237 of General and Subsidiary Rule Book 2008 (Third edition), after item (c) of SR 4.62.04, the following should be added as in English version only;

SR 4.62.05.— (a) A material train shall be worked under the system of working in force.

(i) On double line section, when working on the proper line, on completion of work, it shall proceed to the next block station in advance and return, if required, on the proper line.

**Addendum & Corrigendum No.10 to General and Subsidiary
Rule Book 2008 (3rd edition) of S. E. C. Railway.**

In General and Subsidiary rule Book 2008 (Third edition), the distance of Fog signal post on diagram of position of Fog signalman, after item (e) of SR 3.61.01. should be read as 275 Meters in English version only at page no.118. It is already mentioned as 275 Meters in Hindi version.

**Addendum & Corrigendum No. 11 to General and Subsidiary
Rule Book 2008 (3rd edition) of S. E. C. Railway.**

In General and Subsidiary Rule Book 2008,

1. In table of SR 7.01.01 the following may be added as item No. (iv)

	Section	System of working in force
(iv)	Raipur- Telibandha	One Train Only System

2. In SR 13.01.01 following may be added as item No. 4

4. Raipur- Telibandha

3. The existing item no. (xiv) of SR 4.19.01 (a), (x) of SR 4.19.01 (b), (xv) of SR 4.19.01 (c), (vi) of SR 4.19.01 (d), (xii) of SR15.05.09 and (8) of SR 16.02.01 may be replaced as under:

SR & item No.	Existing	Replaced by
4.19.01(a)(xiv)	Flare Signal (Fusee) – One-(while working on automatic Block/ Suburban/Ghat/Double line sections.)	Battery Operated LED based Flashing Hand Signal Lamp cum Tri color torch- One while working on automatic Block/Suburban/ Ghat/ Double line sections.)
4.19.01(b)(x)		
4.19.01(c)(xv)	Flare Signal (Fusee-one)	Battery Operated LED based Flashing Hand Signal Lamp cum Tri color torch- One
4.19.01(d)(vi)		
15.05.09(xii)	1 fusee (flare signal)while on patrolling duty on Ghat, Suburban, Double/Multiple line and automatic signaling sections.	Battery Operated LED based Flashing Hand Signal Lamp cum Tri color torch- One while on patrolling duty on Ghat, Suburban, Double/Multiple line and automatic signaling sections.
16.02.01(8)	Fusee 1 (3 in Multiple line, double line, parallel lines, suburban section, automatic signaling and ghat section.)	Battery Operated LED based Flashing Hand Signal Lamp cum Tri color torch- One (3 in Multiple lines, double line, parallel lines, suburban section, automatic signaling and ghat section.)

Addendum & Corrigendum No.13 to General and Subsidiary _____Rule Book 2008 (3rd edition) of S. E. C. Railway.

In General and Subsidiary rule Book 2008 (Third edition),

SR 3.78.01 (a) (ii) may be replaced by the following:

In case a staff has not operated on a section for more than six months, he shall be given road learning trips to fresh his knowledge of the road before he is allowed to work on that section as per the schedule given below:

Duration of absence	No. of trips
6 months to 2 years	One trip
Above 2 years	Two trips

However the learning road for engine Crew in respect of siding will be as follows:-

6 months and above- One trip.

**Addendum & Corrigendum No.1 to Operating Manual Book 2008 of
S. E. C. Railway.**

In Operating Manual Book 2008

The existing Rule 17.26(d) 1 (vii) may be replaced by the following:

17.26(d) 1 (vii)- There should be at least three fully air braked vehicles (including brake-van) with operative cylinders out of the last four vehicle at the rear end of the train.

Duties of drivers while passing manned level crossing.

While approaching level crossing following duties and responsibility has been assigned to train driver which has to ensured and complied and no exception in compliance will be tolerated:-

- (i) To whistle intermittently from the whistle boards (SR 4.50.2) to level crossing and be on look out for any obstruction at level crossing (SR 4.50.01).
- (ii) While passing through a manned level crossing Driver shall keep a good look out for gateman's signal and exhibit a green hand signal towards the gateman as an indication that they are alert except when the gateman has exhibited a stop hand signal or in the situation covered by GR(16.08).
- (iii) To look out for gateman of level crossing will found standing attentively at the gate lodge side facing the track with furled red and green flag signal during the day in right and left hand respectively and at night hold the hand signal lamp with the white light in night.
If driver does not find gateman exchanging signal same to be repeated through abnormality register while sign off at crew lobby to repeat to TLC on duty (SR 16.04.01).
- (iv) To be prepared for being shown danger signal by gateman for any unusual condition like hot- axle, chain hanging, vehicle /wagon on fire load shifted etc. and take prompt action (SR 16.04.01).
- (v) To be prepared to stop on banner flag in day and red hand signal (SR16.06.04) while approaching level crossing and then move cautiously on hand signal of gateman (GR 16.06.(c), inserted vide correction slip no.23 dtd.16.9.98) while ensuring that level crossing is not obstructed and on confirmation of no obstruction negotiate level crossing continuously (Para 16.06.04(d))
- (vi) To be cautious about blowing of detonator on approach of level crossing i.e. one detonator about 90 mts. before of gate signal if it becomes defective (para SR 16.06.02) in "Off" condition. The gateman may break green glass in night to the gate signal in case it becomes defective in "Off" condition.
- (vii) To be cautious about blowing of three detonators on approach successively (GR16.07) approximately separated by 10 Mtrs. for stop dead and if such detonating sound is heard then he should immediately control the train and proceed cautiously and be prepared to stop the train ahead of obstruction.
- (viii) To be watchful to gateman about his gesture to indicate train parting of his train (GR 16.08).

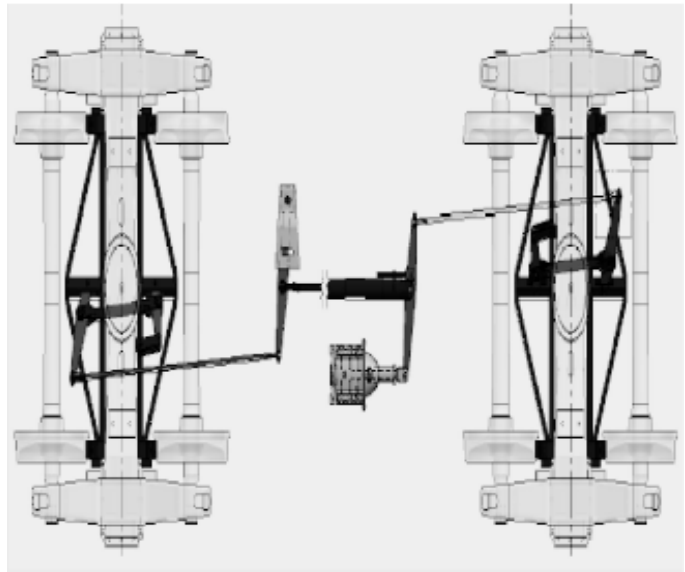
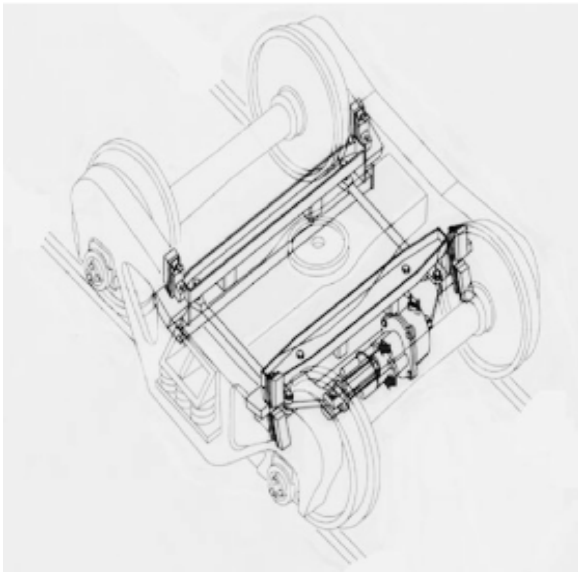
Duties of loco pilot&ALPs in the event of a serious accident (OR Duties of loco pilot & ALPs during golden hour of a serious accident) (ACM 3.03/.3.04):-

1. The first one hour after occurrence of an accident is known as GOLDEN hour.
2. The senior most officer, whether on duty or travelling on leave would act as officer-in-charge at the site of accident.
3. All available officers & staffs will report to the guard of the affected train. Then they will offer their services as per guidance of officer-in-charge at the site of accident.
4. Note the time of accident and km no. & switch "ON" flasher light.
5. Protect the adjacent line in front (in D/L section) & towards the direction of approaching train (in multiple line section) as per GR 6.03.
6. Inform the matter to the SM of adjacent block station by available means of communication.
7. If the adjacent line is obstructed, then advise TPC to switch off the OHE supply.
8. Immediately inform to divisional control office through available means.
9. If any miscreant activities are noticed with track, rail, fish plate, bolt or to any other fittings, ensure nobody is manipulating the same and keep a constant vigil till ART/ARME arrives.
10. To protect the train or engine ensure the required technical precautions. (i.e.: In speedometer, Memory freeze switch to be operated after breaking the glass, Loco hand brake to be operated etc.)
11. Do not involve yourself in expressing views to press or print media unless authorized to do so.
12. In case of guard's inability train to be protected in rear.
13. Offer all possible assistance to guard in relief measure to the injured and assessing loss to rolling stock, p.way, locomotive, OHE & assistance required
14. ALP will man the engine in absence of loco pilot.

OPM 4.34 Haulage of dead locomotive (on goods train)

- i) Need for haulage of a dead loco:-
 - a) To balance engine from one shed to other
 - b) To clear block section and take the locomotive to its destination in case of failure or breakdown in section.
 - c) to send the loco to shed for maintenance schedules
- ii) A dead engine is to be counted as being equivalent to the number of units in terms of 4 wheeler (i; e: WAG-5/WAG-7/WAG-9/WAP-4= 4units)
- iii) "Fit to run certificate" from SE/LI/PC for attaching with passenger or goods or both, to be obtained.
- (iv) If dead loco is attached in rear of brake van or has defect in undergear equipment, then ALP or higher competent person should escort. The hand brake & flasher light of dead loco should be in working order and loco to be protected if uncoupled on run.
- (v) If dead loco is attached next to engine & brakes including proportionate brakes are fully operational, then no escort is required.
- (vi) Maximum permissible speed of dead loco should not less than the booked speed of the train to which it is being attached.
- (vii) Running of double/triple headed trains is to be permissible on the section over which the dead loco is to be hauled.
- (viii) After coupling of a dead loco, the coupled locos should be run for about 500meters for any abnormal rise in the temperature of the wheels of the dead loco due to brake binding.
- (ix) Loco with defects in undergear equipments can be attached only in goods trains.
- (x) Only one dead loco is allowed to run in a goods train with brake operative. Total number of loco shall not exceed '3' including dead one.
- (xi) The total length of train after attaching dead loco should not exceed 725 meters.
- (xii) If the dead loco is attached as piped vehicle with vacuum stock, at least 20 fully vacuum braked 4 wheeler units shall be attached behind the dead loco.
- (xiii) If the dead loco is attached as piped vehicle with air stock, at least 10 fully air braked 4 wheeler units shall be attached behind the dead loco.

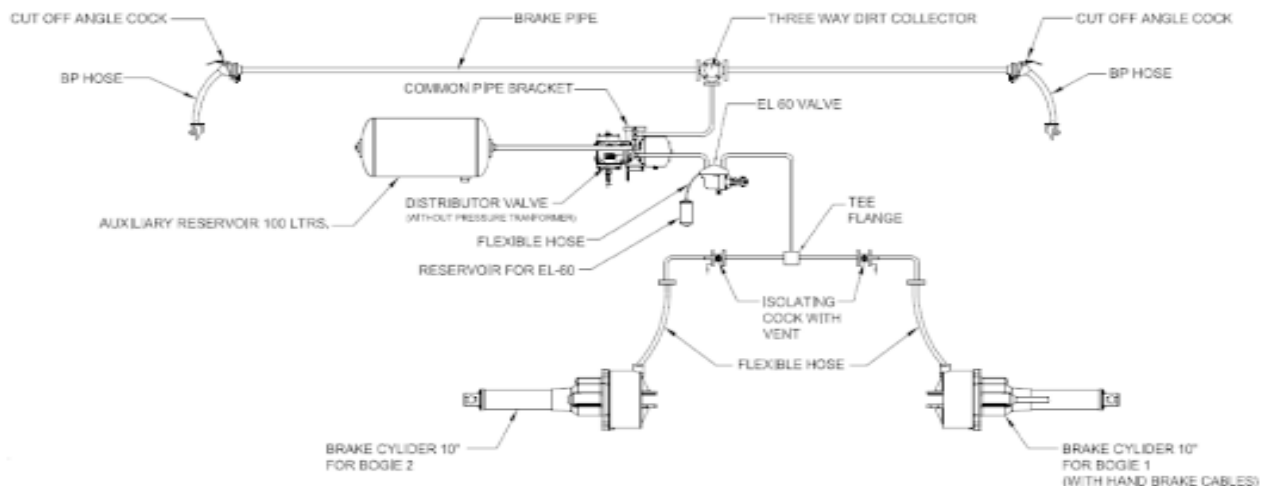
Bogie Mounted Brakes Vs Body Mounted Brakes



Bogie Mounted Brake System	Body Mounted Brake System
Equal brake shoe force - higher performance.	Friction & wear results in unequal shoe force – wheel heating.
Brake Cylinder with built-in double acting Slack Adjuster.	External Slack Adjuster.
Simplifies brake design. Suitable for all types of wagons.	Complex Rigging – more parts/higher weight.
Ease of on-car maintenance; Modular brake cylinder, Removable brake heads.	Lower Brake Efficiency.
Lower installation time and cost.	Higher Maintenance effort/cost.
	Higher installation time and cost.

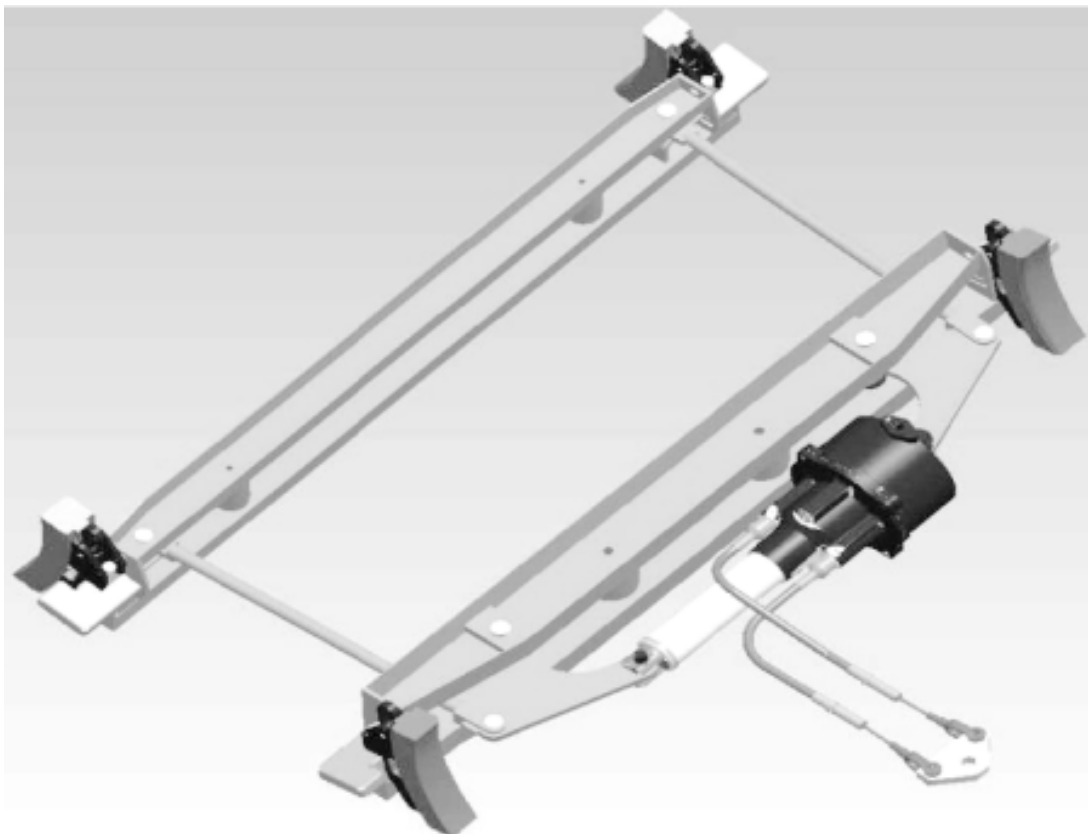
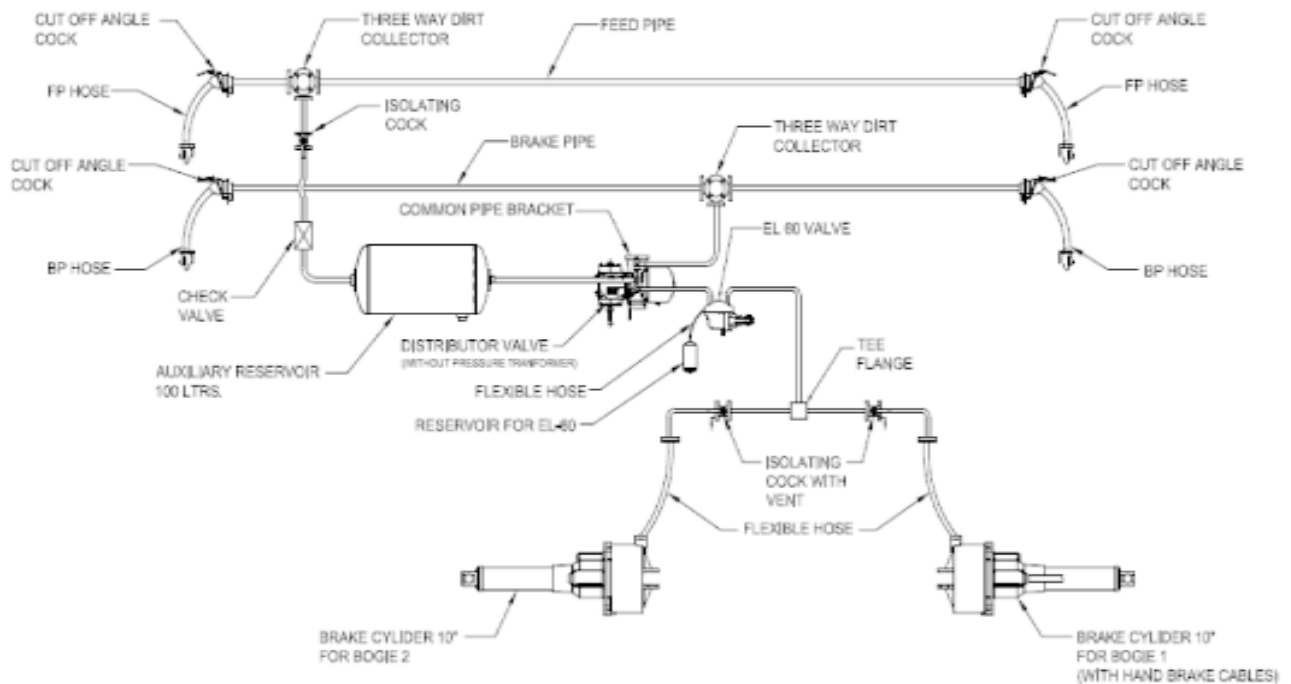
Air Brake Layout with BMBS (Single Pipe System)

SCHEMATIC FOR GRADUATED RELEASE SINGLE PIPE AIR BRAKE SYSTEM WITH BOGIE MOUNTED BRAKE SYSTEM



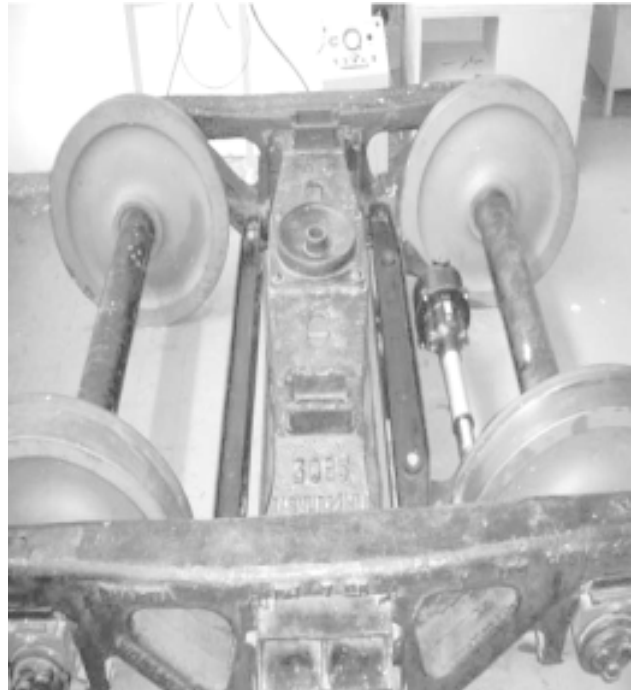
Air Brake Layout with BMBS (Twin Pipe System)

SCHEMATIC FOR GRADUATED RELEASE TWIN PIPE AIR BRAKE SYSTEM WITH BOGIE MOUNTED BRAKE SYSTEM



Working of BMBS

1. Brake cylinder extends on both sides.
2. Rotates the Levers on pivots points.
3. Pushes Secondary Beams to wheels through push rods.
4. Simultaneous reaction force moves the primary brake beam to wheels.
5. On release of brakes, the return spring inside the brake cylinder, brings the system back to original position.
6. Brake cylinder is equipped with a double acting slack adjuster.
7. Automatically compensates wear /slackness in the system.



HOW TO RELEASE BRAKE BINDING

To release brake binding in BMBS wagons the following steps are to be followed by loco pilot:-

1. Ensure B.P pressure in engine is 5 kg/cm² .
2. Release the brakes by operating release handle of C3W DV.
3. If still brakes are not releasing, isolate the angle cock of that particular bogie.
4. If still air is not coming out from brake cylinder, then loosen the check nut connected with cylinder and ensure release of locked air pressure.
5. If still brake block is not releasing then separate it by inserting any hard material between wheel and brake block.
6. Never strike the piston towards inside to release the brake.
7. If there is brake binding in both the trolleys then isolate C3W distributor valve and release it manually .
8. We should confirm the releasing of brake by observing the orange colour knob .If it remains out side means brakes are in applied condition and if it remains inside, it indicates releasing of brakes.

BP-FP Bypassing when BP iron pipe damages by BP-FP coupler:-

This bypassing is possible when the train is equipped with twin pipe air brake system. It is carried out when BP iron pipe or its angle cock is damaged. After bypassing the defective coach, the complete train runs on single pipe operation. It means the FP air supply is to be cut off by isolating the FP angle cock in rear of engine.

- **If 10th coach is affected, procedure to bypass:-**

- 1. Angle cock are to be closed in the following**

- Both angle cock of BP & FP on 9th coach
- All the four angle cocks of 10th coach
- Both angle cock of BP & FP on 11th coach

- 2. All the air hose pipes between 9th, 10th & 11th coach are to be uncoupled.**

- 3. To couple angle cocks:-**

- couple BP air hose of 9th coach/wagon to BP end of Bypass coupler & then couple with FP air hose of 10th coach/wagon
- couple FP air hose of 10th coach/wagon to FP end of Bypass coupler & then couple with BP air hose of 11th coach/wagon

- 4. To open angle cocks**

- First open BP angle cock in front of 11th coach/wagon
- Then both FP angle cock of 10th coach/wagon
- At last BP angle cock in rear of 9th coach/wagon

- 5. The following operations are to be carried out in the affected 10th coach/wagon**

- Release the pistons manually
- Keep the isolating handle of DV in isolated position
- Keep isolating cock of branch pipe of BP,FP & BC in isolated position
- Ensure all the pistons of BC are in released condition.

Now after recreating prescribed BP & FP BP continuity test is to be done. All the above particulars are to be informed to TLC/on duty & to be endorsed in BPC.

दक्षिण पूर्व मध्य रेलवे
SOUTH EAST CENTRAL RAILWAY

क्रम सं. Sr. No. _____

"ऑन" या खराब स्थिति में सिगनलों को पार करने के लिए प्राधिकार
AUTHORITY TO PASS SIGNALS IN 'ON' OR DEFECTIVE POSITION
(ड्राइवर/रिकॉर्ड / Driver/Record)

Station _____ स्टेशन

दिनांक Date _____

समय Time _____

TO

The Driver of Train No. गाड़ी नं. _____ Up/Down अग/डाउन के ड्राइवर को

सिगनल का नं. और विवरण

Description and No. of Signals.

AT STATION _____ स्टेशन पर सिगनल खराब हो गया है/गाए है

is/are OUT OF ORDER.

आपको "ऑन" स्थिति में खराब बाहरी/होम/इनर होम/मार्ग सूचक/स्टार्टर/मध्यवर्ती स्टार्टर/अग्रिम स्टार्टर/आई बी एस सिगनल/सिगनलों को ऐसी रफ्तार से पार करने के लिए प्राधिकृत किया जाता है जो 15 कि.मी. प्र.घं. से अधिक न हो. You are hereby authorized to pass the defective Outer/Home/Inner Home/Routing/ Starter/ Intermediate Starter/ Advanced Starter/IBS signal/s at 'ON' position at a speed not EXCEEDING 15 KMPH.

*गाड़ी लाइन नं. (अंकों में) _____ एवं (शब्दों में) _____ पर प्रवेश करेगी.

*The train will be admitted on line No.(in words) _____ and (figures) _____.

**अग्रिम स्टार्टर सिगनल/अंतिम रोक सिगनल में खराबी होने पर लाइन क्लियर _____ स्टेशन से बिनली संप्रेषण यंत्र पर प्राप्त की जा चुकी है. प्राप्त किया गया ग्राइवेट नं.(शब्दों में) _____ (अंकों में) _____.

**In case of failure of Advanced Starter Signal/Last Stop Signal - Line clear has been obtained on Electrical Communication Instrument from _____ station. Private No. received (in figures) _____ (in words) _____.

ड्राइवर के हस्ताक्षर Signature of Driver

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

दिनांक Date _____

स्टेशन मास्टर की मोहर Station Master Stamp

दिनांक Date _____

. *यदि मार्ग सूचक/इनर होम/होम सिगनल खराब हो जाता है.In case of Routing/Inner Home/Home Signal is defective

**यदि दोहरी लाइन वाले क्षेत्र में अग्रिम स्टार्टर/अंतिम रोक सिगनल में खराबी आ जाती है In case of failure of Advance Starter/Last Stop Signal on Double Line/Single Line Territory.

अनुदेश INSTRUCTIONS

जो लागू न हो उसे काट दें.Strike out whichever is not applicable.

इस प्राधिकार पर चलने वाले ड्राइवर इस फॉर्म के पीछे उल्लिखित अनुदेशों का पालन अवश्य करें. Driver proceeding on this authority must observe instructions mentioned at the back of this form

कृ.पू.उ. PTO

दक्षिण पूर्व मध्य रेलवे

SOUTH EAST CENTRAL RAILWAY

स्टेशन पर त्रुटिपूर्ण सिगनलों को पार करने के लिए अग्रिम प्राधिकार **ADVANCE AUTHORITY TO PASS DEFECTIVE SIGNALS**
AT STATION _____ स्टेशन पर

(ड्राइवर /रिकॉर्ड / Driver/Record)

Station _____ स्टेशन

दिनांक Date _____ 20__

To

The Driver of Train No. गाड़ी नं. _____ Up/Downअप/डाउन के ड्राइवर को

As per advice of Station Master, _____ Station स्टेशन के स्टेशन मास्टर की सूचना के अनुसार _____
 स्टेशन पर, *Up/Down अप/डाउन _____ Signal/s * सिगनल खराब है/हैं*is/are out of order at Station अतः आप
 त्रुटिपूर्ण सिगनल पर हैंड सिगनल दिए जाने पर अधिकतम 15 कि. मी. प्रति घंटे की प्रतिबन्धित गति से त्रुटिपूर्ण सिगनल को पार करने के लिए प्रधिकृत
 किए जाते हैं । On being hand signaled at the foot of the defective signal(s), you are, therefore, authorized that you
 shall pass the defective signal(s) at a restricted speed not exceeding 15 kilometres per hour.

ड्राइवर के हस्ताक्षर Signature of Driver_____
स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

. स्टेशन मास्टर की मोहर Station Master. Stamp

दिनांक Date _____

*जो लागू न हो उसे काट दें.Strike out whichever is not applicable

दक्षिण पूर्ब मध्य रेलवे
SOUTH EAST CENTRAL RAILWAY

फॉर्म नं. टी Form No. T / 509

क्रम सं. Sr. No. _____

अवरोधी लाइन पर गाड़ी को लेने के लिए प्राधिकार
AUTHORITY TO RECEIVE A TRAIN ON AN OBSTRUCTED LINE
(ड्राइवर/रिकार्ड Driver /Record)

Station _____ स्टेशन

दिनांक Date _____ समय Time _____

To

The Driver of Train No. गाड़ी नं. _____ (विवरण Description) _____ Up/Down. अप/डाउन के ड्राइवर को .

आपको "ऑन" स्थिति में प्राधिकारधारी द्वारा मार्गदर्शित होते हुए सावधानीपूर्वक गाड़ी की गति 15 कि.मी. से अधिक न रखते हुए "अप/डाउन" आउटर/होम/रूटिंग सिग्नलों को पार करने के लिए प्राधिकृत किया जाता है। आपकी गाड़ी एक अवरोधी लाइन पर ली जा रही है इसलिए आप अपनी गाड़ी को लाइन नं. _____ (शब्दों में in words) _____ (अंकों में in figures) _____ की तरफ के फेसिंग प्वाइंट पर तब तक खड़ी रखेंगे जब तक आगे बढ़ने के लिए हथ सिग्नल नहीं दिया जाता। You are authorized to pass *Up/Down *Outer/Home/ Routing Signals in 'ON' position cautiously not exceeding 15 kmph speed being piloted by the bearer of this Authority and you should bring your train to a stand at the facing points leading to Line No. (in words) _____ (in figures) _____ until Hand Signaled forward as you are being received on an obstructed line.

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

ड्राइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____

*जो लागू न हो उसे काट दें. Strike out whichever is not applicable.

दक्षिण पूर्व मध्य रेलवे
SOUTH EAST CENTRAL RAILWAY

क्रम सं. Sr. No. _____

सिगनल रहित लाइन से प्रस्थान करने के लिए प्राधिकार AUTHORITY TO START FROM A NON-SIGNALLED LINE

(ड्राइवर/रिकार्ड Driver /Record)

Station _____ स्टेशन

दिनांक Date _____ 20 ____

समय Time _____

To

The Driver of Train No. गाड़ी सं. _____ Up/Down अप/डाउन के ड्राइवर को .

आपको "ऑन"स्थिति में *स्टार्टर/एडवांस्ड स्टार्टर को पार करने और जब तक की आपकी गाड़ी सिगनल रहित (नॉन सिगनलड) लाइन सं. (शब्दों में in words) _____ (अंकों में in figures) _____ से चालू नहीं हो जाती, गति 15 कि.मी. प्रति घंटे से अधिक न रखते हुए सक्षम रेल कर्मचारी द्वारा सिगनल रहित लाइन के प्वाइंटों की तरफ सावधानी पूर्वक विधिवत रूप से चार्ज को छोड़ने के लिए प्राधिकृत किया जाता है। You are authorized to pass *Starter/Advanced Starter in the 'ON' position and leave the Yard cautiously duly piloted out up to last set of points of non signaled line by the competent Railway servant at a speed not exceeding 15 kmph over the points as your train is started from non-sigaled line No. _____

टोकन/लाइन क्लियर टिकट नं. Token or line clear ticket /PIC No. _____

प्राइवेट नं. Private No. received (शब्दों में In words) _____ (अंकों में In figures) _____ प्राप्त किया.

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

ड्राइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____

*जो लागू न हो उसे काट दें Strike out whichever is not applicable.

दक्षिण पूर्ण मध्य रेलवे
SOUTH EAST CENTRAL RAILWAY

क्रम सं. Sr. No. _____

सामूहिक प्रस्थान सिग्नल वाली लाइन से प्रस्थान करने के लिए प्राधिकार
AUTHORITY TO START FROM A LINE WITH COMMON STARTER SIGNAL
(डाइवर/रिकार्ड Driver /Record)

Station _____ स्टेशन दिनांक Date _____ समय Time _____

To

The Driver of Train No. गाड़ी सं. _____ विवरण Description _____ Up/Down अप/डाउन के डाइवर को .

आपको स्टार्टर सिग्नल के 'आफ' होने पर लाइन नं. (शब्दों में) _____ (अंकों में) _____ से आगे बढ़ने की अनुमति दी जाती है You are authorized to start from line No. (in words) _____ (in Figures) _____ on the starter signal being taken off. लाइन नं. Line No. _____ is one of the number of lines governed by the same starter signal. उन लाइन नं. में से एक है, जो उसी सामूहिक प्रस्थान सिग्नल से शासित है।

_____ स्टेशन से _____ फोन पर/ब्लॉक यंत्र पर लाइन क्लियर Line clear is obtained on *Block Instrument / _____ phone from _____ Station. प्राप्त किया है।
प्राप्त किया गया प्राइवेट नं. Private No. received (शब्दों में In words) _____ (अंकों में In figures) _____

* टोकन/लाइन क्लियर टिकट नं. Token or line clear ticket/PIC No. _____.

स्टेशन मास्टर/चार्ज मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

डाइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____

* मूर्त (टैंगिबल) प्रस्थान प्राधिकार या पेपर लाइन क्लियर टिकट के बारे में जो लागू न हो उसे काट दें In case of tangible authority or paper line clear ticket, strike out whichever is not applicable.

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

इंजन अथवा गाड़ी के हिस्से को मध्य सेक्शन से अगले स्टेशन तक ले जाने की अनुमति दिए जाने पर गाई द्वारा ड्राइवर को दी गई लिखित अनुमति Written permission given by the Guard to Driver when the Engine or portion of a train is allowed to proceed to the next station from mid section
(ड्राइवर/रिकार्ड Driver/Record)

To, The Driver of Train गाड़ी (नं. एवं विवरण No. and description) _____ के ड्राइवर को .

आपको आपकी गाड़ी के इंजन के साथ आपकी गाड़ी के अगले हिस्से जिसमें अंतिम वाहन सं. _____ सहित _____ वाहन शामिल है _____ स्टेशन के लिए _____ बजकर _____ मिनट पर अलग करने और प्रस्थान करने के लिए प्राधिकृत किया जाता है । You are authorized to uncouple and proceed at _____ hours _____ minutes to _____ station with * the Engine of your Train / the front portion of your train consisting of _____ vehicles with last vehicle No. _____

आप इयूटी पर तैनात स्टेशन मास्टर को घटना की रिपोर्ट देंगे और यदि स्टेशन मास्टर द्वारा ऐसा करना अपेक्षित हो तो आप अपनी खराब गाड़ी पर/ अपनी गाड़ी के पिछले हिस्से को लेने के लिए जो _____ कि.मी. पर है, स्टेशन मास्टर द्वारा अनुमति दिए जाने के बाद इस प्राधिकार पर वापस आएंगे . You shall deliver the report of occurrence to the Station Master on duty and, if required to do so by the Station Master , you shall return to *your disabled train / take the rear portion of your train which is at KM _____ on this Authority after being permitted by the Station Master.

मैं प्रमाणित करता हूँ कि मैंने मूल प्रस्थान अधिकार** _____ प्राप्त कर लिया है और मेरी गाड़ी का सेक्शन क्लियर हो जाने तक इसे सुरक्षित रखूंगा । यह भी प्रमाणित किया जाता है कि सेक्शन के मध्य में रह गया गाड़ी का हिस्सा नियमों के अनुसार कसकर बांधा हुआ है और सुरक्षित रखा गया है . I certify that I have received the tangible authority to proceed ** _____ and will retain it, until the section is cleared of my train. Also certified that portion of Train left in Mid Section is secured and protected as per rules.

ड्राइवर के हस्ताक्षर Signature of Driver

दिनांक Date _____

समय Time _____

गाई के हस्ताक्षर Signature of Guard

दिनांक Date _____

समय Time _____

क्रम सं Sr.No. _____

Driver of _____ के ड्राइवर को, you are hereby permitted and authorized to enter into the Block Section to clear the disabled / rear portion of your train at Km. _____. आपको अपनी गाड़ी के खराब/पिछले हिस्से को क्लियर करने के लिए ब्लॉक सेक्शन में प्रवेश करने हेतु एतद्वारा अनुमति _____ कि.मी. पर दी जाती है और प्राधिकृत किया जाता है ।

ऑन स्थिति में सिगनलों को पार करने के लिए प्राधिकार AUTHORITY TO PASS SIGNALS IN 'ON' POSITION

1.आपको ऑन स्थिति में सिगनल/सिगनलों का नं एवं विवरण _____ को पार करने के लिए प्राधिकृत किया जाता है जिसमें सिगनल पोस्ट/पॉस्टों पर यदि वह कांटों का बचाव करता है, हथ सिगनलों का पालन करते हुए गति 15 कि.मी. प्र.घं. से अधिक न रहें. You are authorised to pass the Signal/Signals No.& description _____ in 'ON' position, speed not exceeding 15 KMPH observing hand signals at the foot of the signal post/s, if it protects points.

सतर्कता आदेश CAUTION ORDER

1.आपको _____ स्टेशन एवं _____ स्टेशन के बीच _____ कि.मी. तक आपके राहतइंजन/गाड़ी की गति 15 कि.मी. प्र. घ. दिन में जब दृश्यता आगे साफ हो तथा 10 कि.मी. प्र.घ.रात्रि में अथवा जब दृश्यता आगे साफ न हो, रखते हुए चलाने की अनुमति दी जाती है । You are permitted to run your Relief Engine/Train upto km _____ between Station _____ and _____ station with the speed of :--
15 KMPH in day and when view ahead is clear /10 KMPH in night or when view ahead is not clear .

2.आपको विशेष रूप से खचेत किया जाता है कि ब्लॉक सेक्शन के बीच स्थित खपार फाटक/फाटकों के पास पहुँचने समय पूर्ण सावधानी बरतें तथा बार - बार सीटी बजाएं । ब्लॉक सेक्शन में लागू सतर्कता आदेश निम्नलिखित है: You are expressly warned to observe every caution while approaching level crossing gate/s situated in block section and whistle frequently. Caution Orders in force in the Block Section areas under:--

क्र. सं. Sr No	स्टेशन Station between		किलोमीटर दूरी Kilometrage		गति कि.मी. प्र.घ. Speed kmph	सतर्कता संकेतक Indicator	गति संकेतक Speed Indicator	गणपथ संकेतक Termination Indicator	कारण/टिप्पणियां Cause / Remarks
	से From	तक To	से From	तक To					

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

दिनांक Date _____ 20 _____

स्टेशन मास्टर की मोहर Station Master Stamp

ड्राइवर के हस्ताक्षर Signature of Driver

दिनांक Date _____

समय Time _____

* जो लागू न हो उसे फाट दें . Score out whichever is not applicable.

** प्रस्थान करने के लिए प्राधिकार का ध्योरा दर्ज करें . Enter particulars of the Authority to Proceed.

दक्षिण पूर्वी मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

क्रम सं. Sr. No. _____

शंटिंग आदेश SHUNTING ORDER
(डाइवर Driver / गार्ड Guard / रिकार्ड Record)

Station _____ स्टेशन

दिनांक Date _____ 20 ____

समय Time _____ घण्टा hours _____ मिनट minutes.

To

The Driver of Train No. गाड़ी नं. _____ *अप / डाउन *UP / DOWN के डाइवर को.

कृपया निम्नलिखित आदेशों का बिना किसी विलम्ब के तत्परता से अनुपालन करें Please comply with the following order promptly without any type of delay:--

1. _____
2. _____
3. _____
4. _____

*आपको ऑन स्थिति में You are authorized to pass _____ signal in the 'ON' position and सिगनल पार करने के लिए और proceed *up to/beyond ** _____ signal. सिगनल तक/ से आगे पहुंचने के लिए प्राधिकृत किया जाता है।
***इस उद्देश्य के लिए Section between _____ स्टेशन और station and _____ station is *blocked back / blocked forward. स्टेशन के बीच सेक्शन ब्लॉक बैक/ब्लॉक फॉरवर्ड कर दिया है।

*आपको टोकन/टैबलेट/लाइन क्लियर टिकट For this purpose and Token / Tablet / Line Clear Ticket No. _____ *एस एच/ओ सी ई चाबी SH/ SMT / OCE Key is hereby issued to you from _____ hours घण्टा _____ minutes मिनट से to _____ hours घण्टा _____ minutes. मिनट तक एतद्वारा जारी की जाती है।

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

डाइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____ समय Time _____ घण्टा hours _____ मिनट minutes.

गार्ड के हस्ताक्षर Signature of Guard _____ दिनांक Date _____ समय Time _____ घण्टा hours _____ मिनट minutes

*जो लागू न हो उसे काट दें Strike out which ever is not applicable

** कि.मी. या सिगनल नं. दर्ज करें Enter Km or Signal No.

*** लाइन ब्लॉक बैक/ब्लॉक फॉरवर्ड होने पर लागू होता है Applies when Line is Blocked back / Blocked forward.

दक्षिण पूर्व मध्य रेलवे
SOUTH EAST CENTRAL RAILWAY

फॉर्म सं. टी/ए Form No. T/A 409

क्रम सं. Sr. No. ____

(ड्राइवर/गार्ड/रिकॉर्ड ड्राइवर/Guard/ Record)

Station _____ स्टेशन दिनांक Date _____ 20 ____

To

गाड़ी नं. The Driver of Train No. _____ विवरण (Description) _____ के ड्राइवर को

सतर्कता आदेश CAUTION ORDER

कड़ नहीं N I L

Up to _____ स्टेशन तक Station (Name of next Notice Station / Station) (अगला उल्लिखित स्टेशन/ स्टेशन का नाम)

समय Time _____

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की पोहर Station Master Stamp

ड्राइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____ समय Time _____

गार्ड के हस्ताक्षर Signature of Guard _____ दिनांक Date _____ समय Time _____

SOUTH EAST CENTRAL RAILWAY

क्रम सं. Sr. No. _____

अनुस्मारक सतर्कता आदेश **REMINDER CAUTION ORDER**
(ड्राइवर/गार्ड/रिकॉर्ड **Driver/Guard/ Record**)

Station _____ स्टेशन

Date _____

To : The Driver of Train No. गाड़ी नं. _____ विवरण(Description) _____ के ड्राइवर को
आपको एतद्वारा अनुदेश दिया जाता है कि निम्नलिखित गति अवरोध You are hereby instructed to exercise the following speed restrictions observing engineering indicators, hand signals of flagman and banner flag exhibited as necessary :- _____ (अंकों में in figures) _____ (शब्दों में in words) प्रदर्शित इंजीनियरी संकेतकों, फ्लैग मैन के हथ सिगनलों और बैनर फ्लैग का यथावश्यक अनुपालन करें

म.सं. Sr. No.	स्टेशन Stations Between	किलोमीटर Kilometrage	गति कि.मी.प्र. सं. Speed Kmph	सतर्कता संकेतक Caution Indicator	गति संकेतक Speed Indicator	गलत संकेतक Termination Indicator	कारण/टिप्पणियां Cause/ remarks
1.	से From						
	तक To						
2	से From						
	तक To						
3	से From						
	तक To						
4	से From						
	तक To						
5	से From						
	तक To						
6	से From						
	तक To						
7	से From						
	तक To						
8	से From						
	तक To						

समय Time _____

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर का मोहर Station Master Stamp

ड्राइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____ 20 _____ समय Time _____

गार्ड के हस्ताक्षर Signature of Guard _____ दिनांक Date _____ 20 _____ समय Time _____

दक्षिण पूर्व मध्य रेलवे
SOUTH EAST CENTRAL RAILWAY
सतर्कता आदेश **Caution Order**
(ड्राइवर/गार्ड/रिकॉर्ड **Driver/Guard/ Record**)

फॉर्म सं. टी Form No. T/409

क्र.सं. Sr. No. _____

Station _____ स्टेशन
To _____

दिनांक Date _____ 20__

The Driver of Train No. गाड़ी नं. _____ (विवरण Description) _____ के ड्राइवर को ।
एतद्वारा आपको अनुदेश दिया जाता है कि निम्नलिखित गति अवरोध You are hereby instructed to exercise the following Speed restrictions observing engineering indicators, hand signals of flagman and banner flag exhibited as necessary :- _____ (अंकों में in figures) _____ (शब्दों में in words) प्रदर्शित इंजीनियरी संकेतकों, फ्लैग मैन के हथ सिगनलों और बैनर फ्लैग का यथावश्यक अनुपालन करें । Between _____ and _____

म.सं. Sr. No.	स्टेशन Stations Between	किलोमीटर Kilometrage	गति क्र.मो.क्र. सं. Speed Kmph	सतर्कता संकेतक Caution Indicator	गति संकेतक Speed Indicator	गणना संकेतक Termination Indicator	कारण/टिप्पणियां Cause/ remarks
1.	से From						
	तक To						
2	से From						
	तक To						
3	से From						
	तक To						
4	से From						
	तक To						
5	से From						
	तक To						
6	से From						
	तक To						
7	से From						
	तक To						
8	से From						
	तक To						

समय Time _____

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर का मोहर Station Master Stamp

ड्राइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____ 20__ समय Time _____

गार्ड के हस्ताक्षर Signature of Guard _____ दिनांक Date _____ 20__ समय Time _____

अप/डाउन UP/DOWN

फॉर्म सं. टी/ए Form No.T /A 602
क्रम सं.Sr. No.-----

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY
किसी ऑक्कपाइड ब्लॉक सेक्शन में राहत इंजन/गाड़ी ले जाने के लिए प्राधिकार
AUTHORITY TO PROCEED FOR RELIEF ENGINE/TRAIN INTO AN OCCUPIED BLOCK SECTION
(ड्राइवर/गार्ड/रिकार्ड Driver/Guard/ Record).

Station _____ स्टेशन .

दिनांक Date: _____ 20 ____.

To: Driver of Relief Engine/Train No. राहत इंजन/गाड़ी नं. _____ के ड्राइवर को.

लाइन क्लियर के बिना आगे बढ़ने के लिए ब्लॉक टिकट BLOCK TICKET TO PROCEED WITHOUT LINE CLEAR

- 1) आप को _____ कि.मी. तक अप/डाउन लाइन पर _____ स्टेशन से अपने राहत इंजन/गाड़ी के साथ लाइन क्लियर के बिना सावधानी पूर्वक आगे बढ़ने के लिए एतद्द्वारा प्राधिकृत किया जाता है। You are hereby authorized to proceed cautiously without Line clear from station _____ up to km _____ on UP/DOWN Line with your Relief/ Engine/Train.
- 2) आपको अपने इंजन /गाड़ी को _____ कि.मी. पर थिराकर रोकना होगा और इसके बाद साइट पर सक्षम प्राधिकारी के अनुदेशों द्वारा निर्देशित किया जाएगा। You shall bring your Engine/Train to a stop at km _____ and there after be guided by the instructions from the Competent Authority at the site.
- 3) यह आदेश _____ स्टेशन और _____ स्टेशन की बीच _____ कि.मी. पर लाइन पर अवरोध की वजह से दिया जाता है। This order is given due to obstruction on _____ line between _____ Station and _____ station at KM No. _____.
- 4) आप _____ स्टेशन पर सेक्शन क्लियर करेंगे। You shall clear the section at station _____.

ऑन स्थिति में सिगनलों को पार करने के लिए प्राधिकार AUTHORITY TO PASS SIGNALS IN 'ON' POSITION

1. आपको ऑन स्थिति में सिगनल/सिगनलों का नं एवं विवरण _____ को पार करने के लिए प्राधिकृत किया जाता है जिसमें सिगनल पोस्ट/पोस्टों पर यदि यह कांटों का बचाव करता है, इस सिगनलों का पालन करते हुए गति 15 कि.मी. प्र.घं. से अधिक न रहें। You are authorised to pass the Signal/Signals No.& description _____ in 'ON' position, speed not exceeding 15 KMPH observing hand signals at the foot of the signal post/s, if it protects points.

सतर्कता आदेश CAUTION ORDER

1. आपको _____ स्टेशन एवं _____ स्टेशन की बीच _____ कि.मी. तक आपके राहत इंजन/गाड़ी की गति 15 कि.मी. प्र. घ. दिन में जब दृश्यता आगे साफ हो तथा 10 कि.मी. प्र.घ. रात्रि में अथवा जब दृश्यता आगे साफ न हो, रखते हुए चलाने की अनुमति दी जाती है। You are permitted to run your Relief Engine/Train upto km _____ between Station _____ and _____ station with the speed of :-
 15 KMPH in day and when view ahead is clear .
 10 KMPH in night or when view ahead is not clear .
2. आपको विशेष रूप से सचेत किया जाता है कि ब्लॉक सेक्शन के बीच स्थित समथर फाटक/फाटकों के पास पहुंचते समय पूर्ण सावधानी बरतें तथा बार - बार सीटी बजाएं। ब्लॉक सेक्शन में लागू सतर्कता आदेश निम्नलिखित है: You are expressly warned to observe every caution while approaching level crossing gate/s situated in block section and whistle frequently. Caution Orders in force in the Block Section areas under:-

क्र. सं. Sr No	स्टेशन Station between		किलोमीटर दूरी Kilometrage		गति कि.मी. प्र.घ. Speed kmph	सतर्कता संकेतक Caution Indicator	गति संकेतक Speed Indicator	गणतब्य संकेतक Termination Indicator	कारण/टिप्पणियां Cause / Remarks
	से/From	तक/To	से/From	तक/To					
1									
2									
3									
4									

मैंने इसकी विषय-वस्तु को समझ लिया है। I have understood the contents of this Authority

ड्राइवर के हस्ताक्षर
Signature of Driverगार्ड के हस्ताक्षर
Signature of Guardस्टेशन मास्टर के हस्ताक्षर Signature of Station Master
स्टेशन मास्टर की मोहर Station Master Stamp

समय Time _____ घंटा/Hours. _____ मिनट/Minutes

अनुदेश INSTRUCTIONS

1. इस प्राधिकार पर चलने वाले ड्राइवर इस फॉर्म के पीछे उल्लिखित अनुदेशों का पालन अवश्य करें। Driver proceeding on this authority must observe instructions mentioned at the back of this form .
2. ड्राइवर ऊपर दशाए गए कि.मी. (अवरोध) के पहले ठहराव लेगा तथा इसके पश्चात् वह स्थल पर उपस्थित कर्मचारी/पर्यवेक्षक के सिगनलों का पालन करेगा। Driver will stop short of km (obstruction) shown above and thereafter he will obey the signals of employee/ supervisor present at site.

दक्षिण पूर्वी मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

इकहरे लाइन सेक्शन पर पूर्ण संचार-भंग के दौरान संचार चालू करने के लिए प्राधिकार AUTHORITY FOR OPENING COMMUNICATION DURING TOTAL INTERRUPTION OF COMMUNICATION ON SINGLE LINE SECTION.

(ड्राइवर/रिकार्ड Driver/Record)

Station _____ स्टेशन

दिनांक Date: _____ 20 _____

To : Driver of Engine/ Self propelled Vehicle इनन/स्वचालित वाहन के ड्राइवर को।

लाइन क्लियर के बिना आगे बढ़ने का प्राधिकार AUTHORITY TO PROCEED WITHOUT LINE CLEAR.

1. आपको You are hereby authorized to proceed cautiously without Line clear स्टेशन से from station _____ to station _____ स्टेशन से स्टेशन तक अपने इनन/स्वचालित वाहन के साथ लाइन क्लियर के बिना सावधानी पूर्वक प्रस्थान करने के लिए एतद्वारा प्राधिकृत किया जाता है with your Engine/Self propelled Vehicle

2. आप अपने इनन/वाहन को अगले ब्लॉक स्टेशन के पहले रोक सिग्नल के बाहर छोड़ कर देंगे और इसके बाद उस स्टेशन के स्टेशन मास्टर के अनुदेशों द्वारा निर्देशित किया जाएगा। You shall bring your Engine/Vehicle to a stand out side the first stop signal of the next block station there after be guided by the instructions from the Station Master of that Station.

3 गाड़ी सं. Train No. _____ was Last Train over the section. इस सेक्शन पर अन्तिम गाड़ी थी। वह यहाँ से It left/arrived* here at _____ बजे _____ बजे _____ मिनट पर गई/पहुँची पर आई.

4. यह प्राधिकार निम्नलिखित के कारण दिया जाता है This authority is given due to:-

क a) Total interruption of communication with Station _____ from _____ Hours. दिनांक of date _____ 20 _____ को _____ बजे से _____ स्टेशन पर पूर्ण संचार भंग।

और स्थिति में सिग्नलों को पार करने के लिए प्राधिकार AUTHORITY TO PASS SIGNALS IN 'ON' POSITION

1. आपको और स्थिति में सिग्नल/सिग्नलों का नं. एवं विवरण- _____ को पार करने के लिए प्राधिकृत किया जाता है, जिसमें सिग्नल पोस्ट/पोस्टों पर यदि यह कांटों का बचाव करता है, इस सिग्नल को पालन करते हुए गति 15 कि.मी. प्र.घं. से अधिक न रखें।

1. You are authorized to pass the Signal/s No.& description _____, in 'ON' position, speed NOT exceeding 15 KMPH observing hand signals at the foot of the signal post/s, if it protects points.

सतर्कता आदेश CAUTION ORDER

1. आपको _____ स्टेशन से _____ स्टेशन के बीच आपके इनन/स्वचालित वाहन की गति दिन में जब दृश्यता आगे साफ हो 15 कि.मी. प्र.घं. तथा रात्रि में अथवा जब दृश्यता आगे साफ न हो 10 कि.मी.प्र.घं. रखने की अनुमति दी जाती है। You are permitted to run your Engine/ Self propelled Vehicle between Station _____ To _____ station with the speed of :-15 KMPH in day and when view ahead is clear .

2. आपको विशेष रूप से सचेत किया जाता है कि ब्लॉक सेक्शन के बीच स्थित समतल काट/काटों के पास पहुँचते समय पूर्ण सावधानी बरतें तथा बार-बार सीटी बजाएं (You are expressly warned to observe every caution while approaching level crossing gate/s situated between block section and whistle frequently).

3. इसके अलावा निम्नलिखित सतर्कता आदेश ब्लॉक सेक्शन में लागू हैं। In addition following caution orders are in force in block section :-

क्रम सं. Sr No	स्टेशन Station between		किलोमीटर Kilometrage		गति कि.मी.प्र.घं. Speed kmph	सतर्कता संकेतक Caution Indicator	गति संकेतक Speed Indicator	गणतय संकेतक Termination Indicator	कारण/टिप्पणियाँ Cause / Remarks
	से From	तक To	से From	तक To					
1									
2									
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4									

लाइन क्लियर पूछताछ संदेश LINE CLEAR ENQUIRY MESSAGE

To The Station Master _____ स्टेशन मास्टर को.

संदेश सं. Message No _____ वापसी पर क्या लाइन क्लियर होगी तथा इसे चलने के लिए तैयार गाड़ी नं. _____ के लिए क्लियर रखा जाएगा ? On return of _____ will line be clear and kept clear for train no. _____ waiting to proceed ?

संदेश लाइन क्लियर संदेश CONDITIONAL LINE CLEAR MESSAGE

To, The Station Master _____ स्टेशन मास्टर को.

संदेश सं. Message No. _____ आपके यहाँ _____ के पहुँचने पर, गाड़ी के साथ / के बिना लगे _____ गाड़ी इनन या अन्य इनन या स्वाचालित वाहन /ट्रॉली आदि (पूरा ब्लॉक) के वापस आने के लिए लाइन क्लियर होगी और क्लियर रखी जाएगी। On arrival of _____ at yours, line will be clear and kept clear for _____ Train Engine to return with/without attached to a Train or Another Engine or Self Propelled Vehicle/Trolley etc.(Complete particulars.)

प्राइवेट नं. Private Number (शब्दों में in words) _____ (अंकों में in figures) _____.

मैंने इसकी विषय-वस्तु को समझ लिया है। I have understood the contents of this.

ड्राइवर के हस्ताक्षर Signature of Driver

गाई के हस्ताक्षर Signature of Guard

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

समय Time _____ बजे _____ मिनट Minutes

अनुदेश INSTRUCTIONS

- *नो लागू न हो उसे काट दें. *Strike out which ever is not applicable.
- इस प्राधिकार पर चलने वाले ड्राइवर इस फॉर्म के पीछे उल्लिखित अनुदेशों का पालन अवश्य करें। (Driver proceeding on this Authority must observe instructions mentioned at the back of this form.)

दक्षिण पूर्वी मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

दोहरे लाइन सेक्शन पर पूर्ण संचार भंग के दौरान गाड़ियों के संचालन के लिए प्राधिकार AUTHORITY FOR WORKING OF TRAINS DURING TOTAL INTERRUPTION OF COMMUNICATION ON DOUBLE LINE SECTION

Station _____ स्टेशन (ड्राइवर/रिकॉर्ड Driver / Record) दिनांक Date: _____ 20 ____

To: Driver of Train No. गाड़ी नं. _____ के ड्राइवर को

लाइन क्लियर के बिना आगे बढ़ने के लिए प्राधिकार AUTHORITY TO PROCEED WITHOUT LINE CLEAR

- आपको _____ स्टेशन से _____ स्टेशन तक अपनी गाड़ी/स्वाचालित वाहन संख्या _____ के साथ लाइन क्लियर के बिना सावधानी पूर्वक आगे बढ़ने के लिए एतद्वारा प्राधिकृत किया जाता है। You are hereby authorized to proceed cautiously without Line clear. From station _____ to station _____ with your Train/Self propelled Vehicle No. _____.
- आप अपनी गाड़ी को अगले ब्लॉक स्टेशन के राइट लाइन से सम्बन्धित पहले रोक सिग्नल/ रॉग लाइन से सम्बन्धित अन्तिम रोक सिग्नल, जो भी पहले आए, के बाहर खड़ा कर देना होगा और इसके बाद उस स्टेशन के स्टेशन मास्टर के अनूदेशों द्वारा निर्देशित किया जाएगा। You shall bring your Train to a stand out side the first stop signal pertaining to right line / last stop signal pertaining to wrong line whichever comes across first of the next Block Station there after be guided by the instructions from the Station Master of that Station.
- गाड़ी नं. _____ इस सेक्शन पर अंतिम गाड़ी थी। वह वहां से _____ बजकर _____ मिनट पर गई / आई* (Train No. _____ was Last Train over the section. It left/arrived* here at _____ hours _____ minutes)
- परिस्थितियां निम्न में यह आदेश दिया जाता है।
- Circumstances under which this order is given :-
(क) दिनांक _____ को _____ बजकर _____ मिनट से _____ स्टेशन के साथ पूर्ण संचार भंग.
(a) Total interruption of communication with Station from _____ hours _____ minutes of date _____.

ऑन स्थिति में सिग्नलों को पार करने के लिए प्राधिकार AUTHORITY TO PASS SIGNALS IN 'ON' POSITION

आपको ऑन स्थिति में सिग्नल का / सिग्नलों के नं. एवं विवरण _____ को पार करने हेतु प्राधिकृत किया जाता है जिसमें सिग्नल पोस्ट/पोस्टों पर, यदि वह काटों का बचाव करता है, हाथ-सिग्नलों का पालन करते हुए गति 15 कि.मी. प्र.घं. से अधिक न रहें। You are authorised to pass the Signal/Signals No.& description _____, in 'ON' position, speed not exceeding 15 KMPH observing hand signals at the foot of the signal post/s, if it protects points.

सतर्कता आदेश CAUTION ORDER

- आपको _____ स्टेशन एवं _____ स्टेशन के बीच अपनी गाड़ी की गति 25 कि.मी.प्र.घं. दिन में जब दृश्यता आगे साफ हो तथा 10 कि.मी.प्र.घं. रात्रि में अपना जब दृश्यता आगे साफ न हो, रखने की अनुमति दी जाती है। You are permitted to run your Train between Station _____ & _____ station with the speed of:- 25 KMPH in day and when view ahead is clear .
10 KMPH in night or when view ahead is not clear .
- आपको विशेष रूप से सचेत किया जाता है कि ब्लॉक सेक्शन के बीच स्थित समथर फाटक/फाटकों के पास पहुँचते समय पूर्ण सावधानी बरतें तथा बार-बार सीटी बजाएं ! You are expressly warned to observe every caution while approaching level crossing gate/s situated between block section and whistle frequently.

क्रम सं. Sr No	स्टेशन Station between		किलोमीटर Kilometrage		गति कि.मी.प्र.घं. Speed kmph	सतर्कता संकेतक Caution Indicator	गति संकेतक Speed Indicator	गणतय्य संकेतक Termination Indicator	कारण /टिप्पणियां Cause / Remarks
	से From	तक To	से From	तक To					
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मैंने इसको ध्यान -स्मिति को समझ लिया है . I have understood the contents of this Authority.

ड्राइवर के हस्ताक्षर Signature of Driver

गाई के हस्ताक्षर Signature of Guard

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master's Stamp

समय Time _____ बजकर Hours. _____ मिनट Minutes.

अनुदेश INSTRUCTIONS

- *नो लागू न हो उसे काट दें Strike out which ever is not applicable.
- इस प्राधिकार पर चलने वाले ड्राइवर इस फॉर्म के पीछे उल्लिखित अनुदेशों का पालन अवश्य करें Driver proceeding on this Authority must observe instructions mentioned at the back of this form.

अप/डाउन UP/DOWN

क्रम सं S r. N o. _____

दक्षिण पूर्वी मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

दोहरी लाइन पर अस्थाई इकहरी लाइन संचालन के लिए प्राधिकार
AUTHORITY FOR TEMPORARY SINGLE LINE WORKING ON DOUBLE LINE
 (ड्राइवर/गार्ड/रिकार्ड Driver/Guard/ Record)

Station _____ स्टेशन _____ दिनांक Date _____ 20 _____
 To The Driver of Train No. गाड़ी नं. _____ (विवरण Description) के ड्राइवर को

लाइन क्लियर टिकट LINE CLEAR TICKET

लाइन क्लियर है और आपको _____ स्टेशन तक अप/डाउन लाइन पर आगे बढ़ने के लिए प्राधिकृत किया जाता है।
 The line is clear and you are authorized to proceed on Up / Down Line up to station _____
 प्राइवेट नं. Private No. (शब्दों में in words) _____ (अंकों में in figures) _____.

ऑन स्थिति में सिग्नल को पार करने के लिए प्राधिकार AUTHORITY TO PASS SIGNAL IN 'ON' POSITION

*सिग्नल का नं. एवं विवरण Description & No. of Signal _____ स्टार्टर, मध्यम स्तर स्टार्टर, अग्रिम स्टार्टर अथवा अंतिम रोक सिग्नलों के लिए लागू. सिग्नल पर यदि कांटो का बचाव करता है, हम सिग्नल / सिग्नलों का पालन करें. Applicable to Starter, Intermediate Starter, Advanced Starter, or Last Stop Signals. Observe hand signal/s at the foot of the signal, if it protects points.

** आपको उन स्टेशनों के सिग्नलों को पार करने हेतु प्राधिकृत किया जाता है जिन्हें इकहरी लाइन संचालन के लिए अस्थाई रूप में बंद कर दिया गया है, _____ स्टेशन You are authorized to pass the signals of those stations which have been closed temporarily for single line working. Station/s _____.

आपको मार्ग में गेटमैन तथा गैंग मैन इकहरी लाइन संचालन को चालू होने के लिए भी सूचित करना होगा. You have to also inform the Gate men and Gang men enroute for Introduction of single line working.

सतर्कता आदेश CAUTION ORDER

i. आपकी गाड़ी राइट लाइन/रॉंग लाइन पर जा रही है। अवरोध _____ कि.मी. पर है।

*i) Your train is going on Right line/Wrong line. The obstruction exists at km. _____.

ii) आपकी गाड़ी अस्थाई इकहरी लाइन को पार करने के लिए पहली गाड़ी है। लागू अन्य गति प्रतिबंध का पालन करते हुए गति 25 कि.मी. प्रतिघण्टे तक सीमित रखी जाएगी. Your train is First Train to pass over the Temporary Single Line. Speed shall be restricted to 25 KMPH subject to observance of other speed restriction in force.

iii) यदि आपकी गाड़ी रॉंग लाइन पर जा रही है तो आपको आदेश रहित सेक्शन 'न्यूट्रल सेक्शन' (केवल विद्युतिकृत सेक्शन में) का पालन करने के लिए विशेष रूप से सचेत किया जाता है। संबंधित लाइन पर कोई ट्रैप प्वाइंट नहीं है/ट्रैप प्वाइंट को क्लैम्प/स्पाइक कर दिया गया है. You are expressly warned to observe 'Neutral section' (only in Electrified section) if your train is going on wrong line. There are no Trap Points on the line in question/ Trap points have been clamped/spiked.

iv) ब्लॉक सेक्शन में लागू सतर्कता आदेश नीचे दिए अनुसार है. Following are the caution Orders in force in Block Section :-

क्रम सं. Sr No	स्टेशन Station between		किलोमीटर Kilometrage		गति कि.मी.प्र.घ. Speed kmph	सतर्कता संकेतक Caution Indicator	गति संकेतक Speed Indicator	गुणतन्त्र संकेतक Termination Indicator	कारण /टिप्पणियाँ Cause / Remarks
	से From	तक To	से From	तक To					
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मैंने इसको ध्यान से पढ़ा है और समझ लिया है. I have understood the contents of this Authority.

ड्राइवर के हस्ताक्षर Signature of Driver

गार्ड के हस्ताक्षर Signature of Guard

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master's Stamp

समय Time _____ घण्टा Hours _____ मिनट Minutes

* जो लागू न हो उसे काट दें. Strike out whichever is not applicable.

** मध्यम स्तर ब्लॉक स्टेशनों के लिए. For intermediate block stations.

अप/डाउन UP/DOWN

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

क्रम सं Sr.No. _____

इकहरी लाइन सेक्शन पर पूर्ण संचार-भंग के दौरान गाड़ियों को भेजने के लिए लाइन क्लियर पृष्ठ के लिए लाइन क्लियर पृष्ठताछ संदेश
LINE CLEAR INQUIRY MESSAGE ASKING LINE CLEAR FOR DESPATCH OF TRAINS
DURING TOTAL FAILURE OF COMMUNICATION ON SINGLE LINE SECTION.
 (ड्राइवर/रिकार्ड) (Driver /Record)

Station _____ स्टेशन

दिनांक Date _____

समय Time _____ बजकर hours _____ मिनट-minutes.

From Station Master _____ स्टेशन मास्टर से

To Station Master _____ स्टेशन मास्टर को.

संदेश सं. _____. * गाड़ी के साथ/के बिना गाड़ी इंजन/स्वचालित वाहन/अन्य वाहन सं. _____ की वापसी पर क्या अनुगामी गाड़ी /गाड़ियों के लिए जो आपकी तरफ आने की प्रतीक्षा में है, लाइन क्लियर होगी और क्लियर रखी जाएगी ?

Message No. _____. On return of *Train Engine with/without train / self propelled vehicle/other vehicle number _____ will line be clear and kept clear for following train/s:-

1. _____
2. _____
3. _____
4. _____

waiting to proceed to yours.

स्टेशन मास्टर की हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

* जो लागू न हो उसे काट दें. Strike out whichever is not applicable.

फॉर्म सं. टी/एफ Form No. I/F602

*अप /डाउन UP/DOWN

क्रम Sr. No. _____

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

कंडीशनल लाइन क्लियर संदेश **CONDITIONAL LINE CLEAR MESSAGE**

(ड्राइवर/रिकार्ड) (Driver / Record)

दिनांक Date _____ 20 ____.

समय Time _____ बजकर hours _____ मिनट minutes.

Station _____ स्टेशन

From Station Master _____ स्टेशन मास्टर से

To Station Master _____ स्टेशन मास्टर को.

संदेश सं. Message No. _____

आपका संदेश सं. _____. *आपके यहां गाड़ी के साथ /के बिना गाड़ी इंजन/स्वचालित वाहन/अन्य वाहन सं. _____ के पहुंचने पर अनुगामी गाड़ी/गाड़ियों के लिए लाइन क्लियर होगी और क्लियर रखी जाएगी।

Your Message No. _____ On arrival of * Train Engine with/without train/self propelled vehicle/other vehicle number _____ at yours line will be clear and kept clear for following Trains:----

- (1) गाड़ी नं. Train No _____ प्राइवेट नं. Private No. (शब्दों में in words) _____ (अंकों में in figures) _____
- (2) गाड़ी नं. Train No _____ प्राइवेट नं. Private No. (शब्दों में in words) _____ (अंकों में in figures) _____
- (3) गाड़ी नं. Train No _____ प्राइवेट नं. Private No. (शब्दों में in words) _____ (अंकों में in figures) _____
- (4) गाड़ी नं. Train No _____ प्राइवेट नं. Private No. (शब्दों में in words) _____ (अंकों में in figures) _____

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

*जो लागू न हो उसे काट दें' Strike out whichever is not applicable

अप UP

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

फॉर्म सं. टी/जी Form No. T/G 602

क्रम सं Sr.No. _____

कंडीशनल लाइन क्लियर टिकट
(डाइवर/रिकार्ड Driver/Record)Station _____ स्टेशन दिनांक Date _____ 20 _____
समय Time _____ बजकर Hours _____ मिनट Minutes

From Station Master _____ स्टेशन मास्टर से

To

गाड़ी नं. The Diver of Train No. _____ अप UP (विवरण Description) _____ के डाइवर को.

लाइन क्लियर है और आपको _____ स्टेशन के लिए पहुंचने हेतु प्राधिकृत किया जाता है।

The line is clear and you are authorized to proceed to _____ Station.

*अंतिम

गाड़ी इस स्टेशन से *Last train left this station at _____ बजकर hours _____ मिनट minutes. पर चली।

आपको गाड़ी के पीछे-पीछे चलने और _____ स्टेशन तक पहुंचने के लिए एतद्वारा प्राधिकृत किया जाता है।

You are hereby authorized to follow and proceed to _____ Station.

प्राइवेट नं. Private No. (शब्दों में in words) _____ (अंकों में in figures) _____.

**जा चुकी गाड़ियों का ब्योरा

Particulars of trains left

गाड़ी नं. Train No. समय Time

_____	_____
_____	_____
_____	_____

पीछे जाने वाली गाड़ियों का ब्योरा

Particulars of trains to follow

गाड़ी नं. Train No. समय Time

_____	_____
_____	_____
_____	_____

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

विशेष अनुदेश Special Instructions

जब कोई गाड़ी पीछा कर रही है तब कड़ी निगरानी रखें। जब दृश्यता साफ हो तब गति 25 कि.मी. प्र. घं. से अधिक न हो तथा जब दृश्यता किसी भी वजह से साफ न हो तब गति 10 कि.मी. प्र. घं. या इससे कम रखें। When following a Train keep sharp look out at speed not exceeding 25 kmph when view is clear and 10 kmph or less when view is impaired due to any reason.

*अनियमितता के दौरान दूसरी तथा इसके बाद की गाड़ियों की प्रविष्टि को काट दें. Strike out in case of second and subsequent trains during uneven flow.

**अनियमितता के दौरान पहली गाड़ी की प्रविष्टि तथा नियमितता के दौरान सभी गाड़ियों की प्रविष्टि काट दें. Strike out in the case of first train during uneven flow and for all trains during even flow.

डाउन DOWN

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

क्रम सं Sr. No. _____

कंडीशनल लाइन क्लियर टिकट CONDITIONAL LINE CLEAR TICKET
(डाइवर/रिकार्ड Driver/Record)

Station _____ स्टेशन

दिनांक Date _____ 20 ____-

समय Time _____ बजकर Hours _____ मिनट Minutes

From

Station Master _____ स्टेशन मास्टर से

To

गाड़ी नं. The Driver of Train No. _____ डाउन DOWN विवरण Description _____ के डाइवर को.

लाइन क्लियर है और आपको _____ स्टेशन के लिए पहुंचने हेतु प्राधिकृत किया जाता है।

The line is clear and you are authorized to proceed to _____ Station.

*अंतिम गाड़ी

इस स्टेशन से *Last train left this station at _____ बजकर hours. _____ मिनट minutes. पर चली।

आपको गाड़ी के पीछे-पीछे चलने और _____ स्टेशन तक पहुंचने के लिए एतद्वारा प्राधिकृत किया जाता है।

You are hereby authorized to follow and proceed to _____ Station.

प्राइवेट नं. Private No. (शब्दों में in words) _____ (अंकों में in figures) _____.

**जा चुकी गाड़ियों का व्यौरा Particulars of trains left

गाड़ी नं. Train No.

समय Time

पीछे जाने वाली गाड़ियों का व्यौरा Particulars of trains to follow

गाड़ी नं. Train No.

समय Time

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

विशेष अनुदेश Special Instructions

जब कोई गाड़ी पीछा कर रही है जब कड़ी निगरानी रखें। जब दृश्यता साफ हो तब गति 25 कि.मी. प्र. घं. से अधिक न हो तथा जब दृश्यता किसी भी वजह से साफ न हो तब गति 10 कि.मी. प्र. घं. या इससे कम रखें। When following a Train keep sharp look out at speed not exceeding 25 kmph when view is clear and 10 kmph or less when view is impaired due to any reason.

*अनियमितता के दौरान दूसरी तथा इसके बाद की गाड़ियों की प्रविष्टि को काट दें। Strike out in case of second and subsequent trains during uneven flow.

**अनियमितता के दौरान पहली गाड़ी की प्रविष्टि तथा नियमितता के दौरान सभी गाड़ियों की प्रविष्टि काट दें। Strike out in the case of first train during uneven flow and for all trains during even flow.

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

**सिगनलों की दीर्घ कालीन खराबी के दौरान स्वचालित ब्लॉक पद्धति पर आगे बढ़ने के लिए प्राधिकार . AUTHORITY TO
PROCEED ON AUTOMATIC BLOCK SYSTEM DURING PROLONGED FAILURE OF SIGNALS**
(ड्राइवर/रिकॉर्ड Driver/Record)

Station _____ स्टेशन

दिनांक Date _____ 20 _____

समय Time _____ घंटे hrs. _____ मिनट minutes

ई एम यू / इंजन / गाड़ी * सं. _____ विवरण _____ अप / डाउन * के मोटर मैन / ड्राइवर को.
To The Motorman/Driver* of EMU/Engine/Train* No. _____ Description _____ Up / Down *.

_____ स्टेशन और _____ स्टेशन के बीच सभी सिगनल खराब हो गए हैं । अपने प्राइवेट नं. (शब्दों में)
_____ (अंकों में) _____ से _____ स्टेशन से लाइन क्लियर प्राप्त कर लिया गया है . All signals between
_____ Station and _____ Station have failed. Line Clear has been received from
_____ Station under his Private No.(in words) _____ (in figures) ____.

आपको अप / डाउन * लाइन पर _____ स्टेशन से _____ स्टेशन तक 25 कि.मी.प्रति घण्टा से अधिक गति न
रखते हुए सावधानी पूर्वक आगे बढ़ने के लिए एतद्वारा प्राधिकृत किया जाता है . You are, hereby, authorized to proceed cautiously
from _____ station to _____ Station on Up / Down* line at a speed not exceeding 25 Kmph.

आपको समपार फाटक पर सभी सावधानियों को बरतते हुए इस प्राधिकार पर ऑन स्थिति पर प्रस्थान सिगनलों, फाटक सिगनलों और अन्य
मध्यवर्ती सिगनलों को पार करने के लिए भी प्राधिकृत किया जाता है . You are also authorized to pass departure signals, Gate
signals and other intervening signals at 'ON' on this authority observing all precautions at the level crossing
gates.

Distinguishing number _____
Of Signals to be _____
passed at "on" _____

आपको _____ स्टेशन पर पहले रोक सिगनल के बाहर रुकना होगा और इसके बाद उस स्टेशन के स्टेशन मास्टर के अनुदेशों द्वारा
निर्देशित किया जाएगा . You shall stop outside the first Stop Signal at _____ Station and thereafter be
guided by the instructions of the Station Master of that station.

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

मैंने इस प्राधिकार की विषय-वस्तु को समझ लिया है . I have under stood the contents of this authority

गार्ड के हस्ताक्षर Signature of Guard _____ दिनांक Date _____ समय Time _____ घंटे hours _____ मिनट minutes

ड्राइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____ समय Time _____ घंटे hours _____ मिनट minutes

* जो लागू न हो उसे काट दें. Strike out which ever is not applicable.

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

क्रमसं.Sr.No. _____

**AUTHORITY TO PROCEED
FOR RELIEF ENGINE /TRAIN INTO AN AUTOMATIC BLOCK SIGNALLING SECTION**

(ड्राइवर / रिकॉर्ड Driver / Record)

Station _____ स्टेशन

दिनांक Date _____ 20 _____

समय Time _____ बजकर hours _____ मिनट minutes.

राहत इंजन / गाड़ी नं. _____ अप / डाउन के मोटर मैन / ड्राइवर को
To The Motor man / Driver of Relief Engine /Train No. _____ Up / Down.

आपको अप/डाउन *लाइन पर अपने राहत इंजन/गाड़ी * के साथ _____ स्टेशन _____ कि.मी. तक लाइन क्लियर के बिना सावधानी पूर्वक आगे बढ़ने के लिए प्राधिकृत किया जाता है . You are here by authorized to proceed cautiously without line clear from _____ station to up to Km. _____ with your Relief Engine/Train* on UP / DOWN* Line. आप अपनी राहत गाड़ी / इंजन * को _____ कि मी. पर खड़ा कर देंगे और उसके बाद साइट पर सक्षम प्राधिकारी के अनुदेशों द्वारा निर्देशित किया जाएगा . You shall bring your Relief Train / Engine* to a stand at Km. _____ and there after be guided by the instructions from the Competent Authority at site.

यह आदेश _____ कि. मी. पर अप / डाउन * लाइन पर अवरोध होने के कारण दिया जाता है । परन्तु _____ स्टेशन और _____ स्टेशन के बीच संचार व्यवस्था उपलब्ध है . This order is given due to obstruction on UP/DOWN* line at km _____. But communication is available between _____ station and _____ station. इस सेक्शन पर गाड़ी सं. _____ अंतिम गाड़ी थी । यह यहां से / पर _____ बजकर _____ मिनट चली / पहुंची . Train No. _____ was the last train over the section. It left / arrived* here at _____ hours _____ minutes.

सतर्कता आदेश CAUTION ORDER

आपको _____ स्टेशन से _____ स्टेशन तक अपने राहत इंजन / गाड़ी को दृश्यता साफ होने पर सीधी लाइन पर 25 कि. मी.प्रति घण्टा और सुरंग,धुमाव,अवरोध,बारिश,कोहरा या किसी अन्य कारण से दृश्यता साफ न होने पर 10 कि.मी.प्रति घण्टा से अधिक गति न रखते हुए चलाने की अनुमति दी जाती है . You are permitted to run your Relief Engine /Train from _____ station to km _____ with the speed not exceeding 25 kmph over the straight with clear view and 10 kmph when view ahead is impaired due to tunnel, curve, obstruction, rain, fog or any other cause.

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

मैंने इस प्राधिकार की विषय-वस्तु को समझ लिया है . I have under stood the contents of this authority

गार्ड के हस्ताक्षर Signature of Guard _____ दिनांक Date _____ समय Time _____ बजकर hours _____ मिनट minutes

ड्राइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____ समय Time _____ बजकर hours _____ मिनट minutes

*जो लागू न हो उसे काट दें.Strike out whichever is not applicable.

इस प्राधिकार पर चलने वाला ड्राइवर इस फॉर्म के पीछे उल्लिखित अनुदेशों का पालन अवश्य करे ।

Driver proceeding on this authority must observe instructions mentioned at the back of this form.

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

Sr. No. _____

स्वचालित ब्लॉक सिग्नल क्षेत्रों पर लाइन क्लियर के बिना आगे बढ़ने के लिए प्राधिकार AUTHORITY TO PROCEED WITHOUT LINECLEAR ON AUTOMATIC BLOCK SIGNALLING TERRITORIES

(डाइवर / रिकॉर्ड Driver / Record)

Station _____ स्टेशन

दिनांक Date _____ 20 _____

Time समय _____ बजकर hours _____ मिनट minutes

इंजन / गाड़ी / यूनिट नं. _____ अप/डाउन के मोटर मैन/डाइवर को.

To The Motor man / Driver of Engine / Train/ Unit No. _____ Up / Down.

आपको अप/डाउन लाइन पर अपनी गाड़ी के साथ या के बिना _____ स्टेशन से _____ स्टेशन तक लाइन क्लियर के बिना सावधानी पूर्वक आगे बढ़ने के लिए एतद्वारा प्राधिकृत किया जाता है . You are here by authorized to proceed cautiously without line clear from _____ station to _____ station with or without your train on UP / DOWN* Line.

आपको स्टेशन से संबंधित लाइन पर पहले रोक सिग्नल/अंतिम रोक सिग्नल से पहले अपनी गाड़ी/इंजन /यूनिट नं. _____ खड़ी कर देनी होगी और इसके बाद _____ स्टेशन के स्टेशन मास्टर के अनुदेशों द्वारा निर्देशित किया जाएगा . You shall bring your Train / Engine / Unit No. _____ to stand out side first Stop Signal/ Last Stop Signal pertaining to wrong line* of the station and there after be guided by the instructions from the Station Master of _____ station.

सेक्शन पर गाड़ी नं. _____ अंतिम गाड़ी थी. यह _____ बजकर _____ मिनट पर यहां से /पर चली / पहुंची .

Train No. _____ was the last train over the section. It left / arrived* here at _____ hours _____ minutes.

परिस्थितियां जिनमें यह प्राधिकार दिया जाता है . CIRCUMSTANCES UNDER WHICH THIS AUTHORITY IS GIVEN

* (क A) सिग्नलों एवं संचार व्यवस्था का खराब हो जाना . Failure of signals and communications.

* (ख B) _____ स्टेशन और _____ स्टेशन के बीच _____ कि.मी. पर लाइन का अवरोध होना / पर अवरोध होना और संचार साधनों का खराब हो जाना . Blocking of line / obstruction on line at KM _____ between _____ station and _____ station and failure of means of communication.

* (ग C) _____

‘ऑन’ स्थिति में सिग्नलों को पार करने के लिए प्राधिकार AUTHORITY TO PASS SIGNALS IN ‘ON’ POSITION

आपको रेल कर्मचारी द्वारा हाथ से सिग्नल देने पर ‘ऑन’ स्थिति में मध्यस्थी स्वचालित सं. _____, अर्धस्वचालित सिग्नल, हस्तचालित सिग्नल सं. _____ और ऑन स्थिति में फाट रोक सिग्नलों को खमपार तक सावधानी पूर्वक पार करने के लिए प्राधिकृत किया जाता है .

You are authorized to pass the Intervening Automatic Signal Nos. _____ at ‘ON’/Semi Automatic Signal, Manually Operated Signal Nos. _____ on being hand signaled by a railway employee and the Gate Stop Signals in ‘ON’ position cautiously up to the level crossing.

सतर्कता आदेश CAUTION ORDER

1. आपको _____ स्टेशन से _____ स्टेशन तक अपने इंजन / गाड़ी / यूनिट सं. _____ को दृश्यता साफ होने पर सीधी लाइन पर 25 कि.मी. प्रतिघण्टा और सुरंग, धुमाक, अवरोध, बारिश, कोहरा या किसी अन्य कारण से जब दृश्यता साफ न हो 10 कि.मी. प्रति घण्टा से अधिक गति न रखते हुए चलने को अनुमति दी जाती है . You are permitted to run your Engine/Train / Unit No. from _____ station to _____ Station with the speed not exceeding 25 kmph over the straight with clear view and 10 kmph when view ahead is impaired due to tunnel, curve, obstruction, rain, fog or any other cause.

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

मैंने इस प्राधिकार की विषय-वस्तु को समझ लिया है . I have under stood the contents of this authority.

गार्ड के हस्ताक्षर Signature of Guard _____ दिनांक Date _____ समय Time _____ बजकर hours _____ मिनट minutes

डाइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____ समय Time _____ बजकर hours _____ मिनट minutes

*जो लागू न हो उसे काट दें. Strike out whichever is not applicable.

इस प्राधिकार पर चलने वाले डाइवर इस फॉर्म के पीछे उल्लिखित अनुदेशों का पालन अवश्य करें ।

Driver proceeding on this authority must observe instructions mentioned at the back of this form.

दक्षिण पूर्वी मध्य रेलवे SOUTH EAST CENTRAL RAILWAY क्रम सं. Sr. No. _____

स्वचालित/अर्ध-स्वचालित/हस्तचालित/ फाटक रोक सिगनलों को पार करने के लिए प्राधिकार **AUTHORITY TO PASS AUTOMATIC / SEMIAUTOMATIC / MANUALLY OPERATED /GATE STOP SIGNALS**
(ड्राइवर/रिकार्ड Driver / Record)

Station _____ स्टेशन

दिनांक Date _____ 20__

समय Time _____ बजकर hours ____ मिनट minutes.

To

The Driver / Motorman of Train No. गाड़ी नं. _____ Up / Down. अप/डाउन के ड्राइवर/मोटरमैन को.

(क a) स्वचालित सिगनल खराब हो जाने पर आपको _____ स्टेशन और _____ स्टेशन के बीच सभी स्वचालित सिगनलों को पार करने के लिए प्राधिकृत किया जाता है । इसके अलावा आपको बर्दी में रेल कर्मचारी के द्वारा इन सिगनलों को पार करने पर अर्ध-स्वचालित/हस्तचालित/ फाटक रोक सिगनलों को पार करने के लिए भी प्राधिकृत किया जाता है । खण्ड 'ख' में यथा उपबंधित को छोड़कर सिगनल पर हथ-सिगनल दिखाया जाना है . Automatic Signalling has failed and you are, hereby authorized to pass all Automatic signals between _____ station and _____ station. In addition, you are also authorized to pass Semi Automatic / manually operated / gate stop signals on being hand signalled past such signals by a railway servant in uniform, the hand signals being displayed at the foot of signal except as provided for in clause 'b'.

(ख b) जब रांग दिशा में चल रहे हों (i) आप अवश्य पता लगाएं कि बाहरी साइडिंग के कांटे, यहां उपलब्ध कराए गए हों, सही तरीके से सेट करके ताला लगाया गया है और या समपार फाटक सं. _____ पर ताले लगाए गए हैं और (ii) कांटो और समपार फाटकों को पार करने से पहले बर्दी में रेल कर्मचारी द्वारा हथ-सिगनल दिखाए जाते हैं . When running in wrong direction (i) you must ascertain that points of outlying sidings, where provided, are correctly set and locked and/or the level crossing gates Nos. _____ are locked and (ii) hand signals are displayed by the railway servant in uniform at the points/ level crossing gates before passing them.

यहां पार किए जाने वाले सभी सिगनलों ;
के पहचान नं. का उल्लेख करें ;

Here indicates distinguishing)
Number of all signals to be
thus passed.}

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

मैंने इस प्राधिकार की विषय वस्तु को समझ लिया है . I have understood the contents of this authority.

गाड़ी के हस्ताक्षर Signature of Guard _____ दिनांक Date _____ समय Time _____ बजकर hours . _____ मिनट minutes.

ड्राइवर के हस्ताक्षर Signature of Driver _____ दिनांक Date _____ समय Time _____ बजकर hours . _____ मिनट minutes.

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY

क्रम सं. Sr. No. _____

पेपर लाइन क्लियर टिकट PAPER LINE CLEAR TICKET

डाउन Down

(डाइवर / रिकार्ड Driver / Record)

गाड़ी सं. Number of Train _____ डाउन Down विवरण Description _____.

दिनांक Date _____ समय Time _____ बजकर hours _____ मिनट minutes.

Line Clear received by means of (Mode of communication*) _____ के साधन द्वारा (संचार का प्रकार*)

From Station Master _____ स्टेशन मास्टर से

To The Driver of Train No. गाड़ी सं. _____ Down डाउन के डाइवर को

लाइन क्लियर है और आपको The line is clear and you are authorised to proceed to _____ station.
स्टेशन के लिए आगे बढ़ने के लिए प्राधिकृत किया जाता है।अन्तिम गाड़ी सं. Last train No. _____ ने स्टेशन पर सेक्शन क्लियर किया cleared section at _____
Station.

प्राइवेट नं. Private No. (शब्दों में in words) _____ (अकों में in figures) _____.

‘ऑन’ स्थिति में सिगनल पार करने के लिए प्राधिकार AUTHORITY TO PASS SIGNAL AT ON POSITION

* आपको अन्तिम रोक सिगनल को खतरे में पार करने हेतु प्राधिकृत किया जाता है, जब सिगनल ब्लॉक उपकरण के साथ इंटरलाक्ड हो.

*You are authorized to pass Last Stop Signal in danger, when the signal is interlocked with Block Instrument.

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

*जो लागू न हो उसे काट दें Strike out which ever is not applicable.

दक्षिण पूर्व मध्य रेलवे SOUTH EAST CENTRAL RAILWAY क्रम सं. Sr. No. _____

अप Up

पेपर लाइन क्लियर टिकट PAPER LINE CLEAR TICKET
(डाइवर/रिकार्ड Driver / Record)

गाड़ी सं. Number of Train _____ अप UP (विवरण Description) _____.

दिनांक Date _____ समय Time _____ बजकर hours _____ मिनट minute

Line Clear received by means of (Mode of communication*) _____ के साधन द्वारा (संचार का प्रकार*)

From Station Master _____ स्टेशन मास्टर से

To The Driver of Train No. गाड़ी सं. _____ UP अप के डाइवर को.

लाइन क्लियर है और आपको The line is clear and you are authorised to proceed to _____ station. स्टेशन के लिए आगे बढ़ने के लिए प्राधिकृत किया जाता है।

अन्तिम गाड़ी सं. Last train No. _____ ने _____ स्टेशन पर सेक्शन क्लियर किया cleared section at _____ Station.

प्राइवेट नं. Private No.(शब्दों में in words) _____ (अंकों में in figures) _____.

‘ऑन’ स्थिति में सिगनल पार करने के लिए प्राधिकार AUTHORITY TO PASS SIGNAL AT ON POSTION

- * आपको अन्तिम रोक सिगनल को खतरे में पार करने हेतु प्राधिकृत किया जाता है, जब सिगनल ब्लॉक उपकरण के साथ इंटरलाकड हो
- * You are authorized to pass Last Stop Signal in danger, when the signal is interlocked with Block Instrument.

स्टेशन मास्टर के हस्ताक्षर Signature of Station Master

स्टेशन मास्टर की मोहर Station Master Stamp

*जो लागू न हो उसे काट दें. Strike out which ever is not applicable.